

1 INTRODUCTION

Chest X-ray imaging remains central to thoracic disease screening because low cost, rapid acquisition, and broad availability support first-line assessment across emergency care, inpatient care, and outpatient care. Artificial intelligence has therefore become a major research direction in chest radiography, especially for automated detection of common and rare cardiopulmonary findings. Recent development has moved from conventional deep classifiers toward graph-based learning, transformer encoders, anatomy-aware tokenization, and multimodal language-guided models, as shown by Hu et al., [1], Marikkar et al., [2], Serra et al., [3], and Kang et al., [4]. Such development reflects an important change in research scope. Earlier work often focused on classification accuracy alone. Current work increasingly targets richer decision support, including confidence estimation, structured reasoning, and clinically meaningful evidence presentation. A doctoral study in chest radiography therefore requires a broad view of both algorithmic progress and remaining engineering limits.

Thoracic screening from chest X-rays also differs from many standard image- recognition tasks because a single study may contain several co-existing findings. Pulmonary opacity may appear together with pleural effusion, edema may co-exist with cardiomegaly, and one region may contain evidence relevant to more than one label. A single-label formulation therefore fails to match routine radiology practice. Multi-label learning offers a more realistic formulation, yet multi-label learning also raises new challenges related to label dependence, overlapping visual signatures, and competition for shared image evidence. Research on graph learning and label-dependency modeling has addressed part of the challenge through explicit representation of co-occurrence, spatial context, and discriminative label interactions, as reported by Cai et al., [5], Xu et al., [6], Zhang et al., [7], and Lu et al., [8]. A further complication concerns trustworthiness. Reliable screening requires more than strong ranking performance. Uncertain supervision, calibration quality, and stability under data shift also matter, as emphasized by Ghesu et al., [9] and Shekar et al., [10].

Uncertainty-aware and severity-informed modeling has become one of the most important strands in recent chest X-ray research. Conventional dataset construction often simplifies

radiology reports into hard positive or negative targets, yet routine radiology language frequently contains uncertainty, probability wording, and severity variation. Hu et al., [1] addressed a major limitation through a benchmark with clinical uncertainty and severity labels combined with multi-relationship graph learning. Zhang et al., [11] advanced a related direction through expert uncertainty and severity aware graph-based classification. Ghesu et al., [9] focused on explicit uncertainty quantification for chest radiograph assessment and showed value in reducing overconfident decisions. Ihler et al., [12] extended the discussion toward semi-supervised multi-label learning under imbalance, while Lanfredi et al., [13] improved dataset annotation through privacy-preserving large language models for extraction of richer label types, including severity and uncertainty. Taken together, such studies move chest X-ray learning closer to the language and ambiguity of real radiology reporting.

A critical reading of uncertainty-aware literature also reveals unresolved limits. Graph-based uncertainty-aware models provide closer alignment with radiology reasoning, yet graph design often depends on predefined relation structures and benchmark-specific annotation policies, a concern visible in Hu et al., [1] and Zhang et al., [11]. Standalone uncertainty scores can separate confident cases from doubtful cases, yet confidence values without grounded explanatory evidence may remain difficult to audit in clinical practice, a limitation associated with Ghesu et al., [9]. Semi-supervised learning and richer label extraction enlarge usable supervision, but calibration stability still depends on annotation quality, class prevalence, and transfer behavior across institutions, as suggested by Ihler et al., [12], Lanfredi et al., [13], and Shekar et al., [10]. Current uncertainty research has therefore delivered major progress, but full clinical dependability still requires tighter coupling among uncertainty estimation, evidence grounding, and robust representation learning.

Long-tailed disease prevalence forms another major challenge in multi-label thoracic screening. Chest X-ray datasets usually contain many high-frequency labels and a much smaller number of rare but clinically important labels. Standard training procedures often favor head classes and suppress tail sensitivity under such conditions. Holste et al., [14] placed the problem at the center of community benchmarking through the CXR-LT challenge. Huang et al., [15] addressed rare-class recognition together with fairness concerns through LTCXNet. Jeong et al., [16], Mau et al., [17], and Seo et al., [18] explored ensemble design, augmentation, and graph-enhanced long-tailed learning for

stronger performance under imbalance. Lai et al., [19] further examined long-tailed multi-label learning with noisy labels, adding a realistic view of report-derived supervision. Long-tailed research has therefore improved benchmark realism and encouraged direct evaluation of rare-class behavior rather than exclusive reliance on aggregate scores dominated by frequent findings.

Long-tailed learning literature nevertheless shows important tradeoffs. Many competitive solutions rely on ensembles, tailored sampling rules, or augmentation pipelines with higher design complexity and deployment cost, a pattern visible in Jeong et al., [16] and Mau et al., [17]. Some methods improve overall performance while leaving difficult tail classes weakly represented in learned feature space, a concern highlighted by Holste et al., [14] and Lai et al., [19]. Graph-based enhancement and class-balancing strategies improve sensitivity for rare findings, yet stronger rare-class performance does not automatically guarantee clinically meaningful explanations or stable fairness across subgroups, a limitation reflected in Seo et al., [18] and Huang et al., [15]. Long-tailed chest X-ray screening therefore remains both a data-distribution problem and a model-design problem, especially under multi-label conditions where rare findings often appear with common abnormalities.

Research on label dependency provides a direct response to multi-label complexity. Cai et al., [5] used graph convolutional networks to improve label semantics in chest X-ray classification. Xu et al., [6] combined group-wise spatial attention with label-dependency learning, linking local image structure with multi-task classification. Zhang et al., [7] proposed label-correlation-guided feature learning, while Lu et al., [8] combined transformer and graph components in CvTGNet. A common message runs across such studies: chest X-ray models benefit when learned representations reflect clinical co-occurrence and structural context instead of isolated binary outputs. Even so, dependency modeling introduces another engineering challenge. Stronger label coupling may improve macro-level performance, yet stronger coupling may also amplify dataset bias or annotation noise when learned correlations mirror benchmark frequency patterns rather than underlying pathophysiology. Dependency modeling therefore holds clear value, but effective dependency modeling requires anatomy-aware and uncertainty-aware control.

Transformer, anatomy-aware, and multimodal approaches have widened the design space for trustworthy chest X-ray screening. Marikkar et al., [2] showed competitive multi-label performance with a vision transformer. Serra et al., [3] introduced finding-aware

anatomical tokens for automated reporting, demonstrating the value of explicit anatomical structure in representation learning. Kang et al., [4] presented a large language model framework for broad chest X-ray understanding, while Alam et al., [20] connected concept bottlenecks with interpretable radiology report generation. Müller et al., [21] added interactive localization and region description, moving closer to clinician-facing assistance. A shared advantage of such models lies in richer context modeling across image regions and between image evidence and language. Anatomy-aware tokenization deserves special attention because anatomical regions provide a stable intermediate level between pixels and labels, a property with clear relevance for explanation, calibration, and future concept-level reasoning.

Progress in explainability has followed a similar pattern of expansion from simple visual highlighting toward more structured evidence. Bouabdallah et al., [22] proposed a complementary explanation approach for thorax multi-label classification. Talasila et al., [23] combined self-supervision and weak supervision to improve both accuracy and interpretability. Alam et al., [20] exposed intermediate semantic structure through concept bottlenecks for interpretable report generation. Müller et al., [21] demonstrated interactive localization and localized textual description. Explainability research in chest X-ray analysis now covers case-based reasoning, region grounding, concept-level description, and report-linked interpretation. Such movement is important because clinical acceptance depends on more than a colored heatmap. Radiology workflows benefit from explanations connecting image regions, semantic findings, and confidence estimates in a form usable during review, communication, and quality assurance.

Explainability literature still leaves several engineering questions open. Complementary and concept-based explanations improve structure, yet broad clinical validation remains limited, a limitation associated with Bouabdallah et al., [22] and Alam et al., [20]. Interactive localization supports clearer evidence review, yet workflow integration and evaluation at radiologist level remain incomplete, as suggested by Müller et al., [21]. Weakly supervised interpretability methods can improve anatomical focus, yet improved focus does not automatically guarantee faithful reasoning, a caution linked to Talasila et al., [23]. A more dependable explainability strategy therefore requires closer alignment among representation learning, anatomical grounding, concept structure, and calibrated uncertainty. Such alignment becomes especially important in multi-label screening, where overlapping findings can generate plausible but incomplete explanations when model

design lacks explicit control over evidence pathways.

Fairness and domain shift add a further layer of difficulty. A benchmark model with strong internal performance may still degrade across demographic groups, image- quality strata, scanner conditions, or institutional populations. Lin et al., [24] used contrastive learning to improve fairness in automated chest X-ray diagnosis. Moukheiber et al., [25] examined subgroup intersectional fairness in multi-label chest X-ray classification, extending analysis beyond simple one-factor subgroup comparisons. Shekar et al., [10] studied domain shift in chest radiograph classification within a veterans healthcare setting. Huang et al., [15] also linked long-tailed classification with fairness evaluation. Such work confirms a central lesson for trustworthy screening: performance stability across groups and sites deserves equal attention with accuracy. Without fairness-aware evaluation and domain-shift analysis, strong benchmark results may conceal unsafe variation in real deployment environments.

A synthesis across current literature shows major progress but also clear fragmentation. Uncertainty-aware learning has improved realism in label supervision through Hu et al., [1], Zhang et al., [11], Ghesu et al., [9], Ihler et al., [12], and Lanfredi et al., [13]. Long- tailed research has improved rare-class evaluation through Holste et al., [14], Huang et al., [15], Jeong et al., [16], Mau et al., [17], Seo et al., [18], and Lai et al., [19]. Label- dependency modeling has advanced through Cai et al., [5], Xu et al., [6], Zhang et al., [7], and Lu et al., [8]. Anatomy-aware and multimodal modeling has expanded through Marikkar et al., [2], Serra et al., [3], Kang et al., [4], Alam et al., [20], and Müller et al., [21]. Explainability, fairness, and domain-robust evaluation have grown through Bouabdallah et al., [22], Talasila et al., [23], Lin et al., [24], Moukheiber et al., [25], and Shekar et al., [10]. Most contributions, however, still optimize one part of trustworthiness at a time rather than an integrated set of capabilities.

A doctoral thesis on trustworthy multi-label thoracic screening from chest X-rays can therefore occupy a valuable research position. Current literature suggests strong need for frameworks combining anatomy-aware representation learning, counterfactual and clinically meaningful explanation, structured concept-level reasoning, uncertainty- aware prediction, and reliable behavior under long-tailed and shifted conditions. Serra et al., [3] support the value of anatomy-aware structure. Hu et al., [1], Zhang et al., [11], and Ghesu et al., [9] support the value of uncertainty-aware modeling. Alam et al., [20] and Müller et al., [21] support the value of concept and region grounded explanation. Lin et al., [24] and

Shekar et al., [10] support the value of fairness-aware and shift-aware evaluation. A thesis built around such integration can respond directly to the current research landscape, not by seeking another benchmark score alone, but by advancing a coherent engineering framework for clinically dependable chest X-ray decision support.

1.1 Problem Statement

Automated thoracic screening from chest X-rays has advanced quickly, yet dependable clinical deployment remains unresolved. Strong benchmark scores have shown that deep models can detect several thoracic abnormalities from a single image, but screening practice places a different demand on model behavior. A frontal chest radiograph may contain pleural effusion, pulmonary opacity, edema, cardiomegaly, atelectasis, or other findings in overlapping combinations, and each finding may influence the visual expression of another finding. Such co-occurrence creates label dependence, shared visual evidence, and clinically meaningful interaction among findings. Independent binary prediction cannot represent such structure adequately. Multi-label graph and attention models have improved representation of spatial and label relationships, but benchmark improvement alone does not satisfy screening requirements because screening decisions require reliable recognition of multiple findings under uncertainty, not only strong average discrimination, as shown by Xu et al., [6], Hu et al., [1], and Zhang et al., [11].

A second difficulty arises from supervision quality. Large chest X-ray datasets are commonly labeled from radiology reports rather than direct image-level expert annotation. Report language often contains uncertainty terms, severity qualifiers, hedging expressions, and omission of subtle abnormalities. Automated label extraction can add further inconsistency through rule-based or language-model-based simplification. Under such conditions, a hard positive or negative label can conceal clinical ambiguity and can push model training toward overconfident outputs, especially near decision thresholds. Richer annotation strategies and uncertainty-aware learning have improved alignment with radiology reporting, yet remaining label ambiguity continues to weaken calibration and error awareness. A prediction score with high numeric confidence may therefore correspond to weak clinical evidence rather than dependable certainty. Such mismatch matters in triage, prioritization, and second-reader support, as reported by Lanfredi et al., [13], Hu et al., [1], Zhang et al., [11], and Ghesu et al., [9].

A third difficulty arises from disease prevalence. Thoracic abnormality distributions are strongly long-tailed. Common findings dominate training data, whereas rare but clinically

important findings appear far less often. Rare-class weakness is not a minor benchmark issue. In screening and triage, missed rare findings can delay escalation, referral, or follow-up, and a model that performs well only on frequent labels offers limited clinical value. Long-tailed benchmark work has highlighted persistent underperformance on low-frequency labels, especially when noisy supervision and multi-label co-occurrence interact in the same image. A rare finding that appears beside a frequent finding may receive weaker representation because head-class patterns dominate shared features during training. Current balancing, augmentation, and challenge-driven benchmarking have improved awareness of the problem, but rare-class reliability and rare-class calibration remain insufficient for dependable screening, according to Holste et al., [14] and Lai et al., [19].

A fourth difficulty concerns explanation quality. Clinical use requires evidence that radiologists can inspect, question, and relate to anatomy and known thoracic concepts. Many current explainability methods still depend on saliency maps, complementary visualizations, or case-based evidence that may improve transparency without guaranteeing faithful clinical reasoning. Broad heatmaps, weak localization, and limited concept-level interpretation reduce usefulness during audit, disagreement resolution, and workflow adoption. A clinician facing a high-risk output needs more than highlighted pixels; a clinician also needs a plausible link between image regions, abnormal patterns, and final labels. More structured explanation approaches, including complementary explanation and concept bottleneck strategies, have moved beyond simple saliency, yet strong clinical faithfulness and consistent integration with multi-label prediction remain limited, as discussed by Bouabdallah et al., [22] and Alam et al., [20].

A fifth difficulty concerns operational reliability under real deployment conditions. A model with strong internal test performance may still fail under new imaging protocols, scanner conditions, hospital populations, or demographic subgroups. Distribution shift can change image appearance, prevalence patterns, disease presentation, and label extraction behavior. Subgroup disparities can also remain hidden behind global averages. Weak calibration under such conditions is especially dangerous because overconfident error can produce false reassurance in triage, delayed referral, or inappropriate automation bias in decision support. Fairness-focused and domain-shift studies in chest radiography have shown that subgroup gaps and transfer degradation remain substantial, even after accuracy gains on benchmark datasets. Reliable screening therefore requires not only average performance but also stable confidence and stable error behavior across

population slices and institutional settings, as shown by Lin et al., [24], Moukheiber et al., [25], and Shekar et al., [10].

Taken together, current research has produced important advances in label dependency modeling, uncertainty-aware learning, long-tailed benchmarking, explainability, fairness analysis, and robustness evaluation. A major limitation, however, lies in fragmentation across research directions, with uncertainty, explanation, rare-class learning, and robustness often studied separately, as reflected across Hu et al., [1], Holste et al., [14], Bouabdallah et al., [22], and Shekar et al., [10]. One group of studies improves uncertainty handling without clinically grounded explanation. Another group improves explanation without strong calibration. A separate group targets rare-class learning or subgroup fairness without resolving multi-label consistency and faithful evidence presentation. Clinical screening, by contrast, requires simultaneous handling of co-occurring findings, ambiguous supervision, rare-label sensitivity, explanation faithfulness, calibrated confidence, and stability across domains and subgroups. Therefore, the central problem addressed in this doctoral research is the design and validation of an integrated framework for multi-label thoracic screening from chest X- rays that can model dependent and co-occurring findings under weak and long-tailed supervision, produce clinically faithful and anatomically grounded explanations, generate calibrated uncertainty-aware decisions, and maintain reliable performance across shifted domains and patient subgroups.

1.2 Research Gaps

Recent chest X-ray research has achieved strong progress in multi-label classification, graph learning, transformers, explainability, uncertainty quantification, long-tailed learning, and fairness analysis. Progress across research streams, however, remains fragmented. Many studies optimize a single dimension of trustworthiness while leaving other dimensions outside model design. Uncertainty-aware classifiers often stop at confidence scores. Explainability studies often stop at visual evidence. Fairness and domain-shift studies often appear as external audits rather than design principles. Rare-class learning is usually handled as a separate benchmark problem. Clinical screening requires joint performance across accuracy, explanation, calibration, robustness, and operational safety. Benchmark leadership in isolated tasks has therefore not translated into a complete trustworthiness framework for screening pathways. Fragmentation therefore creates a major research gap because deployment decisions depend on integrated behavior

under real workflow pressure, not on isolated metric gains on selected benchmarks.

Explainability research has moved beyond basic saliency maps toward attention visualizations, case-based evidence, concept bottlenecks, counterfactuals, and localized text-grounded outputs. Recent progress has improved transparency and has made model outputs easier to inspect. Even with recent progress, clinical faithfulness remains limited. Broad heatmaps can still highlight non-diagnostic regions. Attention maps do not guarantee causal evidence. Case-based explanations can depend on superficial similarity rather than pathology-relevant similarity. Concept-level outputs often lack strong clinical validation. Counterfactual explanations remain promising but often face questions about plausibility, realism, and stability in multi-pathology images. Clinical adoption needs explanation outputs usable during routine reading, teaching, documentation, and quality assurance. Chest X-ray screening needs explanations linking anatomy, abnormality patterns, and final decisions in support of audit, disagreement review, and safe escalation. Limited clinical faithfulness therefore remains an important gap between model transparency and model trust.

Uncertainty-aware modelling has advanced through evidential learning, severity-aware labels, expert uncertainty annotations, and semi-supervised strategies recognising ambiguity in radiology reports. Research in uncertainty-aware modelling has shown clear value in reducing overconfidence and improving alignment with real reporting language. A major gap remains, however, in complete modelling of uncertainty under weak and ambiguous labels. Many datasets still compress nuanced report language into simplified targets. Severity information, report hedging, and inter-reader disagreement remain unevenly represented across major datasets. Uncertainty is often treated as a training-label issue or as a post hoc score, rather than as a full design principle linked to representation learning, explanation, and decision control. Calibration under distribution shift also remains insufficiently studied. In practical screening, an uncertainty estimate has value only with meaningful evidence alignment and support for safe actions such as abstention, escalation, or conservative reporting. Incomplete uncertainty modelling therefore limits clinical reliability even with strong classification accuracy.

Anatomy-aware learning and structured label reasoning have each produced valuable advances. Anatomical tokens, region-based attention, and anatomy-guided encoders have improved spatial grounding. Graph models and label-correlation learning have improved handling of co-occurring findings and semantic dependence across labels. Integration

between anatomy-aware learning and structured reasoning, however, has not yet been adequately developed. Many anatomy-aware models improve localization without enforcing cross-label consistency. Many graph-based models improve label relations without grounding relational reasoning in explicit thoracic regions. A missing link remains between region-level evidence and concept-level decision logic. Location-aware evidence remains essential for findings such as effusion, basal opacity, and cardiomegaly. The missing link matters because thoracic abnormalities follow anatomical patterns, and clinical interpretation depends on both location and relation. Without combined anatomical grounding and structured reasoning, multi-label systems can produce internally plausible but clinically inconsistent outputs, especially in images with overlapping findings.

Long-tailed learning has become a visible research theme through dedicated benchmarks, adaptive sampling, augmentation, ensemble methods, and noise-aware training. Progress in rare-class recognition has improved awareness of imbalance beyond aggregate performance scores. Benchmark design has improved problem visibility, yet benchmark design alone cannot solve representation bias. Despite progress, unresolved challenges remain in the joint handling of long-tailed prevalence, rare-class detection, label co-occurrence, and noisy supervision. Rare findings often appear together with frequent findings, so head-class features can dominate shared representations. Many competitive solutions also rely on complex ensembles or heavy augmentation pipelines, with reduced interpretability and weaker deployment efficiency as possible costs. Metric gains on tail classes do not automatically confirm clinically reliable detection of uncommon but high-risk conditions. A remaining research gap therefore concerns scalable and interpretable methods improving rare-class sensitivity without loss of calibration, explanation quality, or stability under weak labels.

Fairness and domain-shift research has begun to expose important limits of current chest X-ray systems. Subgroup studies, balanced retraining, contrastive learning, and transfer analysis have shown performance variation across demographic strata, image quality conditions, scanner protocols, and institutions. Even with recent progress, fairness and robustness evaluation remains limited in breadth and depth. Many studies still report results on narrow internal test sets. External validation is often brief, and subgroup reporting is often secondary rather than primary. Intersectional subgroup analysis remains

underexplored. Robustness assessment often focuses on performance drop alone rather than on combined effects on calibration, abstention behavior, and explanation stability. Real-world deployment requires dependable behavior across varied patient populations and imaging conditions, not only good average results on a single benchmark. Limited fairness and domain-shift evaluation therefore leaves a serious deployment gap because hidden subgroup overconfidence can create unequal referral patterns and unsafe automation bias.

Current literature therefore justifies a doctoral research programme built around integrated design rather than isolated optimisation. Anatomy-aware representation learning can ground evidence in thoracic structure. Counterfactual reasoning can provide clinically meaningful healthy-reference explanations. Neuro-symbolic modelling can connect image evidence with concept consistency across related findings. Uncertainty-aware prediction can support calibration, abstention, and conservative label sets for screening workflows. A doctoral research programme centred on combined development and evaluation of anatomy-aware representation learning, counterfactual reasoning, neuro-symbolic modelling, and uncertainty-aware prediction is therefore well justified by current gaps in fragmentation, explanation faithfulness, uncertainty handling, structured reasoning, rare-class reliability, and deployment robustness.

1.3 Motivation

Chest X-ray imaging remains a major entry point for thoracic assessment because low cost, rapid acquisition, and availability support use across care settings. Thoracic screening from chest X-rays also carries high practical value in environments with high case volume, limited specialist availability, and urgent triage needs. A single radiograph may contain more than one clinically relevant finding, including opacity, pleural abnormality, edema, cardiomegaly, or atelectatic change. Multi-label screening therefore matches real radiology practice more closely than single-label classification. Clinical motivation arises from a clear need: support systems for chest radiography must help identify several possible abnormalities in one image while preserving safety, clarity, and usefulness for downstream clinical action.

Clinical adoption of automated screening depends on trust. A chest X-ray model can score highly on a benchmark and still remain difficult to use in practice. Screening environments require more than ranking accuracy because triage, referral, escalation, and quality assurance depend on dependable case-level behavior. A false negative can delay care for

a serious thoracic condition. A false positive can trigger unnecessary follow-up, workload, and reduced confidence in automated support. An unclear model output can slow decision making rather than improve efficiency. Practical motivation therefore comes from workflow reality: radiology support systems must not only detect abnormalities, but must also present evidence in a reviewable form for audit, disagreement analysis, and decision making under uncertainty.

Scientific motivation follows a similar line. Recent chest X-ray research has achieved strong gains in classification performance through graph learning, transformers, ensembles, and multimodal architectures. Progress in benchmark performance has demonstrated strong feature learning capacity, but accuracy-only modelling still leaves major engineering questions unanswered. Average AUROC or macro F1 cannot fully describe screening quality in a setting with co-occurring findings, weak labels, long-tailed prevalence, and variable image quality. A model may detect common labels well and still underperform on rare labels. A model may achieve strong discrimination and still remain poorly calibrated. A model may highlight image regions and still fail to provide faithful clinical reasoning. Scientific motivation therefore centers on a shift in design philosophy, from prediction-only systems toward trustworthy decision-support systems. Such a shift requires joint attention to representation learning, confidence estimation, explanation quality, and behavior under deployment conditions.

Uncertainty quantification forms a central part of the motivation. Chest X-ray labels often come from radiology reports with hedging language, uncertain wording, and incomplete mention of subtle findings. Ambiguous supervision creates a natural risk of overconfident training and brittle threshold decisions. Uncertainty estimation offers a mechanism for separating strong evidence from weak evidence, and confident predictions from doubtful predictions. Such separation matters in triage because low-confidence cases may require escalation, secondary review, or conservative reporting. Interpretability provides a second essential requirement. Radiology practice depends on visible evidence, anatomical reasoning, and understandable justification for abnormality claims. A model output without grounded explanation offers limited value during audit, teaching, or disagreement resolution. Fairness and robustness provide further motivation. Chest X-ray deployment spans different patient groups, hospitals, scanners, and acquisition conditions. Safe clinical AI therefore requires stable behavior across subgroup variation and distribution shift, not only good performance on a single internal test set. Clinical adoption depends on

confidence in reliable model behavior when image quality changes, prevalence changes, or patient mix changes.

Methodological motivation also comes from timing. Separate research streams now provide important building blocks for a stronger integrated framework. Anatomy-aware encoders offer better region grounding. Graph and label-dependency models offer better handling of co-occurring findings. Counterfactual and concept-based explanation methods offer richer evidence than basic saliency maps. Evidential and Bayesian-style uncertainty methods offer stronger control of overconfidence. Fairness and domain-shift studies offer a clearer view of deployment risk. A meaningful opportunity now exists because field maturity has moved beyond proof of basic feasibility. Current research conditions support a new stage focused on integration of complementary strengths into a unified engineering design. Such integration remains necessary because isolated advances rarely produce a complete screening system suitable for dependable clinical use.

The thesis direction follows directly from the opportunity. Anatomy-aware representation learning deserves emphasis because thoracic abnormalities follow strong spatial patterns. Pleural findings, lower-zone opacities, diffuse edema, and cardiomedial enlargement all depend on location-aware evidence. Region-structured learning can therefore improve both recognition and explanation. Counterfactual explanation deserves emphasis because comparison with a plausible healthy reference matches a clinically meaningful radiology question and can produce residual evidence maps with clearer pathological focus than many post hoc heatmaps. Neuro-symbolic consistency modelling deserves emphasis because multi-label thoracic

screening involves related concepts rather than isolated binary events. Concept graphs and soft clinical constraints can reduce contradictory outputs and expose intermediate reasoning states. Uncertainty-aware prediction deserves equal emphasis because screening decisions need more than probabilities. Calibrated uncertainty, abstention, and conservative label sets can support safer automation in workflows with ambiguous evidence, rare presentations, and domain shift.

A doctoral research programme built around anatomy-aware representation learning, counterfactual explanation, neuro-symbolic consistency modelling, and uncertainty-aware prediction offers a coherent response to current clinical and scientific needs. Such a programme aligns model design with thoracic anatomy, aligns explanation with clinically meaningful evidence, aligns label prediction with structured reasoning, and aligns

confidence with operational safety. Current progress across chest X-ray AI has created a timely opening for integrated trustworthy system design. Timeliness, necessity, and likely contribution therefore arise from a simple engineering fact: clinically useful chest X-ray screening demands accurate prediction, reliable uncertainty, grounded explanation, and robust behavior in one unified framework.

1.4 Summary of the Chapter

Chapter One established trustworthy multi-label thoracic screening from chest X-rays as a significant research area at the intersection of medical imaging, machine learning, and clinical decision support. The chapter clarified the multi-label nature of chest X-ray screening, since several thoracic findings may appear together in one image and screening quality depends on reliable recognition of interacting abnormalities. Chapter One also framed trustworthiness as a broader requirement than classification accuracy alone, including calibrated uncertainty, clinically meaningful explanation, robust handling of weak and noisy supervision, sensitivity to rare findings, and stable performance across varied imaging conditions and patient groups.

The central problem identified in Chapter One concerns the absence of an integrated screening framework able to learn from uncertain and long-tailed data, model label dependence, produce anatomically grounded and faithful explanations, and maintain reliable confidence under subgroup variation and domain shift. Gap analysis showed fragmentation across current research streams, with uncertainty modelling, explainability, anatomy-aware learning, structured reasoning, long-tailed classification, fairness, and robustness often addressed in isolation. Clinical motivation arises from the need for screening systems supporting triage, prioritization, and safer decision support in high-volume settings. Methodological motivation arises from recent advances now making integrated framework design feasible, especially through anatomy-aware representation learning, counterfactual explanation, neuro-symbolic consistency modelling, and uncertainty-aware prediction. The remaining chapters therefore move from literature framing toward methodological development, system implementation, empirical evaluation, and final contribution analysis. Subsequent chapters will present the proposed framework, describe experimental design and implementation choices, evaluate accuracy, calibration, explanation, robustness, and fairness, and conclude with the overall doctoral contribution to trustworthy clinical AI for chest X-ray screening.

1.5 Organization of the Thesis

Chapter 1, Introduction, presents the background of chest X-ray based thoracic screening and frames the doctoral problem around trustworthy multi-label prediction. The chapter defines the problem statement, research gaps, motivation, research objective, scope, significance, and the overall structure of the thesis.

Chapter 2, Literature Review, provides a critical review of research on uncertainty-aware classification, long-tailed and noisy-label learning, label-dependency modelling, explainability, anatomy-aware and vision-language methods, fairness, and domain shift in chest X-ray AI. The chapter synthesizes current knowledge and consolidates the gaps that motivate the proposed doctoral contribution.

Chapter 3, NEPTUNE-CXR: A Neuro-Symbolic, Counterfactual, Uncertainty-Aware Model for Multi-Label Thoracic Disease Screening on Chest X-Rays, presents the main methodological contribution of the thesis. The chapter describes the anatomy-aware zoning module, zone-aware transformer encoder, counterfactual healthy normalization, neuro-symbolic concept graph, and uncertainty-aware prediction framework, followed by experimental design, implementation, results, ablation studies, and robustness analysis.

Chapter 4, Summary and Future Work of the Thesis, consolidates the main findings and original contributions of the doctoral research. The chapter also discusses the limitations of the study and outlines future directions such as broader external validation, stronger concept reasoning, and further deployment-oriented evaluation for trustworthy clinical AI.

The References section presents the complete list of scholarly sources cited in the thesis. The section supports academic traceability, verification of evidence, and consistency of the doctoral research document.

2 LITERATURE REVIEW

This chapter provides the scientific foundation for the thesis by examining how chest X-ray research has progressed from conventional disease classification toward trustworthy multi-label thoracic screening. Chest X-ray screening is not a simple single-label task, because one image may contain several related findings at the same time, while labels are often uncertain, incomplete, or noisy. For that reason, the chapter reviews the main technical directions that shape current research, including uncertainty-aware learning, long-tailed disease classification, label-dependency modelling, explainability, transformer and vision-language methods, fairness, and domain shift. The chapter also examines how these directions contribute to clinical reliability, interpretability, and safe decision support in real screening settings. A central aim of the chapter is to move beyond a descriptive list of published studies and build a clear technical argument for the doctoral research. The review therefore identifies what current methods have achieved, where important weaknesses remain, and why those weaknesses matter for clinical deployment. Particular attention is given to gaps in calibration, explanation faithfulness, rare-class performance, subgroup reliability, and integration of anatomical structure with structured reasoning. On that basis, the chapter establishes the need for a more unified framework for trustworthy chest X-ray screening, which directly motivates the proposed thesis direction based on anatomy-aware representation learning, counterfactual reasoning, neuro-symbolic modelling, and uncertainty-aware prediction.

Nie et al. [26] addresses multi-label thoracic disease classification from chest X-rays under limited labeled data and proposes a prior-guided deep reinforcement learning framework instead of a conventional fixed classifier. The main contribution is a multi-agent design that combines semantic prior agents, visual prior agents, a main diagnostic agent, transformer-based multi-information fusion, and epsilon-greedy exploration so that learning can continue as more data arrive. Methodology relies on prior knowledge from pre-trained models or related domains to reduce dependence on target-domain labels, followed by evaluation on NIH ChestXray14 and CheXpert with AUC-based comparison. Reported results are competitive on both benchmarks, with better generalization and stronger few-shot behavior than standard training strategies. A major

merit is the attempt to model diagnosis as a guided sequential decision process, a useful design when inter-class similarity is high. Possible limitations include simple agent interaction rules, higher computational cost, slower training than standard classifiers, and uncertain benefit when transferred prior knowledge is weak or poorly matched to the target task.

Marikkar et al. [2] aims to improve pure vision-transformer performance for multi-label chest X-ray classification and introduces LT-ViT as a label-token transformer tailored to chest radiography. The core contribution is a combined-attention mechanism between image tokens and randomly initialized auxiliary label tokens, allowing label-specific evidence aggregation inside the model rather than through post hoc heatmaps. Methodology uses a vision-only transformer pipeline and tests the design on two public chest X-ray datasets under different pre-training settings, with emphasis on initialization-agnostic behavior. Reported results show that LT-ViT surpasses prior pure ViT baselines and remains effective across multiple pre-training choices. A strong merit is the built-in interpretability, since label tokens expose model attention without Grad-CAM or related add-on methods. Possible limitations include the absence of explicit uncertainty modeling, unclear robustness under domain shift, and the usual transformer burden in memory and data demand, especially when comparison extends beyond pure ViTs to stronger hybrid CNN-transformer models.

Zhang et al. [11] focuses on a central weakness in chest X-ray learning, namely unrealistic hard labels extracted from reports despite frequent severity variation and diagnostic uncertainty in radiology. The main contribution is a two-part framework: rule-based label re-extraction with expert-curated severity and uncertainty terms, and a multi-relationship graph learning model equipped with an expert uncertainty-aware loss. Methodology therefore improves both supervision quality and label-dependency modeling, while also adding interpretability through graph-based disease relationships. Reported results show that severity-aware and uncertainty-aware graph learning outperforms earlier state-of-the-art approaches, indicating that more realistic labels improve both accuracy and explanation quality. A major merit lies in closing the gap between raw report language and model supervision. Possible limitations include dependence on rule-based keyword design, sensitivity to incomplete or institution-specific terminology, and reliance on predefined disease relations that may not transfer cleanly across hospitals or reporting styles.

Kang et al. [4] targets a broader chest X-ray understanding problem and argues that report-

only vision-language pipelines are too narrow for real clinical reasoning. The main contribution is WoLF, a wide-scope large language model framework that brings together multi-faceted patient records, anatomy-structured report decomposition, masked-attention report generation, and an AI-evaluation protocol designed for large language model assessment rather than surface-level text similarity alone. Methodology uses Electronic Health Records to generate instruction-following data and tests the framework on MIMIC-CXR for visual question answering and report generation. Reported results show clear gains, including up to 9.47 percentage points in VQA mean score and 7.3 points in BLEU-1 for report generation. A major merit is the move from image-report pairing toward richer patient context and stronger evaluation logic. Possible limitations include dependence on complete EHR availability, higher data- engineering and compute cost, and possible hallucination or calibration problems that remain common in large language model systems.

Moukheiber et al. [25] addresses fairness in multi-label chest X-ray classification and moves beyond coarse protected attributes toward intersectional subgroup analysis shaped by social determinants of racial health inequities. The key contribution is a fairness framework that retraining the last classification layer of pre-trained models using balanced data across groups, adds fairness constraints, and applies class-balanced fine-tuning for multi-label prediction. Methodology uses MIMIC-CXR and evaluates the tradeoff between diagnostic accuracy and fairness across more granular subgroups than standard race-only or sex-only studies. Reported results indicate an improved balance between fairness and accuracy relative to baseline methods. A major merit is the sharper fairness lens, since subgroup intersections capture bias patterns that single-attribute analysis can miss. Possible limitations include partial correction of bias because last-layer retraining may leave backbone-level representation bias unchanged, strong dependence on subgroup metadata quality, and limited evidence on transfer of the fairness benefit beyond one dataset and one institutional context.

Wang et al. [27] studies robust image classification under variable annotation quality and offers a strong answer to noisy supervision in chest X-ray datasets. The main contribution is a learning-to-vote framework with a label-sampling module that refines labels on the fly through a weighted combination of multiple annotation sources, supported by additional back-propagation during meta-training. Methodology first validates the idea on synthetic noisy CIFAR-10 and then applies the approach to MIMIC-CXR and OpenI in both image-

only and image-text settings. Reported results are concrete and meaningful: average chest X-ray classification AUC rises by about 3– 5%, and performance on OpenI improves by about 4.5% even without direct training on that dataset. A major merit is practical robustness to imperfect NLP-derived labels and annotator disagreement. Possible limitations include extra training complexity, dependence on availability of multiple label sources, and uncertainty about performance when label sources are highly correlated or uniformly poor.

Gao et al. [28] focuses on Acute Respiratory Distress Syndrome detection, a setting where expert disagreement and stage ambiguity create serious label-noise problems. The main contribution is a training strategy that treats multi-annotator label uncertainty as privileged information and injects that information into a Transfer and Marginalized network with knowledge-transfer mechanisms during training. Methodology keeps uncertainty information available during learning but not during inference, then compares the approach against original TRAM, confusion estimation, and mean- aggregated label training. Reported results are strong: mean AUROC reaches 0.850, AUPRC 0.868, and F1 0.797, while removal of equivocal cases lifts performance to AUROC 0.973, AUPRC 0.971, and F1 0.921. A major merit is the direct use of annotator disagreement as useful supervision rather than as unwanted noise. Possible limitations include dependence on the quality and consistency of expert labels, reduced flexibility when privileged uncertainty is unavailable, and narrow clinical scope because ARDS-specific behavior may not extend automatically to broader thoracic disease screening.

Cai et al. [5] aims to improve multi-label chest X-ray classification beyond simple label co-occurrence and introduces a Label Semantic Improvement Graph Convolutional Network for richer disease-relation modeling. The contribution combines an image representation learning module, a graph convolution module with semantic enhancement, and a distance-metric module so that label relations and visual relations are optimized together. Methodology extracts high-level label-specific image features, propagates semantic relations through the graph, and then sharpens feature discrimination with metric learning. Reported results show higher mean AUC than comparative graph-based baselines, including stronger performance on ChestX-ray14 and CheXpert. A clear merit is the attempt to model global label correlation and visual correlation simultaneously, which is important for co-existing thoracic findings. Possible limitations include reliance on graph structures that may reflect dataset statistics more than clinical causality, unclear scalability

cost, and limited evidence on external generalization beyond chest radiography benchmarks.

Qiu et al. [29] tackles interpretability through direct modeling of radiologist visual behavior and proposes joint disease diagnosis and clinical visual attention prediction for chest X-rays. The major contribution is a dual-encoder multi-task U-Net that combines a DenseNet201 backbone, a residual and squeeze-and-excitation encoder for attention-map prediction, a multi-scale feature-fusion classifier for disease classification, and a multi-stage cooperative learning strategy with contrastive pre training to handle asynchronous task optimization. Methodology therefore couples diagnosis with gaze-based saliency prediction instead of treating explanation as a separate post hoc step. Reported results are strong and specific, with AUC reaching 0.93 for diagnosis and correlation coefficient reaching 0.58 for visual attention prediction, both above existing techniques. A major merit is closer alignment between model evidence and clinician attention patterns. Possible limitations include reliance on gaze or saliency supervision, heavier training complexity than single-task models, and uncertain portability to hospitals without eye-tracking data.

Chun et al. [30] targets multi-label lesion detection and localization in abnormal chest X-rays and proposes a detection framework intended to approach radiologist-level performance. The main contribution is an advanced CNN-Swin Transformer module that combines global and local context, moving-window pixel-level self-attention for multi-scale feature capture, and a channel-plus-spatial permutation attention mechanism for stronger emphasis on salient lesion regions. Methodology is evaluated on VinDr-CXR, a dataset suited to localization because radiologist annotations support lesion-level analysis rather than image-level labels alone. Reported results show a 2.3% gain in mean Average Precision over strong baselines, indicating real improvement in lesion detection quality. A major merit is stronger handling of small targets and visually similar lesion categories, both of which are common failure points in chest X-ray detection. Possible limitations include substantial model complexity, uncertain transfer from a lesion-detection benchmark to routine screening workflows, and lack of explicit uncertainty calibration for safety-critical deployment.

Du et al. [31] studies a central problem in chest X-ray learning, namely the loss of useful clinical information when uncertain labels are ignored or forced into hard binary targets. The main contribution is a transformer-based multi-disease diagnostic framework that learns

from both certain and uncertain labels so that label dependencies are modeled under a more realistic supervision setting. Methodologically, the study uses a transformer architecture for multi-label chest X-ray diagnosis and treats uncertain and definite labels as complementary signals rather than as conflicting noise. Reported results show better performance than approaches trained only on definite labels, which indicates that uncertainty-aware supervision can strengthen dependency learning and improve prediction quality. A major merit is the direct alignment with radiology practice, where ambiguous labels are common in report-derived datasets. Possible limitations include limited evidence on calibration quality, limited discussion of explanation faithfulness, and unclear robustness under external domain shift, especially because conference-length papers often provide only compact experimental detail.

Talasila et al. [23] aims to improve chest X-ray classification accuracy and interpretability at the same time by reducing dependence on fully supervised training. The main contribution is a self-supervised and weakly supervised pretraining pipeline with an auxiliary loss, followed by supervised fine-tuning, so that learned features remain useful across datasets and also produce more anatomically meaningful activations. Methodologically, the study uses ChestX-ray14 for pretraining and CheXpert for fine-tuning, then evaluates both classification performance and explanation quality. The reported results are specific and strong, with the model outperforming supervised state-of-the-art baselines across 14 findings and improving Intersection over Union by 31% on the NIH chest X-ray dataset. A major merit is the combination of performance gain with better localization behavior under limited bounding-box supervision. Possible limitations include dependence on large pretraining data, limited validation beyond chest radiography, and limited analysis of reliability factors such as calibration, fairness, and out-of-distribution stability.

Holste et al. [14] addresses the long-tailed and multi-label nature of chest X-ray diagnosis by creating a benchmark and challenge rather than proposing a single classifier. The central contribution is the CXR-LT challenge, built on a very large public dataset of more than 375,000 chest X-rays, enriched with new disease findings, a human-annotated gold-standard subset, and a synthesis of successful solutions for rare-class recognition. Methodologically, the work combines large-scale dataset curation, community evaluation, challenge leaderboard analysis, and comparison of winning strategies such as high-resolution training, ensemble learning, and loss re-weighting. The main result is not only

a benchmark resource but also a clear picture of what works for long-tailed chest X-ray classification, along with a path toward vision-language foundation models for few-shot and zero-shot generalization. A major merit is strong field-level impact, because rare disease detection cannot improve without a realistic benchmark. Possible limitations include label noise from report mining, incomplete coverage of the true clinical long tail, and the high cost of building large consensus-labeled test sets.

Dalla Serra et al. [3] focuses on automated chest X-ray report generation and argues that global image tokens are too coarse for clinically faithful reporting. The main contribution is a reporting pipeline built around finding-aware anatomical tokens, extracted by a customized Faster R-CNN that localizes anatomical regions and detects findings inside those regions before passing structured tokens into a multimodal report generator. Methodologically, the study uses a two-stage design with token extraction, triples extraction, and report generation, trained with a multi-task loss that combines anatomy classification, box regression, and finding detection. Results on MIMIC-CXR show state-of-the-art report generation quality, improved clinical accuracy, and fewer errors such as hallucinated findings and region misassignment. A major merit is the explicit link between anatomy and language, which makes generated reports more grounded than conventional whole-image report generators. Possible limitations include dependence on reliable anatomy localization, higher system complexity than end-to-end captioning baselines, and possible sensitivity to reporting styles outside the training domain.

Pellegrini et al. [32] tackles zero-shot chest X-ray diagnosis with a strong emphasis on explainability rather than direct disease prediction alone. The main contribution is Xplainer, a classification-by-description framework that asks a contrastive vision-language model to detect descriptive radiographic observations first and then derives diagnosis scores from those descriptor probabilities. Methodologically, the study adapts contrastive prompting to multi-label medical diagnosis, designs prompts around radiology-style descriptors, and evaluates the framework on CheXpert and ChestXray14. Reported results show that Xplainer significantly outperforms earlier zero-shot methods, while also offering intrinsic explainability because final disease predictions can be traced back to specific visual descriptors. A major merit is the practical value for rare or unseen findings, where fully supervised labeling is costly and often unavailable. Possible limitations include strong prompt sensitivity, dependence on the quality and completeness of the descriptor vocabulary, and likely performance gaps relative to strong supervised

models on well-represented labels.

Öztürk et al. [33] aims to improve multi-label chest X-ray classification by reducing the weakness of standard deep models in small abnormal regions and cluttered backgrounds. The main contribution is an end-to-end focal modulation based framework that combines background-independent feature map construction with focal modulation so that feature relationships are adapted more effectively for multi-disease recognition. Methodologically, the study uses a conventional convolutional backbone to build cleaner feature maps and then applies focal modulation to improve discriminative focus for 14 chest disease labels under both single-label and multi-label settings. Reported results show superior performance over traditional deep learning architectures on the evaluated chest X-ray dataset. A major merit is the attempt to improve spatial focus without relying on a heavy transformer stack, which can be useful for subtle and visually confusing thoracic findings. Possible limitations include limited reporting on uncertainty handling, limited external validation, and unclear comparison against stronger recent graph-transformer hybrids or clinically grounded explanation methods.

Park et al. [34] addresses the underuse of multi-task capacity in multimodal large language models for chest X-ray interpretation. The main contribution is M4CXR, a single multi-modal large language model that performs medical report generation, visual grounding, and visual question answering within one conversational framework, supported by a bridging module between vision and language embeddings and a chain-of-thought prompting strategy. Methodologically, the study pretrains the bridging module, fine-tunes the full system with visual instruction tuning, and evaluates the framework across single-image, multi-image, and multi-study settings. Reported results show state-of-the-art clinical accuracy in report generation together with strong performance in grounding and question answering. A major merit is unified multi-task design, which moves closer to a clinically useful assistant rather than a narrow single-task model. Possible limitations include substantial computational demand, dependence on diverse multimodal training data, and persistent risk of clinically unsafe language generation such as hallucination or overconfident narration.

Ma et al. [35] studies trustworthiness in medical image analysis more broadly and proposes TRUDLMIA as a framework that treats model trust as a design objective rather than a side effect of accuracy improvement. The central contribution is the combination of self-supervised feature learning with a novel surrogate loss so that model performance

and trustworthiness improve together, especially under limited and imbalanced data. Methodologically, the framework is validated on benchmark datasets for pneumonia, COVID-19, and melanoma through ablation analysis, cross-validation, and result visualization. Reported results are concrete, with improvements of up to 21% in model performance and up to 5% in trust-related evaluation. A major merit is the explicit engineering view that trustworthy medical AI needs both predictive strength and dependable model behavior. Possible limitations include weak chest X-ray specificity, limited evidence for multi-label thoracic screening use, and unclear translation of the trust metric into clinically validated trust by radiologists or other end users.

Wang et al. [36] addresses weakly supervised abnormality localization in chest X-rays, where image-level labels are available but dense lesion annotation is expensive. The main contribution is a localization framework built on a VMamba backbone with three tightly linked modules: non-linear modulation for refining foreground prediction maps, fusion of foreground prediction maps for stronger foreground-background separation, and a foreground control loss for regulating feature maps toward abnormal regions. Methodologically, the system is evaluated on the NIH chest X-ray dataset and the RSNA dataset, with additional robustness checks on three more datasets of varying modality and image quality. Reported results show better localization accuracy than six state-of-the-art methods and stronger robustness across datasets. A major merit is improved lesion localization from weak supervision alone, which has direct practical value because pixel-level annotation is expensive in radiology. Possible limitations include lack of detailed failure-case analysis in the article summary, possible sensitivity to weak-label noise, and an expected performance gap relative to fully supervised detection systems with expert bounding boxes.

Li et al. [37] focuses on longitudinal chest X-ray analysis, where the diagnostic value lies not only in a single image but also in temporal change across multiple studies. The main contribution is a longitudinal representation learning framework that enriches inconsistent sequential imaging records through two-dimensional principal component analysis, a robust adaptive loss function, and an efficient optimization algorithm with both objective and sequence convergence guarantees for a non convex problem. Methodologically, the study models temporal imaging evolution rather than isolated snapshots and evaluates the framework on CheXpert for disease detection and localization. Reported results show significant improvement in capturing indicative abnormalities and achieving satisfactory

diagnostic performance. A major merit is the shift from static image classification to progression-aware analysis, which matches follow-up radiography workflows more closely. Possible limitations include dependence on longitudinal data availability, possible difficulty with irregular scan intervals and missing prior studies, and limited clarity on how strongly localization performance was validated against detailed expert annotations. Ketabi et al. [38] studies multimodal chest X-ray classification by combining images, radiology reports, and eye-gaze signals for both prediction and explanation. The article objective is to test whether added clinical context and human attention data improve CNN diagnosis and attention fidelity. The main contribution is a multimodal CNN design plus a quantitative explainability metric measuring overlap between model attention and radiologist regions of interest. The methodology combines chest X-ray images, report text, and eye-gaze heatmaps, then compares resulting attention maps with GradCAM. Reported results show a 12.85% improvement in pneumonia and congestive heart failure detection after report fusion, plus roughly 11% better explainability for eye-gaze-based heatmaps. Major merit lies in grounding explanation in human viewing behavior rather than only in post hoc saliency. Possible limitations include heavy dependence on scarce eye-gaze annotations, narrow evaluation around selected diseases, and limited evidence on cross-hospital robustness.

Shekar et al. [10] analyzes domain shift in chest radiograph classification between MIMIC-CXR and a large Veterans Healthcare Administration dataset. The objective is to measure how dataset source, labeler choice, demographic factors, and study year affect transfer performance. The contribution is a careful shift analysis using DenseNet121 pretraining, CheXpert and CheXbert report labelers, and subgroup evaluation over more than 259,000 VA-CXR images collected from 2010 to 2022. Findings show lower label disagreement in VA-CXR, meaningful AUC variation between label extraction tools, limited shift for most labels, and clear instability for Enlarged Cardiomedastinum, with study year producing the strongest subgroup variation. Major merit lies in moving beyond average accuracy toward operational questions about label quality and temporal drift. Possible limitations include unquantified differences between inpatient and outpatient clinical settings, exploratory ICD code mapping, and limited guidance on concrete mitigation strategies.

Yaligar et al. [39] addresses multi-label diagnosis of 14 thoracic disorders in NIH chest X-rays through a pragmatic ensemble strategy. The objective is to improve diagnostic

accuracy by combining complementary CNN families rather than relying on a single backbone. The contribution is a comparison of MobileNet, EfficientNet, and Extreme Learning Machine models, followed by average, weighted-average, and max-vote ensembling. The methodology favors architectures with different efficiency and representation properties, then measures multi-label performance with AUC, F1, and Hamming loss. Reported results identify EfficientNet as the strongest single model and the weighted-average ensemble as the best overall system, with AUC 0.8721, F1 0.8245, and Hamming loss 0.0987. Major merit lies in practical robustness and easy reproducibility. Possible limitations include limited novelty at the modeling level, no explicit treatment of uncertainty or fairness, and likely added inference cost from maintaining several models.

Jeong et al. [16] targets the long-tailed nature of chest X-ray disease labels, where rare abnormalities remain the hardest part of multi-label screening. The article objective is to build an optimized ensemble framework for the MIMIC-CXR-LT setting and the CXR-LT competition. The main contribution is an ensemble pipeline tailored through repeated experiments on architecture choice and data augmentation for rare-class prediction. The methodology focuses on class-imbalanced medical images, overlapping pathologies, and long-tail label frequencies, then evaluates performance on the workshop benchmark. Reported findings show clear improvement over existing approaches and a top-ranking result in the CXR-LT competition. Major merit lies in direct attention to rare-disease failure modes, a major weakness in many chest X-ray systems. Possible limitations include high system complexity, reduced interpretability relative to single-model designs, and unresolved questions about scalability and generalization across other medical datasets.

Ihler et al. [12] adapts FixMatch for semi-supervised multi-label chest X-ray learning on CheXpert, a setting with imbalance and partial labels. The objective is to test whether simple semi-supervised learning remains effective after adaptation from multi-class vision tasks to multi-label radiography. The main contribution is ML-FixMatch+DA, a distribution-aware FixMatch variant that uses pseudo-label selection and distribution alignment without adding new hyperparameters. The method directly targets class imbalance and missing labels, two core problems in medical data. Reported results show a 2.6% AUC gain in semi-supervised experiments and a 1.9% AUC gain in missing-label scenarios. Major merit lies in strong performance from a relatively simple design, a useful property for reproducible medical engineering. Possible limitations include sensitivity to pseudo-label errors, reliance on similarity between labeled and unlabeled data

distributions, and limited discussion of calibration or explanation quality.

Lai et al. [19] focuses on long-tailed multi-label chest X-ray learning under noisy labels, a setting closer to real report-mined supervision. The objective is twofold: create a stronger benchmark and propose a baseline that improves rare-disease detection. The contribution begins with LTML-MIMIC-CXR, an extension of MIMIC-CXR containing 26 additional rare diseases, and continues with a baseline that combines adaptive negative regularization with large-loss reconsideration for noisy-label correction. The methodology directly attacks over-suppression of tail-class negative logits and label errors from automated annotation. Reported results show meaningful advances in rare-disease detection and establish a foundation for more robust long- tailed evaluation. Major merit lies in treating imbalance and noise as a coupled problem instead of two separate problems. Possible limitations include continuing dependence on automated annotations, limited evidence on external validation, and probable sensitivity of tail-class performance to label quality.

Cheng et al. [40] proposes a curriculum-learning paradigm for chest X-ray abnormality classification that follows a local-to-global training order. The objective is to train models first on simpler or higher-priority abnormality groups, then gradually expand toward more complex multi-label patterns. The main contribution is ML-LGL, a multi- label local-to-global framework driven by three clinical-knowledge-based selection functions and a dynamic loss. The methodology iteratively builds new training subsets containing selected abnormality categories, then retrains the network with gradually increasing abnormality complexity. Reported results show better initial training stability, performance above baseline methods, and results comparable to state-of-the-art approaches across three open-source datasets. Major merit lies in aligning training sequence with diagnostic progression used in clinical reasoning. Possible limitations include unclear scalability to larger label spaces, dependence on expert-designed priority functions, and limited evidence on robustness to image quality variation and demographic shift.

Hu et al. [1] tackles a major weakness in chest X-ray benchmarks by adding radiologist-defined uncertainty and severity information to multi-label supervision. The objective is to build a clinically richer dataset and a learning framework that can use spatial relations, semantic relations, and expert uncertainty at the same time. The main contribution is a new benchmark plus a multi-relationship graph learning model with a dedicated uncertainty-aware loss. The methodology assembles chest X-ray labels from clinical text with uncertainty and severity assessments, then trains a graph model that links related

findings instead of treating labels as isolated outputs. Reported results show better diagnostic accuracy and stronger interpretability than prior state-of-the-art approaches. Major merit lies in converting uncertainty and severity from annotation noise into structured supervision. Possible limitations include dependency on reliable clinical note extraction, remaining variation across radiologist judgments, and a need for broader real-world validation before routine deployment.

Rocha et al. [41] introduces CLARE-XR, an explainable chest radiograph classifier built around regression in a 2D latent space rather than conventional direct multi-label classification. The objective is to provide pathology predictions alongside transparent visual reasoning that resembles radiologist case comparison. The main contribution is a label-embedding scheme that maps binary pathology annotations to 2D reference coordinates, plus a regression model that places new images near those reference points and retrieves similar training cases. The methodology measures pathology assignment by distance in latent space and supports explanation through explicit classification rules and image retrieval. Reported results show better performance than a multi-label ResNet50 baseline on NIH ChestXray14 across several evaluation settings. Major merit lies in inherent interpretability without heavy reliance on post hoc heatmaps. Possible limitations include possible oversimplification from a 2D pathology space, unclear handling of ambiguous or overlapping diseases, and limited evidence on scalability to larger or more diverse clinical datasets.

Chen et al. [42] addresses noisy multi-label chest X-ray classification, a critical problem when training labels come from radiology reports or automated extraction. The objective is to improve robustness under label noise while preserving multi-label semantic structure. The main contribution is BoMD, a bag-of-multi-label-descriptors method that detects noisy samples, smoothly relabels suspicious annotations, and aligns image descriptors with language-model-derived semantic descriptors. The methodology evaluates training on noisy multi-label sets against clean test data and introduces a benchmark dedicated to systematic assessment of noisy multi-label learning. Reported results show state-of-the-art accuracy and strong robustness across chest X-ray benchmarks. Major merit lies in combining relabeling and semantic alignment inside one pipeline, a strong match for medical datasets with imperfect supervision. Possible limitations include dependence on descriptor quality from language models, extra complexity from relabeling procedures, and limited evidence on performance under external domain shift or institution-specific

reporting styles.

Katona et al. [43] examine multi-label chest X-ray classification under strong label overlap and pursue higher diagnostic accuracy through ensemble design rather than backbone novelty alone. The article contributes an online ensemble strategy that combines member-specific losses with ensemble loss, then fuses several pre-trained CNNs through feature-level linkage or a trainable label-weighted average head. Methodology includes backbone screening on ChestX-ray14, transfer learning from MIMIC-CXR, five-fold cross-validation, and official-split testing under a controlled memory budget. Reported results show strong gains from the modified loss and transfer learning, with the strongest weighted-average setting reaching validation AUC 0.7924 after MIMIC-CXR pretraining, and the best ensemble reported as stronger than earlier state-of-the-art systems on the official split. A major merit lies in disciplined engineering around ensemble diversity, transfer learning, and practical compute limits. Possible limitations include high computational cost, weak interpretability, and unclear robustness beyond the evaluated datasets.

Shaaban et al. [44] target a harder setting in which chest X-rays must be interpreted together with incomplete electronic health records. The central contribution is MedPromptX, a multimodal framework that joins a pre-trained multimodal large language model with few-shot prompting, dynamic few-shot refinement, and visual grounding, together with release of the MedPromptX-VQA dataset for interleaved image-EHR reasoning. Methodology uses the pretrained model to complement missing EHR content, few-shot examples to reduce retraining burden and hallucination risk, and region-focused grounding to narrow attention toward abnormal zones. Reported results show state-of-the-art performance with an 11% F1 improvement over baselines. A major merit lies in moving chest X-ray diagnosis toward richer clinical context instead of image-only prediction. Possible limitations include strong sensitivity to example selection, implementation complexity, dependency on EHR quality, and residual risk of generative error in difficult cases.

Xiao et al. [45] address annotation scarcity through Single Positive Multi-label Learning for chest X-rays, a setting in which each image carries only one annotated positive label even when several findings coexist. The main contribution is the Multi-level Pseudo-label Consistency framework, which combines image-level perturbation consistency, Random Elastic Deformation, feature-level perturbation consistency, and a Transformer-based

Batch-level Transformer-based Correlation regularizer to recover hidden positives and suppress false negatives created by naive negative filling. Methodology evaluates the framework on CheXpert and MIMIC-CXR under weak supervision rather than full annotation. Reported summaries describe extensive experiments on both datasets and show clear effectiveness of the MPC design for the SPML-CXR setting, although accessible summaries do not expose a headline final AUC. A major merit lies in matching real annotation practice, where exhaustive labeling is costly and often unavailable. Possible limitations include dependence on pseudo-label quality, sensitivity to augmentation design, and limited evidence on fairness, calibration, or cross-domain stability.

Meena et al. [46] focus on weakly supervised localization of thoracic abnormalities rather than classification alone, with the objective of helping radiologists see where model evidence lies across 13 disease categories. The article introduces CAPCAM, a Confidence-Aware Probabilistic Class Activation Map framework built on a CX- Ultraret backbone and a joint Confidence Aware Network plus Anomaly Detection Network, thereby avoiding explicit box-level training for each component. Methodology uses large public chest X-ray datasets and compares CAPCAM pooling against existing weakly supervised pooling strategies. Reported results are strong and concrete, with Intersection over Bounding Box scores in the 85% to 94% range and F1 scores in the 88% to 90% range, alongside state-of-the-art localization performance. A major merit lies in extracting clinically useful localization from weak supervision, which is valuable in data-poor radiology settings. Possible limitations include unclear behavior in overlapping multi-disease scenes, limited discussion of calibration, and uncertain deployment speed under routine clinical throughput.

Alam et al. [20] attempt to solve a different trust problem: fluent radiology report generation without transparent clinical reasoning. The article contributes an interpretable framework that uses concept bottleneck models to turn visual features into concept vectors and then feeds those concepts into a multi-agent retrieval-augmented generation system for report creation. Methodology models relations between image features and clinical concepts, uses multi-agent retrieval and generation to assemble reports, and evaluates both classification and report quality on COVID-QU with large- language-model judging of report usefulness. Reported results show 81% classification accuracy and report-quality values between 84% and 90% across key metrics. A major merit lies in connecting

prediction, concept reasoning, and report text inside one pipeline, which raises explainability far above saliency-only systems. Possible limitations include retrieval noise, redundancy, risk of clinically inconsistent generated sentences, and restricted evidence for generalization beyond chest X-rays and the evaluated dataset.

Arora et al. [47] pursue multi-label thoracic disease classification through architectural efficiency and tailored optimization rather than graph reasoning or language modeling. The article proposes a Separable Convolution Neural Network using a modified Mish activation called Swmish and a compound loss that combines Dice coefficient with binary cross-entropy, with the objective of handling both multi-label chest radiographs and COVID-19 CT cases. Methodology studies separable convolution for spatial and depth-wise feature extraction, then validates the framework on a multi-label CXR dataset and a COVID19-CT dataset. Reported results include AUC 0.91 with loss 0.13 on the multi-label CXR task and testing accuracy 0.97, AUC 0.99, and loss 0.11 on the CT task. A major merit lies in clear engineering choices that remain lightweight and numerically strong. Possible limitations include lack of explicit uncertainty modeling, limited interpretability support, and unclear resilience under low-quality scans or external institutional shift.

Zhang et al. [7] address a recurring weakness in chest X-ray multi-label learning, namely poor use of correlations among co-occurring findings. The article contributes the LCFL framework, which learns global label correlations with self-attention, decomposes image features into label-level features, enforces consistency between global and local label correlations, and adds a label-correlation-guided supervised contrastive loss to sharpen discriminative features. Methodology centers on end-to-end training and ablation on the CheXpert dataset, with explicit isolation of label- correlation learning and label-level feature enhancement modules. Reported results reach an average F1 score of 44.6% and an AUC of 76.5%, representing gains of 7.7% and 1.3% over earlier state-of-the-art results. A major merit lies in turning label correlation from a passive co-occurrence prior into an active feature-learning signal. Possible limitations include no visible treatment of uncertainty or fairness and a likely dependence on dataset-specific correlation structure, which may weaken transfer to new hospitals.

Arya et al. [48] provide an extensive survey rather than a new diagnostic model, with the objective of mapping how deep learning is being adopted across chest X-ray analysis. The main contribution lies in a broad synthesis that connects classification with single-level and multi-level localization and segmentation, thereby showing field movement from pure

disease detection toward more precise spatial reasoning. Methodology consists of literature review and technical comparison across chest X-ray analysis pipelines. Reported conclusions in the available summary indicate that deep learning is already improving diagnostic accuracy and reducing misinterpretation, while localization and segmentation methods are increasing precision beyond label- only output. A major merit lies in broad technical coverage, which makes the survey useful for identifying research directions rather than single-algorithm winners. A likely limitation, by survey design, lies in absence of standardized re-implementation or unified quantitative benchmarking across compared studies.

Ali et al. [49] review AI methods for automated thoracic disease assessment from chest X-ray imaging with a structured emphasis on datasets, model architectures, evaluation metrics, and task expansion beyond classification. The article contribution is a systematic synthesis showing that convolutional neural networks remain dominant for classification, while newer work branches into localization, segmentation, report generation, and multitask or multimodal modeling. Methodology relies on literature search, extraction of technical characteristics from eligible studies, and critical identification of recurring challenges. Reported conclusions indicate meaningful progress in automated chest X-ray analysis, promising potential for multitask and multimodal approaches, and persistent obstacles around data quality, evaluation practice, clinical adoption, and real-world validation. A major merit lies in connecting algorithmic performance with deployment readiness. Possible limitations include conference-paper brevity, high-level evidence aggregation, and lack of quantitative meta-analysis across reviewed systems.

Akhter et al. [50] deliver a structured review of AI-based radio diagnosis from chest X-rays across disease categories such as tuberculosis, pneumonia, pneumoconiosis, COVID-19, and lung cancer, with objective coverage of tasks, datasets, evaluation metrics, patents, and open research problems. The article contribution is a disease- oriented survey that joins enhancement, segmentation, detection, classification, image generation, and report generation under one analytical frame, then links technical progress with practical barriers such as small datasets, label scarcity, poor image quality, adversarial vulnerability, privacy constraints, and weak interpretability. Methodology is literature synthesis rather than experimental model design. Reported conclusions argue for large scope in automatic chest X-ray analysis but stress the need for novel diverse datasets and responsible, interpretable, privacy-friendly algorithms. A major merit lies in explicit open-problem framing around

trustworthy deployment rather than accuracy alone. Possible limitations include absence of head-to-head experimental comparison and dependence on the quality and completeness of the surveyed literature.

Kufel et al. [51] examine multi-label classification of 14 chest X-ray abnormalities on NIH ChestXray14 with a transfer-learning pipeline built around EfficientNet. The study objective is practical and well focused: improve multi-label diagnostic accuracy under severe class imbalance without relying on a highly complex architecture. The main contribution is a strong engineering baseline that combines EfficientNet feature extraction, GlobalAveragePooling, Dense and BatchNormalization layers, a custom split for fairer disease representation, and binary cross-entropy for label-wise optimization. The methodology stays simple and reproducible, with emphasis on careful data handling rather than graph reasoning or multimodal fusion. Reported results show an average AUC-ROC of 84.28%, outperforming earlier deep learning baselines in the reviewed source. A major merit lies in the balance between performance and implementation simplicity, since the framework demonstrates that carefully configured transfer learning remains competitive for multi-label thoracic screening. Possible limitations include limited modeling of label dependencies, no explicit uncertainty estimation, no explanation mechanism, and validation centered mainly on a single benchmark, which leaves external robustness and clinical trust less developed than in newer trustworthy-AI designs.

Mau et al. [17] address the long-tailed nature of chest X-ray data, where common diseases dominate training and rare diseases remain weakly learned. The objective is to improve multi-label classification on long-tailed chest X-rays, especially for tail classes that usually suffer under standard training. The core contribution is an ensemble strategy that combines EfficientNetV2 and ConvNeXt with image-size tuning, basic and advanced augmentation, mosaic augmentation, Binary Focal Cross-Entropy, Stratified K-Fold validation, and Test Time Augmentation. The methodology is explicitly competition-oriented and engineered for difficult imbalance conditions rather than for ordinary balanced classification. Reported results reach a mean Average Precision of 0.354 and place the method within the top five of the CXR-LT multi-label long-tailed competition, which indicates strong rare-class handling under a demanding benchmark. A major merit is the direct attack on real-world class imbalance through coordinated design choices rather than through a single

isolated trick. Possible limitations include high training and inference complexity, reduced interpretability because of ensemble dependence, and limited evidence on calibration, fairness, and portability beyond the competition setting.

Bouabdallah et al. [22] focus on explainability for thorax multi-label classification and propose a black-box complementary explanation framework called DAG-CNN. The objective is not only disease prediction but also clinically useful explanation, especially in settings where attribution maps alone are too weak or too unstable for medical use. The main contribution is a case-based explanation mechanism that combines global image features with local lesion features and retrieves similar cases to support the final diagnosis. The methodology uses a discriminative attention-guided convolutional architecture and compares extracted features with stored disease-specific cases, thereby producing example-based explanations instead of relying only on saliency. Experiments on ChestX-ray14 show competitive classification behavior, with especially strong performance for pneumonia and hernia, while the explanation phase reaches an accuracy score of 0.93. A major merit lies in explanation style: retrieved similar cases can be easier for clinicians to inspect than abstract heat maps. Possible limitations include dependence on the quality and diversity of the reference case library, sensitivity to lesion localization quality, and lack of explicit uncertainty estimation, which means strong-looking explanations may still be attached to uncertain predictions.

Sun et al. [52] tackle a major weakness in multi-label medical AI, namely the inability of common explainers to produce class-specific evidence when several findings coexist in one image. The study objective is to build an inherently interpretable multi-label classifier, named Attri-Net, that produces class-specific counterfactual attribution maps instead of generic post hoc heat maps. The key contribution is a model where explanation is built into the prediction process, allowing each finding to receive a separate counterfactual explanation that better matches clinical reasoning. The evaluation compares Attri-Net with five post hoc explanation methods and one interpretable baseline across three chest X-ray datasets. Reported findings in the review source show that Attri-Net generates explanations more consistent with clinical knowledge while keeping classification performance comparable to state-of-the-art models. A major merit is conceptual sharpness: class-specific counterfactuals directly address the confusion created when one heat map is forced to explain many labels at once. Possible limitations include likely computational

overhead from counterfactual generation, the need for strong clinical validation of counterfactual maps, and possible difficulty in extremely subtle or heavily overlapping disease patterns where small visual changes may not map cleanly to one finding.

Lin et al. [24] study fairness in automated chest X-ray diagnosis and use supervised contrastive learning to reduce bias across demographic subgroups. The objective is to produce fairer image embeddings for chest X-ray classification without losing diagnostic value. The main contribution is a contrastive learning strategy that carefully chooses positive and negative samples so that the learned representation becomes less dependent on subgroup-specific shortcuts. The methodology uses two datasets, MIDRC for COVID-19 diagnosis and NIH-CXR for thoracic abnormality classification, and evaluates fairness through marginal AUC difference with statistical testing. Reported findings show statistically significant fairness improvements, with paired t-test $p < 0.0001$, and subgroup-gap measures reported as 0.0116 for sex, 0.2102 for race, and 0.1000 for age on MIDRC, plus 0.0090 for sex and 0.0512 for age on NIH-CXR. A major merit lies in moving fairness from discussion to measurable optimization within representation learning. Possible limitations include dependence on reliable subgroup annotations, uncertain behavior for unseen intersectional subgroups, and limited evidence on whether fairness gains remain stable under domain shift or across other chest X-ray populations.

Ozturk et al. [53] propose HydraViT, an adaptive multi-branch transformer for multi-label chest X-ray disease classification, with the objective of handling strong variability in pathology size, position, appearance, and co-occurrence. The main contribution is a transformer backbone combined with a multi-branch output structure and a novel loss that adaptively weights branch outputs to better learn pathology co-occurrences. The methodology targets long-range context modeling, which is a known weakness of many plain CNN systems in chest radiography. Reported results show that HydraViT outperforms attention-guided methods by an average of 1.2%, region-guided methods by 1.4%, and semantic-guided methods by 1.0%. A major merit is the adaptive branch design, which gives the network multiple complementary views of pathology evidence instead of forcing one uniform decision pathway for all labels. Possible limitations include transformer-related computational cost, limited discussion of calibration and uncertainty behavior, and uncertain robustness when moving from benchmark data to noisier or shifted clinical environments.

Pham et al. [54] develop a controllable and interpretable chest X-ray diagnosis system

designed to decode intense radiologist focus during reading. The objective is unusually clear and clinically meaningful: determine where a radiologist looks, how long attention stays in each area, and what finding is diagnosed from that focus pattern. The main contribution is the I-AI pipeline, which combines eye-gaze supervision with a vision-language model to produce controllable attention maps and to suppress irrelevant image regions during diagnosis. The methodology uses an eye-gaze dataset to build anatomical gaze information and ground-truth heatmaps, then trains the model to classify findings from focused regions rather than from the whole image. Reported findings show that the generated attention heatmaps preserve enough diagnostic information for accurate classification even when only part of the chest X-ray is used. A major merit lies in close alignment with human diagnostic behavior, which gives the framework stronger interpretability than ordinary saliency-based systems. Possible limitations include dependence on expensive eye-gaze data, limited scalability across institutions without gaze annotations, and absence of a formal uncertainty mechanism for deciding when focus-driven interpretation remains unreliable.

Xu et al. [6] address multi-task thoracic disease classification by jointly modeling spatial attention and label dependence, two issues that strongly affect multi-label chest X-ray prediction. The objective is to improve recognition under insufficient data and imbalanced disease prevalence. The main contribution is a DenseNet121-based framework enhanced with a Group-Wise Spatial Attention module to strengthen within-group spatial features and a Label Co-occurrence Dependency module to refine probabilities using disease correlations. The methodology is technically coherent because local feature enhancement and label dependency modeling are trained together rather than as separate post-processing steps. Reported results achieve a state-of-the-art average AUC of 0.818 on ChestX-ray14. A major merit lies in the dual treatment of visual structure and label relation, which is well matched to chest radiography where several abnormalities often overlap or co-occur. Possible limitations include reliance on a CNN backbone that may be weaker than newer transformer systems for long-range context, potential learning of dataset-specific correlations rather than causal clinical relations, and limited evidence on calibration, fairness, and external-domain reliability.

Chen et al. [55] propose a two-stage hierarchical chest X-ray diagnosis framework for COVID-19, built around Multi-task Supervised Contrastive Learning. The objective is to learn robust disease-related features under limited samples and data bias, then use those

features for more reliable separation of normal cases, pneumonia, and COVID-19. The main contribution is the MTSCL training strategy for lung abnormality detection, followed by a second-stage tree-based classifier that uses lung masks, MTSCL lesion masks, and wavelet-based preprocessing to distinguish COVID-19 from typical pneumonia. The methodology is specific and carefully structured, with contrastive learning used to suppress shortcut learning and to improve localization before the final disease classification stage. Reported results show average AUC gains of 1.03% to 5.33% in abnormality detection and a further 12.38% average AUC improvement in COVID-19 versus pneumonia classification, with overall accuracy of 75.19% across two external datasets. A major merit lies in stronger generalization and abnormality localization under a clinically difficult task. Possible limitations include narrow disease scope, high pipeline complexity, and moderate overall accuracy that still leaves significant room before reliable real-world deployment.

Naz et al. [56] present an explainable AI framework for interpreting pulmonary diseases from chest radiographs using transfer learning with ResNet50 and explanation through LIME. The objective is broad pulmonary disease interpretation, including edema, tuberculosis, nodules, and pneumonia, with particular emphasis on COVID-19 pneumonia. The main contribution is a practical explainable pipeline that couples a high-performing transfer-learned classifier with feature-level explanations that highlight image regions driving the output. The methodology uses extensive training on COVID-CT and COVIDNet-related datasets and then applies LIME to generate image-region explanations for class decisions. Reported results show classification accuracy of 93% and 97% on the two evaluated datasets, and the highlighted regions are reported to align with radiologist-marked areas. A major merit lies in practical accessibility, since ResNet50 and LIME are widely understood and relatively easy to implement in engineering settings. Possible limitations include evidence concentrated on COVID-focused datasets rather than a broad multi-institutional pulmonary benchmark, explanations that may remain coarse despite LIME visualization, and uncertain generalization to more diverse thoracic disease patterns and acquisition conditions.

Huang et al. [15] addresses a hard but clinically important problem: long-tailed multi-label chest X-ray classification under fairness constraints. Objective centers on improving rare-disease detection without allowing performance gains to hide demographic imbalance. Main contribution is LTCXNet, a framework that combines the Pruned

MIMIC-CXR-LT dataset, a ConvNeXt backbone, an ML-Decoder head, data augmentation, and ensemble learning, followed by fairness analysis across demographic groups. Methodology explicitly separates long-tail behavior from standard average-score reporting, which sharpens the study focus on minority classes. Results show notable gains for rare labels, with reported improvements of 79% for Pneumoperitoneum and 48% for Pneumomediastinum, while the study also warns that some accuracy-oriented methods can worsen fairness. Major merit lies in treating tail performance and equitable behavior as linked engineering requirements. Possible limitations include ensemble complexity, likely sensitivity to dataset-specific prevalence, and limited evidence on calibration or external transfer across hospitals.

Termritthikun et al. [57] focuses on on-device chest X-ray classification, where real-time inference and low hardware capacity make large models impractical. Objective centers on compressing multi-label chest X-ray classifiers while keeping diagnostic utility and visual interpretability. Main contribution is an explainable knowledge-distillation pipeline in which larger CNN or transformer teachers transfer knowledge to a compact student model, followed by XAI-based visual inspection of decision regions. Methodology compares several teacher alternatives and reports the strongest student setting with DenseNet161 supervising EEEA-Net-C2. Results are concrete and deployment-relevant: EEEA-Net-C2 reaches AUC values of 83.7%, 87.1%, and 88.7% on ChestX-ray14, CheXpert, and PadChest, with only 4.7 million parameters and 0.3 billion FLOPs. Major merit lies in practical edge deployment potential. Possible limitations include dependence on teacher quality, limited treatment of calibration and uncertainty, and possible loss of subtle rare-class knowledge during compression.

Alam et al. [58] studies interpretability quality rather than classifier novelty and asks whether common local explanation tools produce medically meaningful evidence in chest radiography. Objective centers on systematic comparison of explanation methods for pathology prediction in both single-label and multi-label settings. Main contribution is SHAMSUL, a holistic evaluation framework that compares LIME, SHAP, Grad-CAM, and LRP against expert annotation through both quantitative and qualitative analysis. Methodology uses transfer learning on a multi-label, multi-class chest radiography dataset and then inspects explanation maps with medical-significance criteria rather than visual appeal alone. Results show Grad-CAM as strongest on quantitative evaluation, while LIME heatmap segmentation offers the highest perceived medical significance. Major

merit lies in exposing the mismatch that can exist between numerical explanation scores and clinically useful evidence. Possible limitations include dependence on one evaluation protocol, limited extension beyond chest radiography, and unresolved disagreement between metric-based ranking and human interpretation.

Devasia et al. [59] targets active pulmonary tuberculosis with a zone-wise multi-label design rather than a simple tuberculosis-versus-normal classifier. Objective centers on identifying image-wise and lung-zone-wise manifestations of active tuberculosis from chest X-rays, a design with direct screening and triage value in resource-constrained care. Main contribution is a transfer-learned EfficientNetB4 framework combined with Grad-CAM visualization for radiological inspection of disease patterns. Methodology progresses from grouped abnormalities to a 44-label setting covering left-right and upper-middle-lower lung regions, then evaluates both internal and geographically distinct external test data. Results show strong but class-dependent performance: 52.2% of internal-test manifestations achieve AUC above 0.90, 88% of evaluated external-test abnormalities achieve AUC above 0.80, and most abnormalities show high specificity, although very rare manifestations remain unstable. Major merit lies in clinically useful zone-level tuberculosis characterization. Possible limitations include sparse data for several labels, exclusion of extremely rare manifestations, and reliance on largely single-radiologist reading.

Aimen et al. [60] addresses a realistic few-shot setting in which training data are scarce, source and target datasets come from different domains, and evaluation classes may only partially overlap with training classes. Objective centers on multi-label chest X-ray abnormality recognition under simultaneous few-shot, cross-domain, and partial-overlap conditions. Main contribution is the Generalized Cross-Domain Multi-Label Few-Shot Learning framework together with Generalized Episodic Training, a training strategy designed to simulate those deployment difficulties during meta-learning. Methodology compares the proposed design with transfer learning, hybrid transfer learning, and multi-label meta-learning across multiple datasets. Reported results show superiority over those established baselines, indicating that GenET learns more transferable and overlap-tolerant representations than standard few-shot alternatives. Major merit lies in strong alignment with real deployment conditions, where local labeled data are often limited and external shift is common. Possible limitations include meta-learning complexity, persistent bias toward pretrained classes, and limited discussion of calibration, uncertainty, or

explanation fidelity.

Sadashivaiah et al. [61] addresses concept-based explanation for black-box chest X-ray classifiers without requiring concept-labeled image datasets. Objective centers on replacing vague saliency with concise textual concepts that explain how an image would need to change for a classifier to move from “No Finding” to a target pathology. Main contribution is CoCoX, a conceptual counterfactual framework that builds a concept bank from PadChest report terms and additional pathology attributes, projects classifier embeddings into a chest vision-language space, and learns sparse concept perturbations that flip predictions. Methodology evaluates Cardiomegaly, Pleural Effusion, and Atelectasis with CheXpert for backbone training and MIMIC-CXR for projector training. Results are strong and specific: pathology-name concepts rank first in 98–100% of targeted perturbations, primary findings achieve very high recall, including recall@10 of 1.00 for Cardiomegaly and Pleural Effusion in the VLM+Linear setting, and 99% of 600 images receive explanations in under 30 seconds. Major merit lies in semantically faithful explanation without manual concept annotation. Possible limitations include restriction to three pathologies, dependence on concept-bank quality, and incomplete handling of non-visual clinical factors.

Balmuri et al. [62] proposes a full engineering pipeline for automatic multi-disease chest X-ray diagnosis with the dual goal of stronger classification and lower computational cost. Objective covers automated recognition of several chest diseases, including asthma, lung cancer, pneumonia, and COVID-19. Main contribution is an optimized ensemble transfer learning framework that joins ODeepLabv3 segmentation with an ensemble of VGG16, ResNet, MobileNet, GoogleNet, Inception, and Xception, while a Mutation Rate-based Lion Algorithm tunes segmentation and classifier parameters. Methodology follows a staged flow of preprocessing, optimized segmentation, and optimized ensemble classification. Reported results show better sensitivity, precision, specificity, and overall classification accuracy than comparison models in the study. Major merit lies in end-to-end coupling of segmentation and classification rather than treating both stages separately. Possible limitations include high implementation complexity, likely sensitivity to image-quality variation, and weak emphasis on calibration, fairness, or explanation faithfulness despite strong raw performance claims.

Rahimiaghdam et al. [63] focuses on a core weakness in explainable medical AI: explanation quality cannot improve when explanation quality lacks rigorous

measurement. Objective centers on objective evaluation of visual explanations for thoracic disease classification instead of proposing another explainer. Main contribution is a metric framework built around five criteria—informativeness, localization, coverage, multi-target capturing, and proportionality—and a comparative analysis that identifies Jensen–Shannon divergence as the most effective evaluation metric. Methodology applies the framework to LIME explanations generated from a classifier trained on CheXpert and tests explanation behavior on an unseen VinDr-CXR subset under different segmentation strategies. Results show that JS_DIV best captures desirable explanation properties and supports more principled optimization of LIME settings. Major merit lies in moving explanation assessment away from purely subjective visual judgment. Possible limitations include focus on visual explanation only, dependence on proxy ground truth for quality assessment, and limited extension to concept-level, counterfactual, or interactive explanation frameworks.

Seo et al. [18] studies long-tailed multi-label chest X-ray classification through targeted augmentation of ML-GCN rather than replacement of graph learning itself. Objective centers on better tail-class recognition while preserving multi-label label-dependency modeling. Main contribution is an enhanced ML-GCN that combines multiple experts, class-balanced cycle sampling, Log-Sum-Pooling, an additional attention layer, and KL-divergence-based RIDE regularization. Methodology keeps graph-based label reasoning but strengthens feature diversity and imbalance control through four experts and specialized sampling. Results show clear gains over vanilla ML-GCN, with a final reported test mAP of 0.279 and additional improvements from expert modeling, LSE pooling, and regularization. Major merit lies in combining long-tail learning and label- correlation learning inside one design. Possible limitations include remaining instability on tail classes, overfitting risk from over-sampling, and uncertain scalability beyond the long-tailed CXR distribution used in the paper.

Pawar et al. [64] is a comparative review rather than a new diagnostic model, and objective centers on organizing the rapidly growing literature on AI and explainable AI for pulmonary disease detection and severity classification. Main contribution is a structured comparison of pulmonary screening pipelines with emphasis on deep learning performance and post hoc explanation methods such as SHAP, LIME, and Grad-CAM. Methodology follows comparative literature synthesis rather than experimental model development, with attention to both disease detection and severity classification. Main

result is conceptual rather than numerical: deep learning offers strong promise for pulmonary screening, but black-box prediction and poorly grounded explanation can mislead clinical judgment and weaken adoption. Major merit lies in broad landscape mapping that helps identify where accuracy, explanation, and severity assessment meet. Possible limitations include survey-style dependence on prior studies, lack of unified reimplementation or meta-analysis, and limited direct evidence on which XAI method performs best under controlled head-to-head evaluation.

Aasem et al. [65] addresses weakly supervised thoracic disease localization, with the objective of reducing dependence on bounding-box annotation while still producing clinically useful visual evidence. The main contribution is Ensemble-CAM, a class-activation-map ensemble that combines transfer learning, weak supervision, and model fusion to localize disease regions from image-level labels alone. The methodology trains several classifiers on three separate chest X-ray tasks, then converts activation maps into masks and bounding boxes, followed by ensemble consolidation. The evaluation uses RSNA pneumonia, ChestX-ray14 cardiomegaly, and a COVID-19 chest X-ray dataset, with reported single-model classification results reaching 0.96 accuracy, 0.95 recall, and 0.94 precision for cardiomegaly, plus 0.98 accuracy, 0.96 recall, and 0.97 precision for COVID-19 in the strongest configuration. Jaccard-index analysis shows that ensemble fusion can recover useful localization cues even when individual CAM maps are inconsistent. Major merit lies in practical explainability under weak supervision. Main limitations include lower localization fidelity than fully supervised detectors, high computational cost from multiple backbones, and unstable localization when pathology is diffuse or weakly expressed.

Wang et al. [66] focuses on multi-label thoracic disease recognition and targets a specific weakness in many graph-based chest X-ray systems: poor alignment of pathology relationships across different data representations. The main contribution is MRChexNet, a three-part framework composed of a representation learning module, a multi-modal bridge module, and a pathology graph learning module. The representation module captures image-level pathological cues, the bridge module aligns cross-modal pathology relationships, and the graph module models directed disease relationships rather than simple undirected co-occurrence. The methodology therefore combines feature extraction, cross-modal relation alignment, and directed graph reasoning inside one pipeline. Reported results are strong and concrete, with mean AUC values of 0.8503 on ChestX-Ray14 and

0.8649 on CheXpert, both described as state of the art in the article. A major merit lies in the move from static co-occurrence modeling to directed pathology reasoning, which is closer to clinical dependency structure. Possible limitations include added graph complexity, reliance on accurately learned pathology relations, and limited treatment of calibration, uncertainty, or explanation faithfulness beyond classification performance.

Rafferty et al. [67] studies explainability in lung cancer detection from chest X-rays and sets a sharper goal than many post-hoc XAI papers: clinically meaningful explanation that matches radiologist reasoning rather than visually attractive heat maps. The main contribution is XpertXAI, an expert-guided concept bottleneck framework that preserves human-interpretable clinical concepts while scaling beyond a narrow binary setting. The methodology uses an InceptionV3-based classifier on a curated multiclass MIMIC-CXR dataset, compares XpertXAI with post-hoc explainers and an unsupervised concept bottleneck baseline, and validates explanation quality through expert radiologist review. Reported results are strong at both label and concept level. The best XpertXAI variant reaches an F1 score of 0.866 for label prediction on a 3,590- image test set, while concept prediction reaches an F1 score of 0.842. Expert review on 60 chest X-rays found XpertXAI explanations clinically relevant in all healthy scans and in most cancer cases, with only occasional confusion between masses and nodules. Major merit lies in replacing weak heatmap logic with expert-chosen concepts. Possible limitations include narrow expert validation focused on lung cancer, small radiologist review size, and dependence on a single expert-defined concept vocabulary.

Lanfredi et al. [13] addresses a foundational but often underestimated problem in chest X-ray AI: poor supervision caused by weak report labelers. The objective is to improve dataset annotation quality, not by minor rule tuning, but by extracting richer supervision from reports in a privacy-preserving way. The main contribution is MAPLEZ, a locally executable large language model pipeline that extracts not only finding presence but also location, severity, and radiologist uncertainty. The methodology uses a knowledge- driven prompt tree and evaluates annotation quality across eight abnormalities and five test sets, then measures downstream classification impact. Reported gains are substantial: macro F1 improves by 3.6 percentage points for categorical presence labels, location-label F1 rises by more than 20 percentage points, and downstream image classification improves by 1.1 percentage points in AUROC when MAPLEZ-generated multi-type annotations are used for supervision. Major merit lies in moving chest X-ray datasets from coarse binary tags

toward richer clinical labels without exposing reports to external APIs. Possible limitations include noisy severity extraction, lack of direct clinical deployment testing, and practical dependence on local hardware and carefully defined abnormality vocabularies.

Wainwright et al. [68] takes a different research direction by arguing that chest X-ray trustworthiness cannot be judged from accuracy alone. The objective is to uncover hidden problems in data and model training that standard benchmark reporting can hide. The main contribution is a diagnostic audit framework built around attention visualization, uncertainty measures, and low-dimensional data representations. The methodology applies that audit to the COVID-19 Radiography Database, COVIDx, and ChestX-ray14, all datasets with previously reported concerns such as spurious correlations and incorrect labels. Reported findings show that attention maps, uncertainty summaries, and latent-space views can expose hidden shortcuts and poor data quality that would remain invisible under headline accuracy scores alone. Major merit lies in reframing evaluation from “how accurate is the model?” to “why does the model appear accurate, and can model behavior be trusted?” Possible limitations include the absence of a direct correction mechanism, limited emphasis on building a new classifier, and a mainly diagnostic rather than solution-oriented outcome. Even with that limitation, the study is highly relevant for trustworthy thoracic screening because hidden dataset defects can produce false confidence in clinically unsafe models.

Sharma et al. [69] studies vision-language models for chest X-ray interpretation and focuses on a core clinical problem in report generation: fluent language can still be unsafe when confidence is misplaced. The main contribution is CXR-Agent, an agent-based reporting framework that combines linear probes from CheXagent with BioViL- T phrase grounding to generate uncertainty-aware radiology reports with localized pathology descriptions. The methodology evaluates foundational vision-language models across several datasets and benchmarks, probes components such as the vision transformer and Q-former, then adds an agentic reporting layer and a user-study evaluation platform with respiratory specialists. Reported findings show that CheXagent components outperform Torch X-ray Vision baselines, while the final agentic system improves report accuracy, interpretability, and safety. A key result in the paper is not only positive performance but also a warning: many vision-language models hallucinate confident but wrong medical language, which slows clinical reading and can mislead diagnosis. Major merit lies in

treating uncertainty-aware reporting as a safety problem rather than a text-generation problem. Possible limitations include over fitting risk from limited paired scan-report data, lack of longitudinal ICU evaluation, and remaining dependence on benchmark-style evaluation alongside clinical review.

Ifty et al. [70] focuses on explainable lung disease classification from chest X-rays, with a task definition centered on five categories: viral pneumonia, bacterial pneumonia, COVID-19, tuberculosis, and normal lungs. The contribution is not a single novel architecture, but a broad and systematic engineering comparison of CNNs, hybrid models, ensembles, transformers, and Big Transfer models, followed by explanation analysis with Grad-CAM and LIME. The methodology includes transfer learning, fine-tuning, hyper parameter tuning, stratified cross-validation, and multi-model comparison. Reported results identify Xception as the strongest model, reaching 96.21% accuracy after five-fold cross-validation. The study therefore contributes a strong empirical baseline and shows that explanation overlays can help expose which image regions drive predictions. Major merit lies in pairing high classification accuracy with practical XAI tools that clinicians can visually inspect. Possible limitations include limited dataset transparency, incomplete comparison with broader state-of-the-art thoracic screening literature, and a task formulation closer to multiclass pulmonary diagnosis than to full multi-label thoracic screening with overlapping findings.

Mahapatra et al. [71] addresses multi-label active learning for chest X-rays, with the objective of reducing annotation cost while preserving classification quality. The main contribution is a joint framework that combines graph-transformer-based informative sample selection with informative data augmentation. The methodology treats disease labels as graph nodes, uses graph attention transformers to model inter-label relationships, selects the most informative unlabeled samples, and then generates additional informative transformations of those samples to accelerate learning. Reported experiments on public chest X-ray datasets show improvement over state-of-the-art multi-label active learning methods in classification performance, learning rate, and robustness, while qualitative analysis is used to judge realism of generated images. Major merit lies in solving two practical bottlenecks together: label-efficiency and data scarcity. Possible limitations include sensitivity to image noise, strong dependence on realistic synthetic augmentation, and higher engineering complexity than standard pool-based active learning. The framework is especially relevant for medical AI settings where expert annotation time is

limited and label co-occurrence makes naive active learning less effective.

Lu et al. [8] proposes CvTGNet for multi-label chest X-ray classification, with the objective of improving thoracic disease recognition by combining strong visual representation learning with explicit pathology co-occurrence modeling. The main contribution is a hybrid architecture that joins a Convolutional Vision Transformer with a Graph Convolutional Network and then fuses both streams for final prediction. The methodology uses CvT for image representation, a pathology co-occurrence graph generation module for disease relationships, and asymmetric loss for imbalanced multi-label optimization. Reported results are strong and detailed. Mean AUC reaches 0.847 on ChestX-Ray14 and 0.840 on CheXpert. On ChestX-Ray14, CvTGNet reaches 0.927 for cardiomegaly, 0.900 for effusion, 0.927 for edema, and 0.962 for hernia. On CheXpert, strong gains appear in enlarged cardiomedastinum at 0.793, pleural other at 0.925, lung lesion at 0.842, and pneumonia at 0.868. Major merit lies in effective fusion of transformer context and graph-based label reasoning, supported by ablation results showing value from pathology co-occurrence modeling. Possible limitations include weaker scores on some labels such as lung opacity, pneumothorax, pleural effusion, and support devices, plus no explicit treatment of uncertainty, explanation, or fairness.

Shetty et al. [72] presents a survey of deep learning methods for detection and classification of thoracic disease in chest radiography, with a focus on CNN-driven computer-aided diagnosis. The objective is to organize the fast-growing literature on automated chest X-ray analysis, identify major model families, and discuss design factors that matter in multi-disease classification. The contribution is therefore synthetic rather than algorithmic: the survey maps early deep learning progress, reviews breakthrough applications in multi-pathology detection, and highlights practical design questions relevant to real-time CAD systems. The methodology is narrative review rather than benchmark experimentation. Reported conclusions emphasize that deep learning, especially CNN-based models, has markedly improved chest radiography analysis and has made simultaneous detection of several thoracic diseases more feasible. Major merit lies in giving a structured early view of a field moving from handcrafted pipelines toward modern deep models. Possible limitations include limited coverage of later trustworthiness themes such as uncertainty calibration, fairness, domain shift, neuro-symbolic reasoning, and multimodal vision-language systems, which reduces usefulness for current trustworthy-AI research questions. The review captures the breadth of current chest X-ray AI research well, but the strongest

conclusion is not clinical readiness; the strongest conclusion is fragmentation across supervision, modeling, explanation, and deployment. Hu et al. [1] and Lanfredi et al. [13] make clear that uncertainty, severity, and report-derived label quality remain foundational bottlenecks, yet many later studies still claim progress on weak labels without proving improved calibration, stronger uncertainty meaning, or cleaner clinical supervision. Holste et al. [14] makes the long-tailed nature of thoracic screening explicit, but many solutions built around rare-class improvement still depend on ensembles, challenge-specific tuning, and heavy augmentation, a pattern that raises score tables while also increasing inference cost, reducing transparency, and weakening deployment practicality. Bouabdallah et al. [17] shows a useful shift from generic saliency toward case-based explanation, but the broader explainability body in the review still leans too heavily on visual plausibility, proxy localization, and qualitative inspection rather than radiologist-centered studies that test whether explanation improves error detection, trust calibration, or reporting quality. Lin et al. [24] and Shekar et al. [10] expose another major weakness: fairness and domain shift are still treated more often as audit topics than as core architectural constraints, so strong internal AUC can still coexist with subgroup disparity, labeler sensitivity, and temporal instability. A second critical issue lies in the review design itself, because multi-label classification papers, lesion- localization papers, report-generation systems, visual-question-answering models, and multimodal prompting frameworks are discussed within one trustworthiness narrative; broad scope is useful for thesis framing, but direct comparison across such different task definitions can blur engineering conclusions about what actually improves trustworthy multi-label screening. A third issue concerns evidence quality, because peer-reviewed papers, workshop papers, arXiv preprints, and review articles appear on nearly equal footing, while external validation, threshold policy, uncertain-label handling, and calibration protocols remain highly inconsistent across studies. A sharper reading therefore is that current literature is mature in components but immature in integration. Label reliability, rare-class sensitivity, explanation faithfulness, subgroup stability, and uncertainty-aware decision control remain separate achievements rather than parts of one clinically dependable pipeline. A strong doctoral contribution therefore needs unified design rather than another accuracy-only model, and the NEPTUNE-CXR direction is valuable precisely because anatomy-aware grounding, counterfactual evidence, neuro-symbolic consistency, and calibrated uncertainty are treated as coupled requirements rather than optional add-ons.

3 NEPTUNE-CXR: A NEURO-SYMBOLIC, COUNTERFACTUAL, UNCERTAINTY-AWARE MODEL FOR MULTI-LABEL THORACIC DISEASE SCREENING ON CHEST X-RAYS

3.1 Overview

Trustworthy multi-label thoracic screening from frontal chest X-rays remains difficult because radiographic findings often co-occur, supervision from routine reports is frequently weak or uncertain, deployment settings introduce domain shift, and clinical use demands explanations aligned with anatomy and confidence-aware decisions. Common co-occurring patterns such as opacity, pleural change, edema, and cardiac enlargement create non-independent labels and raise the risk of internally inconsistent outputs under independent binary prediction. NEPTUNE-CXR addresses the combined problem through a unified framework for screening-oriented prediction, explanation, and uncertainty control. The proposed design integrates anatomy-aware representation learning through APZ + Z-ViT, counterfactual healthy normalization through CDN, neuro-symbolic consistency modeling through NSCG, and uncertainty-aware outputs through evidential, GP-like, abstention, and conformal prediction. Anatomical masks and six lung zones provide region-grounded structure for feature formation, while a zone-aware encoder captures whole-image context together with zone-level evidence. A healthy-reference counterfactual generator then produces a closest-healthy reconstruction and residual evidence map, allowing abnormality evidence to be expressed as explicit image differences rather than unstable gradient highlights. A concept graph then combines global features, zone summaries, residual evidence, and morphometric descriptors to encourage clinically compatible outputs and expose intermediate concept activations. A final uncertainty layer combines evidence concentration, feature-space novelty, and distance-to-healthy information to support calibrated probabilities, selective rejection of uncertain cases, and conservative set-valued outputs when single-label thresholding is unreliable. The resulting output space includes anatomically grounded attention summaries, residual evidence maps, and concept-level rationales, supporting inspection at spatial, structural, and decision levels during case review. Evaluation used frontal-only CheXpert data under a shared patient-level protocol and compared NEPTUNE-CXR against CuSeCXR and

DEED across discrimination, calibration, uncertainty ranking, selective prediction, and conformal prediction. All compared methods used the same 14-label task definition, the same frontal-view preprocessing family, and the same validation-driven threshold strategy, allowing performance differences to be attributed mainly to model design rather than evaluation protocol. Experimental results show the strongest overall screening performance for NEPTUNE-CXR, including higher macro and micro discrimination, lower calibration error, better error ranking by uncertainty, lower decision risk at matched coverage, and smaller conformal label sets at comparable empirical coverage. Headline improvements include macro AUROC of 0.912 versus 0.898 and 0.889, expected calibration error of 0.027 versus 0.034 and 0.036, risk of 0.156 at 90% coverage versus 0.171 and 0.176, and average conformal set size of 3.10 at target coverage 0.900 versus 3.43 and 3.61. The combined result pattern suggests that anatomically grounded evidence, healthy-reference counterfactual reasoning, structured concept consistency, and calibrated uncertainty can be brought together within a single chest X-ray model for more dependable screening-oriented clinical decision support.

3.2 Prologue

Chest X-ray imaging remains a primary examination for first-line thoracic assessment because low cost, rapid acquisition, and broad availability support routine use across many care settings. Automated screening from frontal chest X-rays can strengthen triage, prioritization, and quality assurance in high-volume workflows, especially under limited access to specialist review. A frontal chest radiograph, however, rarely carries only one clinically relevant finding. Opacity, pleural change, edema, cardiac enlargement, and other thoracic patterns frequently appear together, often with overlapping visual expression. A multi-label formulation therefore matches practical thoracic screening more closely than single-label classification. Early multi-label studies already showed that modelling dependencies among thoracic findings can improve clinically meaningful prediction, Pham et al., [73].

Despite strong progress in deep learning, trustworthy thoracic screening on open chest X-ray corpora remains difficult for several technical reasons. Label co-occurrence is not random. Some findings frequently appear together, whereas other combinations are uncommon or clinically inconsistent. Graph-based dependency modelling has addressed part of the multi-label problem by learning relationships among labels, Chen et al., [74]. Another difficulty arises from supervision. Large chest X-ray datasets often rely on report-

derived labels, heuristic extraction rules, or mixed-confidence sources. Hard binary targets under such conditions can hide ambiguity, magnify annotation noise, and encourage overconfident predictions when evidence is weak. A further difficulty arises from domain shift across institutions, scanners, acquisition protocols, and patient cohorts. Strong within-dataset performance under one distribution therefore may not transfer cleanly to another distribution. Clinical trust also depends on explanation quality, yet many post-hoc heat maps remain unstable and may highlight non-pathologic regions rather than disease-relevant evidence.

Recent research provides useful components for a stronger screening system, but current components remain separated across different modelling directions. Transformer-based encoders improve contextual modelling by capturing long-range visual interactions across the chest image, Du et al., [31]. Multi-relationship graph learning extends dependency modelling beyond simple co-occurrence by incorporating spatial and semantic relations, Zhang et al., [1]. Expert uncertainty and severity-aware learning adds richer supervision than ordinary binary labels and aligns model training more closely with radiology reporting, Zhang et al., [11]. Hybrid designs that combine transformer representations with structured relation learning further show the value of joint visual and label-context modelling, Lu et al., [8]. Such directions improve modelling capacity, but practical screening requires more than stronger feature extraction. Practical screening also requires reliable confidence, anatomically meaningful evidence, and decision outputs that remain useful under ambiguity.

Uncertainty quantification forms another essential direction because multi-label screening requires more than ranking accuracy. Evidential learning represents uncertainty through distributions over class probabilities and reduces overconfidence under ambiguous supervision, Ashfaq et al., [75]. Sparse Bayesian modelling offers an additional route to epistemic uncertainty under distribution shift, Abboud et al., [76]. Yet uncertainty values alone do not satisfy clinical trust when uncertainty remains disconnected from visible evidence. Screening workflows benefit from outputs that connect confidence to anatomically meaningful regions, plausible alternatives, and clinically coherent concept states. Confidence-aware heatmap methods and CAM ensembles improve localization stability, but explanation by highlighted pixels still remains limited for cases with overlapping findings or diffuse evidence, Rocha et al., [77] and Aasem et al., [65]. Counterfactual explanation addresses such a limitation by asking what image content must

change for a different decision. Generative counterfactual methods for chest X-rays already show that plausible healthy or alternative images can support richer explanation than gradient saliency alone, Atad et al., [78] and Thiagarajan et al., [79]. Latent-space smoothing further improves counterfactual plausibility and stability, Li et al., [80], and class-specific counterfactual reasoning demonstrates that explanation can be integrated into the prediction process rather than added after prediction, Sun et al., [52].

Structured reasoning remains equally important because multi-label chest X-ray screening can produce clinically inconsistent outputs when each label is predicted in isolation. Logic-regularized learning incorporates background knowledge to discourage contradictory label states, Ledaguenel et al., [81]. Concept bottleneck approaches expose intermediate concept variables that can support interpretable reasoning and counterfactual concept analysis, Dominici et al., [82]. A clinically meaningful screening model benefits from both visual evidence and structured consistency. Without such structure, one part of a model may respond to local image cues while another part of the same model produces a label combination that does not fit anatomical location, morphometric evidence, or known radiographic relations. Current research therefore shows clear progress in separate directions, but current research still treats label dependency modelling, uncertainty estimation, counterfactual explanation, and logic-guided consistency as largely separate objectives.

A clear methodological gap therefore remains. Current multi-label chest X-ray models rarely link anatomical grounding, healthy-reference counterfactual evidence, concept-level consistency, and uncertainty-aware decision outputs within one architecture. Many approaches improve discrimination without producing clinically meaningful explanation. Other approaches improve explanation without providing calibrated uncertainty or conservative decision options. A further group of approaches improves uncertainty modelling without grounding uncertainty in explicit image evidence or structured concept states. Practical thoracic screening requires a more integrated design. A suitable screening framework should ground evidence in thoracic regions, produce realistic healthy-reference alternatives, enforce compatibility among related findings, and convert uncertainty into operational outputs such as abstention and conservative label sets for triage. Such requirements motivate the design of NEPTUNE-CXR.

NEPTUNE-CXR addresses the methodological gap through a modular design built for screening-oriented prediction, explanation, and decision control. An anatomic prior and

zoning module defines lung and heart regions and partitions the lung field into six anatomically meaningful zones. A zone-aware transformer encoder then forms a structured latent representation with one global token and six zone tokens, allowing whole-image context and zone-level evidence to coexist in the same feature space. A counterfactual diffusion normalizer generates a closest-healthy reconstruction and a residual evidence map, turning abnormality explanation into explicit image differences relative to a healthy reference. A neuro-symbolic concept graph then combines global context, zone evidence, residual summaries, and morphometric descriptors to encourage clinically compatible concept and label states. A final uncertainty layer combines evidential uncertainty, GP-like last-layer uncertainty, and healthy-manifold plausibility into decision outputs that support calibrated probabilities, abstention, and set-valued prediction. Such integration directly targets the practical demands of multi-label thoracic screening, where dependable decision support depends on evidence quality, prediction consistency, and uncertainty-aware action rather than on accuracy alone.

The chapter contributions are fivefold. Contribution one is a unified multi-label thoracic screening architecture that brings anatomy-aware encoding, healthy-reference counterfactual normalization, concept-graph reasoning, and uncertainty-aware decision control into one pipeline. Contribution two is zone-aware representation learning that supports region-structured evidence aggregation and anatomically aligned attribution. Contribution three is healthy-reference counterfactual normalization that produces residual evidence maps and a global distance-to-healthy signal for explanation and plausibility assessment. Contribution four is neuro-symbolic concept-graph integration that improves cross-label consistency and exposes interpretable intermediate concept activations. Contribution five is an uncertainty-aware decision layer that combines complementary uncertainty signals and supports both abstention-based selective prediction and calibrated conformal set outputs for conservative screening.

This provides a selective review of research directions that most directly motivate the NEPTUNE-CXR design. Therefore narrows attention to the immediate methodological context of the proposed framework. The focus falls on five connected areas: multi-label dependency modelling, uncertainty estimation under weak supervision, explainability for screening, neuro-symbolic consistency modelling, and robustness under shift and open-set conditions. The goal is not to repeat a general survey, but to identify what prior work has already contributed, what remains technically insufficient, and why those

unresolved issues justify the integrated design of NEPTUNE-CXR.

Multi-Label Thoracic Screening and Label Dependency Modelling: Multi-label thoracic screening requires explicit modelling of relations among findings because chest X-ray abnormalities often co-occur and influence each other. Graph-based methods address this requirement by treating findings as structured entities rather than independent outputs. Cai et al., [5] improved multi-label chest X-ray classification through graph convolution with label semantic enhancement, showing that structured label relations can strengthen feature learning. Zhang et al., [1] extended this direction through multi-relationship graph learning, where different edge types represent richer clinical structure than simple co-occurrence. Zhang et al., [11] further showed that uncertainty and severity information can be integrated into relation learning, making dependency modelling more compatible with ambiguous clinical supervision.

Transformer-based models provide a different route to dependency modelling by capturing long-range contextual interactions across image regions. Du et al., [31] used a transformer-based design for multi-disease chest X-ray diagnosis and showed that contextual reasoning can help when labels are uncertain. Hybrid graph-transformer designs attempt to combine strong image-context modelling with explicit label-relation learning. Lu et al., [8] reported such a hybrid framework for multi-label chest X-ray classification, indicating that visual context and structured dependencies can complement each other. Another related direction enriches label semantics with language-derived information from radiology text, allowing image modelling to benefit from clinical wording and report structure, Efimovich et al., [83].

These developments improve classification quality in important ways. Graph models reduce the weakness of independent label prediction. Transformer models improve contextual feature aggregation. Hybrid models bring both ideas into one pipeline. Language-enriched supervision adds semantic detail that image-only training may miss. Even with these gains, several limitations remain. Correlation learning can become brittle when learned relations reflect dataset composition rather than clinical causality. Fixed relation graphs can remain incomplete because thoracic dependencies vary with patient condition, imaging quality, and acquisition setting. Weak labels from reports can distort dependency learning, especially when uncertainty is compressed into hard targets. Most dependency models also stop at label interaction and do not connect structured relations to explicit uncertainty control or clinically meaningful explanation. These

limitations motivate a design that preserves dependency modelling, but anchors representation in anatomy, links evidence to a healthy reference, and supports uncertainty-aware decision outputs. That design need directly motivates the APZ, Z- ViT, and NSCG components in NEPTUNE-CXR.

Uncertainty Modeling Under Weak and Ambiguous Supervision: Weak and ambiguous supervision is a central problem in chest X-ray screening because many labels come from clinical reports rather than direct image-level expert annotation. Report language often contains uncertainty, hedging, omission of subtle findings, and variation in severity description. Under such conditions, multi-label prediction needs uncertainty estimates that reflect both data ambiguity and model uncertainty. Aleatoric ambiguity refers to uncertainty already present in the image-label pair, such as faint evidence, overlapping findings, poor image quality, or ambiguous report wording. Epistemic uncertainty refers to uncertainty arising from limited model knowledge, especially when a test image differs from the training distribution because of scanner changes, protocol differences, or uncommon patient presentations.

Evidential learning has become an important approach for modelling uncertainty without repeated Bayesian sampling. Ashfaq et al., [75] showed that evidential formulations can reduce overconfidence by representing predictions through evidence concentration rather than only point probabilities. Bayesian and sparse Bayesian methods provide another route to epistemic uncertainty by attaching uncertainty to feature-space regions with weak training support, Abboud et al., [76]. Uncertainty-aware benchmark design has also become more realistic. Zhang et al., [1] introduced a benchmark that includes clinical uncertainty and severity information, while Zhang et al., [11] further demonstrated that expert uncertainty and severity labels can improve both prediction and reliability. Du et al., [31] also showed that transformer-based multi-disease modelling benefits from learning under certain and uncertain labels together rather than ignoring ambiguous supervision. Important gaps remain. Many uncertainty methods provide a scalar score without strong connection to anatomically grounded evidence. Calibration may degrade under domain shift even when in-distribution uncertainty looks useful. Most prior work also stops at uncertainty estimation and does not fully convert uncertainty into safer screening actions such as abstention or conservative label-set reporting. In addition, some methods capture one form of uncertainty but do not combine weak-evidence uncertainty, feature-space novelty, and global abnormality plausibility in one decision layer. These limitations

motivate the uncertainty design in NEPTUNE-CXR, where evidential uncertainty, GP-like uncertainty, and healthy-manifold plausibility are combined and used for selective prediction and conformal set-valued outputs.

Explainability for Chest X-Ray Screening: Explainability remains necessary for trustworthy chest X-ray screening, but conventional heatmap-based approaches have clear weaknesses in multi-label settings. Class activation maps and related saliency tools can produce diffuse highlights, unstable visual patterns, or responses over non-pathologic image regions, especially when several findings overlap in one radiograph. Such behavior weakens clinical usefulness because a highlighted region does not always explain what abnormal pattern matters or how the decision would change under a different image state. Confidence-aware heatmap methods improve part of this problem by attaching reliability to localization cues. Rocha et al., [77] showed that heatmap interpretation becomes more useful when confidence is considered together with visual explanation. Aasem et al., [65] further improved heatmap stability through CAM ensembles and weak supervision, demonstrating that multiple localization views can reduce some single-method instability. Counterfactual explanation addresses a different and often more clinically meaningful question: what should change in the image for the decision to change? Atad et al., [78] used generative counterfactuals for chest X-rays, showing that alternative image states can support more intuitive explanation than gradient-only saliency. Thiagarajan et al., [79] connected counterfactual generation with calibration-aware training, while Li et al., [80] improved counterfactual trustworthiness through latent-space smoothing to reduce unrealistic artifacts. Sun et al., [52], [81] moved further by integrating class-specific counterfactual reasoning into multi-label classification itself, rather than adding explanation after prediction. Counterfactual explanation is more clinically meaningful than gradient-only saliency for several reasons. A gradient map indicates sensitivity of a score to local perturbation, but a counterfactual reference expresses a plausible alternative state of the same case. A gradient map often highlights where a model looked. A counterfactual can show what image content must change to move toward a healthy or different diagnostic state. Such a formulation fits radiological reasoning more closely, especially in screening where clinicians often compare current appearance with an expected normal appearance. Concept-based explanation offers another useful direction. Dominici et al., [82] showed that counterfactual concept bottlenecks can expose intermediate concept variables and connect concept intervention to explanation. Fang et

al., [84] also demonstrated that counterfactual-powered decision reasoning can expose decisive image patterns in a more actionable way than ordinary heatmaps. Kumar et al., [85] further highlighted that counterfactual explanations must be protected against spurious correlations, otherwise explanation quality can remain misleading even when visual realism looks strong.

A remaining gap still separates current explainability work from trustworthy screening needs. Explanation requires plausibility, anatomical grounding, and alignment with multi-label consistency. Explanation also needs connection to uncertainty so that weak or unreliable evidence is expressed as uncertainty rather than as a confident but unstable explanation artifact. These unresolved issues directly motivate the CDN module and the integration of counterfactual residuals with anatomy-aware tokens, concept reasoning, and uncertainty-aware outputs in NEPTUNE-CXR.

Neuro-Symbolic Reasoning and Logic-Based Consistency in Medical AI: Independent prediction of many thoracic labels can produce clinically inconsistent outputs. A model may assign strong probability to a label combination that does not fit anatomical distribution, morphometric evidence, or known radiographic relations. Neuro-symbolic reasoning addresses this weakness by combining learned image features with structured constraints. Ledaguenel et al., [81] showed that logic-based regularization can improve neural classification by penalizing contradictions during learning or inference. Such methods are useful in multi-label screening because co-occurrence, exclusion, and implication relations often matter clinically even when image evidence is imperfect.

Concept bottleneck modelling supports a related goal by introducing interpretable intermediate states between image features and final predictions. Dominici et al., [82] showed that counterfactual concept bottlenecks can expose concept variables that are both interpretable and intervention-ready. Multi-relationship graph learning also moves toward structured clinical reasoning by encoding relationships among findings, Zhang et al., [1]. Severity- and uncertainty-aware relation learning further suggests that consistency modelling in medical AI should not behave as rigid symbolic logic, because chest X-ray interpretation often remains uncertain, context-dependent, and partly subjective, Zhang et al., [11].

Soft, differentiable constraints are therefore preferable to rigid rules in ambiguous clinical settings. Rigid rules can suppress valid atypical cases or amplify annotation error when evidence is weak. Soft constraints allow evidence and structure to interact gradually. A

model can still express uncertainty, while structured penalties discourage obvious contradictions. Such behavior is well suited to chest X-ray screening, where evidence may be incomplete, noisy, or only weakly visible. A practical neuro-symbolic screening model should therefore connect learned visual evidence, zone-level evidence, morphometric cues, and concept relations inside one differentiable framework.

Robustness, Domain Shift, Open-Set Settings, and Multimodal Trends:

Generalization beyond one dataset remains a major barrier for trustworthy chest X-ray screening. Robustness research increasingly focuses on reducing dependence on spurious correlations and on improving behavior under acquisition and population shift. Carloni et al., [86] explored causality-inspired contrastive disentanglement to separate clinically meaningful factors from nuisance variation. Roschewitz et al., [87] used counterfactual contrastive learning to build features less sensitive to domain shift. Kumar et al., [85] also addressed spurious correlations from the explanation side by debiasing counterfactual reasoning. Such directions are valuable because chest X-ray models can otherwise learn shortcuts linked to scanner style, support devices, image borders, or dataset-specific prevalence patterns rather than true pathology.

Open-set behaviour creates a related challenge. Ko et al., [88] studied difficulty-aware multi-label diagnosis in settings where some cases contain patterns outside the training taxonomy. Open-set recognition is closely tied to uncertainty because unknown or weakly supported patterns should not receive overconfident standard predictions. A screening model therefore needs not only strong in-distribution classification, but also reliable novelty signals that can trigger conservative decisions.

Multimodal trends have expanded the design space further. Efimovich et al., [83] showed that image modelling can benefit from language-derived label semantics, especially when supervision is weak. Sharma et al., [69] extended the reliability discussion to uncertainty-aware radiology reporting, highlighting the need to reduce unsupported statements in language outputs. Lu et al., [8] demonstrated that hybrid architectures remain useful when global context modelling and structured dependency learning must coexist. Fang et al., [84] added a robustness-related view through counterfactual-powered decision reasoning, where decisive evidence patterns can be inspected for spurious reliance.

These directions are relevant, but current progress remains fragmented. Robustness methods often focus on representation learning without structured explanation. Multimodal methods often improve semantic richness without calibrated uncertainty

control. Open-set approaches often remain separate from multi-label dependency reasoning. A unified screening model should connect anatomically grounded evidence, realistic explanation, structured concept consistency, and calibrated uncertainty in one framework. Such a requirement motivates the integrated design of NEPTUNE-CXR rather than another single-objective chest X-ray classifier.

This section shows a consistent pattern across prior research. Multi-label dependency modelling has improved through graphs, transformers, and hybrid architectures. Uncertainty modelling has become more realistic through evidential learning, Bayesian methods, and severity-aware supervision. Explainability has moved beyond unstable heatmaps toward counterfactual and concept-based reasoning. Neuro-symbolic methods have shown value for consistency control, and robustness research has highlighted the need for protection against shift, spurious correlations, and unknown patterns. Even with this progress, current methods still address most trustworthiness components separately. A fully integrated design that links anatomical grounding, healthy-reference counterfactual evidence, concept-level consistency, and calibrated uncertainty-aware outputs remains largely absent. Section 3.3 therefore presents NEPTUNE-CXR as a unified framework designed to connect those components within one multi-label thoracic screening architecture.

3.3 Methods

This section presents NEPTUNE-CXR as a unified pipeline for trustworthy multi-label thoracic screening from frontal chest X-rays. The pipeline starts from one normalized frontal CXR and ends with calibrated label probabilities, per-label uncertainty estimates, abstention decisions, conformal label sets, and explanation artifacts suitable for screening-oriented review. Four methodological elements are integrated within one architecture. Anatomy-aware encoding begins with an anatomic prior and zoning module that estimates lung and heart masks and divides the lung field into six clinically meaningful zones. A zone-aware transformer encoder then builds one global representation and six zone representations through mask-guided attention, preserving whole-image context together with region-specific evidence. Healthy-reference counterfactual normalization follows through a counterfactual diffusion normalizer that generates a closest-healthy reconstruction, a residual evidence map, and a distance-to-healthy score. Neuro-symbolic reasoning is then introduced through a concept graph that combines transformer features, residual summaries, and morphometric descriptors under

soft consistency constraints. A final uncertainty layer fuses evidential uncertainty, GP-like epistemic variance, and generative plausibility to produce calibrated probabilities and conservative decision outputs. Figure 3.1 provides the architecture overview and the data flow across all modules. Section 3.3.1 defines the learning problem and the end-to-end mapping, while later subsections describe the anatomical encoder, counterfactual module, concept graph, and uncertainty mechanisms in detail.

NEPTUNE-CXR: Neuro-symbolic, Counterfactual, Uncertainty-Aware Multi-Label CXR Screening

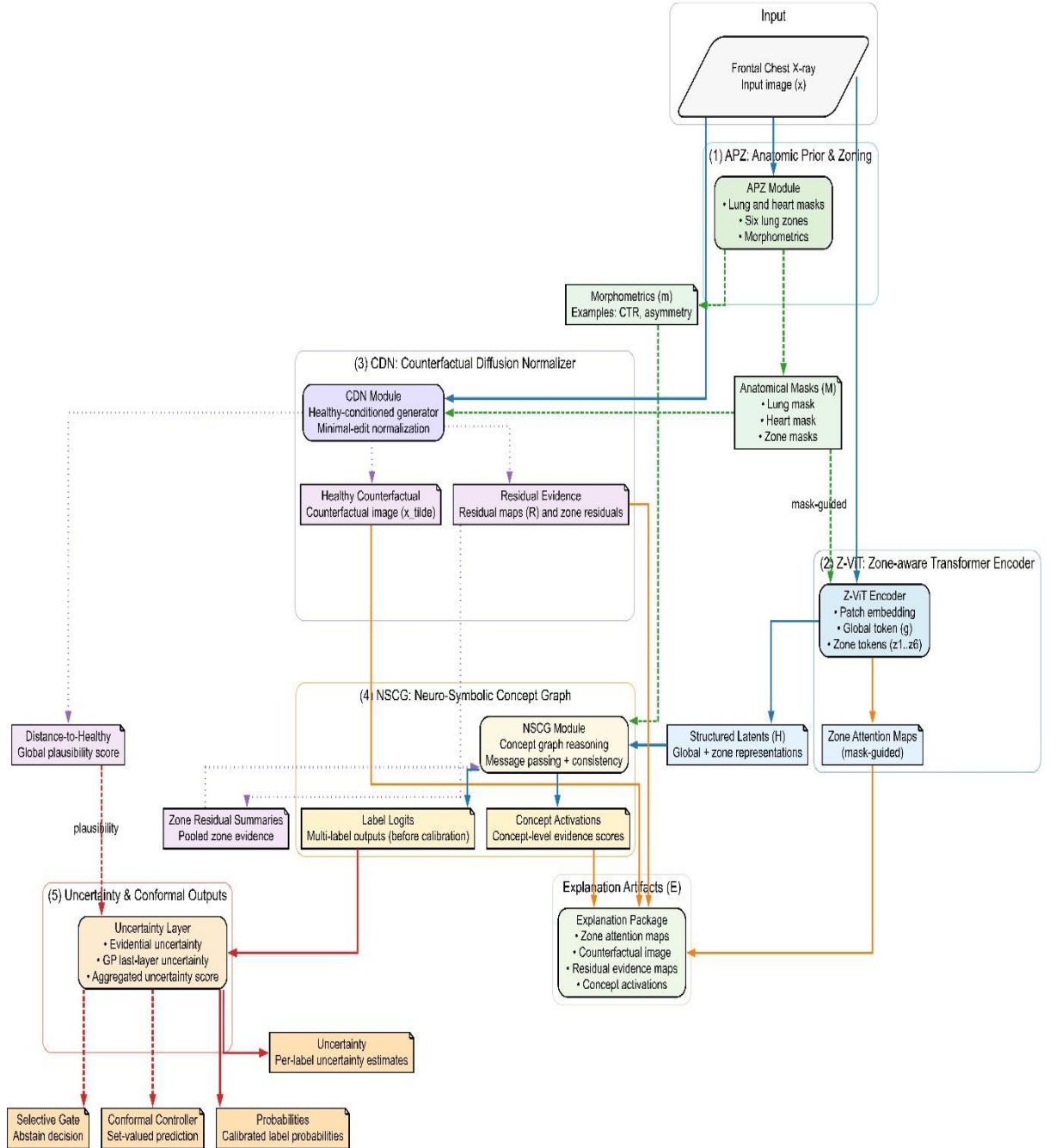


Figure 3.1: NEPTUNE CXR architecture for uncertainty aware, explainable multi label thoracic screening from frontal chest X rays

3.3.1 Problem Formulation and Overall Architecture

Multi-label thoracic screening from chest radiographs is formulated as prediction of several clinically relevant observations from a single frontal chest X-ray. Let $x \in \mathbb{R}^{H \times W}$ denote a normalized frontal CXR image, and let $y \in \{0,1\}^C$ denote the corresponding multi-label target vector for C thoracic observations. For each class index $c \in 1, \dots, C$, $y_c = 1$ indicates presence of observation c , and $y_c = 0$ indicates absence of observation c . The learning objective is estimation of calibrated probabilities for all labels under weak supervision, label ambiguity, and deployment shift. Such a formulation differs from ordinary multi-label image tagging because thoracic labels are clinically related, region dependent, and often unequally reliable across images and across classes.

Large-scale CXR supervision is commonly derived from routine radiology reports or weak labeling pipelines. Report-derived supervision introduces several error sources, including ambiguous wording, inter-reader variation, omission of subtle findings, and label extraction errors. Ambiguous supervision creates a setting in which a hard binary target may hide clinically meaningful doubt. Screening-oriented prediction under such conditions requires more than strong ranking performance. A screening model must also return calibrated probabilities, explicit uncertainty signals, and clinically meaningful explanation artifacts. Calibrated probabilities are required because triage and escalation decisions depend on risk level, not only on label order. Explicit uncertainty is required because low-evidence or novel cases should not receive overconfident predictions. Clinically meaningful explanation is required because a screening output without anatomically grounded evidence offers limited value for review, audit, or disagreement resolution.

NEPTUNE-CXR addresses the problem through a modular pipeline that links anatomical grounding, healthy-reference explanation, concept-level reasoning, and uncertainty-aware decision control. The first stage is the anatomic prior and zoning module, denoted APZ. APZ estimates anatomical masks for the lung fields and cardiac silhouette, constructs six lung zones using a right/left by upper/middle/lower partition, and derives low-dimensional morphometric descriptors such as cardiothoracic ratio and area asymmetry. The second stage is the zone-aware transformer encoder, denoted Z-ViT. Z-ViT converts the image into a structured latent representation with one global token for whole-image context and six zone tokens for localized evidence. The third stage is the counterfactual diffusion normalizer, denoted CDN. CDN generates a plausible closest-healthy reconstruction of the

same case under anatomical conditioning, then computes a residual evidence map by subtracting the healthy reconstruction from the original image. CDN also produces a scalar distance-to-healthy score that summarizes global deviation from the healthy manifold. The fourth stage is the neuro- symbolic concept graph, denoted NSCG. NSCG integrates global features, zone features, residual summaries, and morphometric descriptors into a graph of radiographic concepts and diagnostic relations, then performs message passing under soft consistency constraints. The final stage is the uncertainty layer, which combines evidential uncertainty, GP-like last-layer uncertainty, and generative plausibility into calibrated probabilities, abstention decisions, and conformal set-valued outputs. Figure 3.1 summarizes the full data flow across the five stages. The overall mapping of the proposed system is written as follows:

$$(\hat{p}, U, E) = f_{\theta}(x) \rightarrow (\text{Eq 3.1})$$

In Eq 3.1, $\hat{p} \in [0,1]^C$ denotes the vector of predicted probabilities for the C thoracic labels, $U \in \mathbb{R}^C$ denotes the vector of per-label uncertainty estimates, and E denotes the explanation artifacts returned by the pipeline. The probability vector $\hat{p}$ provides screening scores for presence or absence of each observation. The uncertainty vector U separates prediction strength from confidence by quantifying lack of evidence, representation-space novelty, and low healthy-manifold plausibility. The artifact set E packages evidence in forms that are useful for inspection at different levels, including anatomically grounded zone attention summaries, counterfactual images, residual evidence maps, and concept activations. Eq 3.1 therefore expresses NEPTUNE-CXR as a screening model that predicts, explains, and quantifies decision uncertainty within one end-to-end mapping. Notation used throughout Section 3.3 follows the architecture in Figure 3.1. Symbol x denotes the input frontal chest X-ray, and symbol y denotes the multi-label target vector. Symbol M denotes anatomical conditioning information produced by APZ, with $M = M_{\text{lung}}, M_{\text{heart}}, M_1, \dots, M_6$, where M_{lung} is the lung mask, M_{heart} is the heart mask, and $M_1, \dots, M_6$ are the six zone masks. Symbol g denotes the global transformer token that aggregates whole-image context, and symbols z_i denote zone tokens that aggregate evidence from anatomically constrained regions for $i \in 1, \dots, 6$. Symbol $\bar{x}$ denotes the healthy counterfactual image generated by CDN. Symbol R denotes the residual evidence map defined by $R = x - \bar{x}$. Symbol $d(x)$ denotes the distance-to-healthy score that summarizes global deviation from the healthy conditional manifold. Later subsections

also use pooled zone residual summaries and morphometric descriptors, but Eq 3.1 and Figure 3.1 already capture the central flow of information from raw image to prediction, uncertainty, and explanation. Integration of the modules within one system follows a direct engineering rationale. Anatomical zoning alone can improve region grounding, but anatomical zoning alone does not produce a healthy reference, concept consistency, or decision uncertainty. Counterfactual normalization alone can localize abnormal edits, but counterfactual normalization alone does not enforce compatibility across related findings. Concept- graph reasoning alone can discourage contradictory outputs, but concept-graph reasoning alone depends on structured evidence from image regions and from counterfactual residuals. Uncertainty scoring alone can rank risky cases, but uncertainty scoring alone offers limited clinical value when uncertainty is not connected to visible evidence and concept states. Joint design is therefore necessary because each module addresses a different failure mode of screening from weakly labeled chest radiographs. APZ and Z-ViT address anatomical grounding, CDN addresses healthy-reference evidence extraction, NSCG addresses concept-level consistency, and the uncertainty layer converts model confidence into conservative decision policies. The combined design also allows feature-level interaction among modules rather than late combination of unrelated post-hoc tools.

Figure 3.1 presents the architecture as a sequence of evidence transformations rather than as a single black-box classifier. APZ converts raw image geometry into anatomical priors. Z-ViT converts anatomy-aware inputs into structured global and zonal features. CDN converts structured features and masks into a healthy-reference view and residual evidence. NSCG converts multimodal evidence into concept activations and label logits under soft clinical constraints. The uncertainty layer converts logits, evidence concentration, feature-space novelty, and healthy-manifold plausibility into risk-aware screening outputs. Such a staged architecture is well matched to practical thoracic screening because each stage yields a diagnostically meaningful intermediate representation that can be inspected, calibrated, and stress-tested under weak labels and distribution shift.

3.3.2 Anatomic Prior and Zone-Aware Encoder (APZ + Z-ViT)

Thoracic findings in frontal chest X-rays follow strong anatomical structure. Pleural fluid usually accumulates in dependent lung regions, many opacity patterns show zone preference, and cardiac enlargement is tied to global thoracic proportions rather than to isolated texture patches. A front-end that ignores anatomical layout can mix unrelated evidence, weaken

localization, and increase reliance on dataset-specific shortcuts. For multi-label screening, such behavior is especially problematic because several findings can appear together in different parts of the chest. The APZ + Z-ViT front-end in Figure 3.1 addresses that problem by imposing explicit anatomical structure before higher-level reasoning and uncertainty modeling. The main objective is not only stronger feature extraction, but also formation of region-aligned evidence carriers that remain useful for explanation, consistency checking, and uncertainty-aware decision making later in the pipeline.

The anatomic prior and zoning module begins with estimation of major thoracic masks. Let $M \in \{0,1\}^{H \times W}_{lung}$ denote the lung field mask and $M \in \{0,1\}^{H \times W}_{heart}$ denote the cardiac silhouette mask. An optional central-structure mask can also be included when available, denoted by $M \in \{0,1\}^{H \times W}_{med}$, to separate mediastinal or thoracic-cavity structures from peripheral lung parenchyma. Such masks act as spatial priors rather than as final predictors. The role of the masks is to define where region-specific features should be aggregated and where counterfactual edits should be constrained in later modules. Mask generation can be obtained from a standard segmentation model or from weak supervision, but exact mask-generator choice is not required for the present method definition. The key requirement is availability of anatomically meaningful binary regions that support downstream zoning and morphometric extraction.

Inside the lung field, APZ constructs a fixed six-zone partition that reflects common radiographic reading practice. The right lung and left lung are treated separately, and each side is divided into upper, middle, and lower regions. Let $\{M\}^6$ denote the six zone masks, where each $M_i \subseteq M_{lung}$. The zone system is defined so that the masks do not overlap and, up to minor boundary effects, jointly cover the lung field Eq 3.2:

$$M_i \cap M_j = \phi \text{ for } i \neq j, \bigcup_{i=1}^6 M_i \approx M_{lung} \dots (\text{Eq 3.2})$$

A stable construction rule follows two simple steps. The first step separates right and left lung regions using the estimated lung mask. The second step divides each lung side into three height-based bands according to the vertical extent of that side. Such a rule avoids dependence on pixel-spacing metadata, avoids fragile hand-tuned landmarks, and preserves consistency across datasets with different resolutions, scanners, and acquisition settings.

A stable zone definition is important because later modules use zone identity as part of explanation and reasoning, not only as a preprocessing convenience. APZ also derives a morphometric feature vector $m \in \mathbb{R}^D$ from the estimated masks. Morphometrics provide

low-dimensional, clinically interpretable measurements that complement image features. A primary example is cardiothoracic ratio, defined as $CTR = w_{\text{heart}} / w_{\text{thorax}}$, where w_{heart} and w_{thorax} are width measures obtained from cardiac and thoracic boundaries. Additional features can include total lung area, left-right lung area asymmetry, relative heart area, and zone-wise density summaries based on intensity statistics inside the six zone masks. Lung area can support recognition of reduced expansion or gross cropping. Left-right asymmetry can capture imbalance associated with unilateral disease or imperfect positioning. Relative heart area and CTR can regularize cardiac-related predictions. Zone-wise density summaries can provide coarse intensity trends linked to focal or diffuse parenchymal changes. The vector m does not replace image features. The vector m acts as structured side information that later modules can use for concept consistency and plausibility checks.

The zone-aware transformer encoder converts the image into a structured $\mathbb{R}^d$ latent representation that preserves both global context and anatomically localized evidence. The input image x is partitioned into N non-overlapping patches. Each patch is projected through a linear embedding into a token $p \in \mathbb{R}^d$, for $j=1, \dots, N$. Positional encoding is then added so that token order retains spatial information across the chest image. Two special token types are introduced before transformer processing. A global token g is used to aggregate whole-image context such as diffuse bilateral opacity, overall silhouette shape, or broad exposure-related effects. Six zone tokens $\{z\}^6$ are used to aggregate evidence from the six anatomically defined lung zones. The initial token sequence is therefore formed as $[g, z_1, z_2, z_3, z_4, z_5, z_6, p_1, \dots, p_N]$. Such a token design separates chest-wide reasoning from region-specific reasoning while keeping both in the same transformer backbone.

Localization of each zone token is enforced directly in attention computation. Let $Z(i) \subseteq \{1, \dots, N\}$ denote the set of patch indices whose patch centers fall inside zone mask M_i . In one multi-head self-attention head, the attention from zone token z_i to patch token p_j is restricted through an additive mask:

$$a_{ij} = \text{Soft max} \left(\frac{q_i^T k_j}{\sqrt{d}} + \begin{cases} 0, & j \in Z(i) \\ -\infty, & j \notin Z(i) \end{cases} \right) \dots \text{(Eq 3.3)}$$

In Eq 3.3, q_i is the query vector for zone token z_i , and k_j is the key vector for patch token p_j . The additive mask has a precise effect. For patches inside the anatomical zone, $\Delta_{ij} = 0$, so the raw attention score remains unchanged. For patches outside the anatomical zone, Δ_{ij}

$= -\infty$, so the corresponding probability becomes zero after the softmax operation. As a result, zone token z_i can only pool information from its own anatomical region. The global token g remains unmasked and can attend across all patches. Patch-to-patch attention also remains unconstrained. Such a combination is important because cross-zone interactions can still propagate through the patch stream even when each zone token remains anatomically localized.

After stacked transformer blocks, the encoder returns a compact structured latent state composed of one global token and six zone tokens Eq 3.4:

$$H = \{g, z_1, z_2, z_3, z_4, z_5, z_6\} \dots (\text{Eq 3.4})$$

The output H forms the main representation passed to later modules. The global token carries chest-wide context that can support recognition of diffuse findings and broad anatomical proportions. Each zone token carries region-constrained evidence that can support localized abnormality reasoning. The representation H is also compatible with later fusion of morphometric features m , especially in modules that need both learned image features and explicit anatomical measurements. A key advantage of the structured output is interpretive clarity. Instead of one undifferentiated embedding vector, the encoder returns evidence carriers with known spatial meaning. Such structure later supports zone-wise residual analysis in CDN and concept-level aggregation in NSCG.

Zone tokens also provide direct explainability hooks before any post-hoc saliency method is applied. Each z_i summarizes evidence from a constrained anatomical region by design, not by retrospective interpretation. The masked attention weights used by each zone token can therefore be visualized as zone-conditioned attribution maps.

A clinician or reviewer can inspect whether evidence for a predicted finding concentrates in a lower lobe region, an upper-zone region, or a bilateral pattern distributed across several zones. Layer-wise or head-wise aggregation can provide stable regional summaries and reduce sensitivity to small local perturbations. Such zone-level attribution is complementary to the pixel-level residual evidence generated later by the counterfactual module. Zone-level attribution answers a region question, namely where structured evidence accumulates. Residual evidence answers a change question, namely what image content must change to approach a healthy reference. Concept activations in NSCG later add a third view, namely how structured evidence supports consistent radiographic reasoning. Together, the zone tokens create an interpretable intermediate interface between

raw image appearance and later decision modules.

The overall design rationale of APZ + Z-ViT is therefore straightforward. Thoracic screening benefits from explicit anatomical structure because many findings have characteristic spatial distributions and because multiple findings can coexist in different chest regions. A region-aware encoder reduces reliance on background cues, scanner-specific artifacts, and unstable global shortcuts. A global token preserves full-image context for diffuse disease patterns and overall thoracic geometry. Zone tokens preserve local structure in a clinically meaningful coordinate system. Morphometric features add a compact source of interpretable measurement evidence. Mask-guided attention binds zone tokens to anatomy at the representation level, which improves interpretability and gives later modules cleaner inputs for counterfactual normalization, concept-graph reasoning, and uncertainty estimation. Zone-aware encoding therefore serves two linked purposes inside NEPTUNE-CXR: stronger robustness through anatomically constrained feature learning and stronger interpretability through explicit region-grounded evidence carriers.

3.3.3 Counterfactual Diffusion Normalizer (CDN)

Pixel-level saliency methods usually explain a classifier by showing image regions that change a score under small perturbations. Such visualizations can be unstable under minor appearance variation and can also respond to non-pathologic cues, especially in multi-label chest X-rays with overlapping findings. A healthy-reference counterfactual provides a different explanation logic. Instead of asking which pixels affect a logit most strongly, a healthy-reference counterfactual asks which image content must change for the same case to become a plausible healthy chest X-ray. That change-based view is more clinically meaningful because radiographic review often depends on comparison against expected healthy appearance. Recent counterfactual work in chest imaging has

already shown the value of plausible alternative images, calibration-aware counterfactual learning, and class-specific counterfactual reasoning, Atad et al., [78], Thiagarajan et al., [79], Li et al., [80], and Sun et al., [52]. Within NEPTUNE-CXR, the Counterfactual Diffusion Normalizer (CDN) operationalizes that idea by generating a closest-healthy reconstruction under anatomical constraints and then using the difference between the original image and the healthy reconstruction as explicit abnormality evidence. Figure 3.1 places CDN after zone-aware encoding and before neuro-symbolic reasoning, which allows counterfactual evidence to support both explanation and uncertainty-aware screening outputs.

Conditional healthy generative model. CDN is defined as an anatomy-conditioned generative model of healthy frontal chest X-rays. Anatomical conditioning is inherited from APZ and is written as

$$M = \{M_{lung}, M_{heart}, M_l, \dots, M_6\} \dots (\text{Eq 3.5})$$

In Eq 3.5, M_{lung} denotes the lung-field mask, M_{heart} denotes the cardiac silhouette mask, and $M_1, \dots, M_6$ denote the six lung-zone masks. Conditioning on M constrains generation to the anatomical layout of the input case. Such conditioning reduces unnecessary structural change, discourages background editing, and keeps the healthy reconstruction aligned with patient-specific geometry rather than producing a generic normal-looking radiograph. The learned target distribution is the healthy conditional distribution $p_\phi(x| \text{healthy}, M)$, where the label “healthy” denotes membership in a healthy subset and M provides structural constraints derived from the same input anatomy. In practical terms, the learned distribution represents the set of plausible healthy chest X-rays that match the observed thoracic layout closely enough to permit meaningful change analysis. A diffusion formulation is adopted for the healthy conditional model because diffusion models offer stable optimization and strong conditional sample quality under complex image distributions. Frontal chest X-rays contain subtle texture variation, broad intensity overlap across tissues, acquisition artifacts, and nontrivial dependence between global thoracic geometry and local pathology.

A diffusion model is well suited to such structure because iterative denoising supports flexible modeling of high-dimensional conditional image distributions without requiring a single direct deterministic reconstruction rule. In the CDN formulation, the diffusion component is trained on healthy samples using a standard denoising objective, denoted abstractly by $L_{\text{diff}}(\phi)$. The present chapter keeps the diffusion definition model-agnostic, which preserves methodological focus on the healthy conditional objective rather than on one fixed denoising architecture. Under that formulation, the generative model learns healthy radiographic appearance constrained by anatomy, and later inference uses that learned healthy manifold as a reference space for abnormality explanation.

Minimal-edit healthy counterfactual inference. Given an input chest X-ray x , CDN generates a healthy counterfactual $x^\dagger$ that preserves patient-specific structure while removing disease-related appearance as far as possible. Counterfactual inference is

formulated as a constrained projection toward the healthy conditional manifold. Let $E_\phi(x'| \text{healthy}, M)$ denote an energy associated with the learned healthy model. Lower energy indicates stronger compatibility with healthy appearance under the same anatomical conditioning. The counterfactual is then obtained by solving

$$\tilde{x} = \arg \min E_\phi(x'| \text{healthy}, M) + \lambda \|x - x'\|_1 + \beta \int (x - x')^2 (1 - M) \dots (\text{Eq3.6})$$

Eq 3.6 has three parts with distinct roles. The healthy energy term drives the candidate image x' toward the healthy conditional distribution. The second term, weighted by λ , limits edits inside the lung field and therefore enforces a minimal-change principle in the main disease-bearing area. The third term, weighted by β , penalizes changes outside the lung mask and therefore preserves background, borders, and non-thoracic image content. The resulting optimization seeks the smallest anatomically plausible set of changes needed to convert the observed case into a healthy-consistent version.

The healthy energy term is the main mechanism that prevents the counterfactual from collapsing into a cosmetically smoothed version of the input. Without the healthy energy, Eq 3.6 would favor $x' \approx x$ because the edit penalties alone reward minimal change. With the healthy energy included, a clearly abnormal image cannot remain unchanged unless the unchanged image is already compatible with the healthy conditional model. The coefficient λ controls the extent of lung-field editing. A very small λ would allow aggressive editing inside the lungs and could erase too much patient-specific appearance. A very large λ would resist clinically necessary change and could leave major abnormalities in place. The coefficient β has a complementary role outside the lungs. A very small β would permit irrelevant changes around image borders, markers, or external structures. A sufficiently strong β keeps the counterfactual focused on the thoracic region and reduces explanation through background manipulation. The combined objective therefore implements a healthy projection that is anatomically anchored, region selective, and edit conservative.

Residual evidence maps and zone decomposition. The central explanation artifact produced by CDN is the residual evidence map. After obtaining the healthy counterfactual $\tilde{x}$, the abnormality evidence is defined directly as the difference between the observed image and the healthy reconstruction:

$$R = x - x, R_i = R \oslash M_i, i \in \{1, \dots, 6\}. \quad R_i \oslash \dots \text{(Eq 3.7)}$$

In Eq 3.7, R denotes the full residual evidence map and R_i denotes the residual restricted to anatomical zone i . The map R highlights image content that must change to make the input anatomically compatible with healthy appearance. Zone-wise decomposition provides localized evidence channels aligned with the APZ partition. A lower-zone pleural abnormality, for xample, should produce stronger residual mass in the basal zone masks than in upper zones. Zone residuals also support compact numerical summaries such as $\sum R_i$, trimmed means, or bounded pooling statistics, which later serve as inputs to the concept graph in Section 3.3.4. CDN therefore produces both pixel-space evidence and region-structured evidence from the same healthy-reference mechanism. Residual evidence differs from gradient saliency in both meaning and stability. A gradient map measures local score sensitivity and therefore answers a discriminative question: which pixels would change the logit under infinitesimal perturbation? A residual map answers a generative question: what image content must change to reach a plausible healthy state under anatomical constraints? The two explanations are fundamentally different.

Gradient saliency is tied to local derivatives of a classifier and can vary sharply with logit scaling, local noise, or small changes in feature geometry. Residual evidence is tied to a constrained healthy reference and therefore reflects finite image edits under an explicit optimization objective. A residual map also inherits anatomical conditioning from M , which reduces irrelevant emphasis outside meaningful thoracic regions. The resulting evidence map is therefore more interpretable for screening because the map corresponds to pathology removal relative to a healthy reference rather than to raw classifier sensitivity.

Distance-to-healthy score. CDN also produces a scalar summary of global abnormality called the distance-to-healthy score. The score combines healthy-model energy with anatomy-masked edit magnitude:

Eq 3.8 has a clear operational meaning. The first term measures how strongly the unterfactual aligns with the healthy conditional model under the same anatomy. The second term measures how much lung-field editing was required to produce the healthy reference. A larger $d(x)$ therefore indicates lower healthy plausibility and larger abnormal deviation. A smaller $d(x)$ indicates that the input image already lies close to the

healthy manifold or requires only limited lung-field modification. The score complements the localized evidence map R by providing a global abnormality signal. Later in the pipeline, the uncertainty layer uses $d(x)$ as a generative plausibility component, so that global deviation from healthy structure contributes to uncertainty-aware decision control. The score is especially useful in screening settings because borderline cases, poor-quality scans, or unusual appearances may have only diffuse local evidence while still showing low global healthy plausibility.

Learning objective. CDN training combines two requirements: faithful modeling of healthy chest X-ray appearance and counterfactuals that differ from observed images only where necessary. A high-level objective is written as

$$d(x) = E_{\phi}(x|healthy, M) + \eta \quad (x - x) \quad M_{lung1} \dots \dots \dots (Eq 3.8)$$

Eq 3.8 has a clear operational meaning. The first term measures how strongly the counterfactual aligns with the healthy conditional model under the same anatomy. The second term measures how much lung-field editing was required to produce the healthy reference. A larger $d(x)$ therefore indicates lower healthy plausibility and larger abnormal deviation. A smaller $d(x)$ indicates that the input image already lies close to the healthy manifold or requires only limited lung-field modification. The score complements the localized evidence map R by providing a global abnormality signal. Later in the pipeline, the uncertainty layer uses $d(x)$ as a generative plausibility component, so that global deviation from healthy structure contributes to uncertainty-aware decision control. The score is especially useful in screening settings because borderline cases, poor-quality scans, or unusual appearances may have only diffuse local evidence while still showing low global healthy plausibility.

Learning objective. CDN training combines two requirements: faithful modeling of healthy chest X-ray appearance and counterfactuals that differ from observed images only where necessary. A high-level objective is written as

$$\min_{\phi} E_{x_0 \sim D_{healthy}} [L_{diff}(x_0; \phi)] + \lambda E_{x \sim D} [R_{edit}(x, x_{\phi}(x, M))] \dots (Eq 3.9)$$

In Eq 3.9, $D_{healthy}$ denotes a healthy subset used to learn the conditional healthy manifold, D denotes the full training distribution, and R_{edit} denotes an edit penalty aligned with the anatomy-masked change terms used in Eq 3.6. The diffusion term teaches the

model what healthy appearance looks like under anatomical conditioning. The edit term teaches the model to preserve patient-specific structure and avoid unnecessary change when generating x . The two-term design is important. A healthy-only generative term without edit control could produce counterfactuals that are visually normal but not close enough to the input for trustworthy explanation. An edit-only term without healthy conditioning could preserve the input too strongly and fail to remove pathology. The combined objective therefore balances plausibility and minimal modification.

Safeguards against trivial solutions. Several safeguards are built into the CDN formulation to prevent degenerate behavior. The first safeguard is healthy conditioning itself. A clearly abnormal image cannot satisfy the healthy conditional model without meaningful change, so identity mapping is disfavored when the healthy energy is high. The second safeguard is the anatomy-masked edit design. Lung-field edits are allowed but penalized, which discourages unnecessary rewriting of the image while still permitting clinically relevant corrections. Outside-lung edits receive an additional penalty, which reduces background manipulation and discourages explanation through irrelevant image regions. The third safeguard is the combined use of global and local evidence. A candidate counterfactual must be both healthy-compatible and close to the original anatomy. Degenerate solutions such as excessive smoothing, border editing, or large non-thoracic changes therefore become costly under Eq 3.6 and Eq 3.9. Together, these safeguards preserve the interpretability of the generated counterfactual and improve the stability of downstream residual evidence.

Role as explanation artifacts. CDN contributes four complementary explanation outputs to the full NEPTUNE-CXR pipeline. The counterfactual image x provides a patient-specific healthy reference under the same anatomical layout. The residual map R provides localized abnormality evidence defined through minimal healthy edits rather than through discriminative sensitivity. The zone-specific residuals $\{R_i\}_{i=1}^6$ provide anatomically aligned evidence summaries that can support both human inspection and concept-level reasoning in the NSCG module. The scalar score $d(x)$ provides a global plausibility signal that helps distinguish near-healthy images from strongly abnormal or weakly supported cases. Taken together, x , R , R_i , and $d(x)$ turn counterfactual generation into a functional screening component rather than an isolated visual explanation add-on. CDN therefore serves two roles at once: explanation through healthy-reference evidence

and reliability support through global healthy-manifold plausibility.

3.3.4 Neuro-Symbolic Concept Graph (NSCG)

Thoracic findings rarely behave as independent binary events. Pleural change, edema, cardiac enlargement, and opacity patterns often share anatomical context or clinical co-occurrence. Independent label heads can therefore produce radiographically implausible combinations, especially under weak supervision and overlapping evidence. The Neuro-Symbolic Concept Graph, abbreviated NSCG, addresses such inconsistency by inserting structured reasoning between evidence extraction and final label prediction. The graph combines global context from the zone-aware encoder, localized abnormality evidence from the counterfactual module, and morphometric cues from the anatomical prior module. The resulting design supports two goals at the same time: improved consistency across related labels and an interpretable intermediate concept layer for inspection before final decision formation.

Let $G = (V, E)$ denote the concept graph, where V is the set of concept nodes and E is the set of typed edges. Let $|V| = K$, with $C \leq K$, where C is the number of target labels in the screening task. Node set V is divided into two groups. The first group contains target-label nodes corresponding directly to the thoracic observations predicted by the model. The second group contains higher-level concept nodes representing intermediate radiographic states, grouped abnormalities, or diagnostically meaningful latent concepts. A shared graph over both node groups allows joint reasoning over direct findings and broader diagnostic structure. Edge set E carries relation types that encode clinically meaningful connections among nodes.

Two relation families are used in NSCG. The first family captures anatomical adjacency or spatial association. Edges in such a family connect concepts with typical involvement of the same hemithorax, neighboring lung zones, or central thoracic structures. A basal pleural concept and a lower-zone opacity concept provide one example of spatial association. The second family captures clinical co-occurrence or exclusion. Edges in such a family connect concepts with frequent joint occurrence or rare joint occurrence. Supportive relations can strengthen compatible concepts, whereas exclusion relations can suppress incompatible concepts. Let $t(u, v) \in \{1, \dots, T\}$ denote the edge type for edge $(u, v) \in E$, where T is the number of relation types. Typed edges therefore give SCG enough flexibility to represent both positive and negative clinical structure without collapsing all dependencies into one

undifferentiated adjacency matrix.

Each concept node receives initialization from three evidence sources: structured tokens from Z-ViT, zone residual summaries from CDN, and morphometric features from APZ. Zone residual summaries are extracted by pooling statistics from each zone residual map R_i . For zone $i \in \{1, \dots, 6\}$, the pooled residual summary is defined as

$$r_i = \text{pool}(R_i), i \in \{1, \dots, 6\}. \quad \dots(\text{Eq 3.10})$$

In Eq 3.10, $\text{pool}(\cdot)$ denotes a robust aggregation operator such as mean absolute residual, trimmed mean, or a bounded norm. The main purpose of r_i is compression of pixel-level residual evidence into a low-dimensional summary aligned with anatomical zones. Such a summary allows graph reasoning to use counterfactual evidence without processing a dense image tensor at every graph node.

The initial embedding of concept node $v \in V$ is formed by a concept-specific projection of global, zonal, residual, and morphometric evidence:

$$h_v^{(0)} = \psi_v^{(g, z_1, \dots, z_6, r_1, \dots, r_6, m)} \quad \dots \dots (\text{Eq 3.11})$$

In Eq 3.11, g denotes the global token, $z_1, \dots, z_6$ denote the six zone tokens, $r_1, \dots, r_6$ denote pooled residual summaries, and m denotes the morphometric feature vector. Mapping $\psi_v(\cdot)$ is a learnable projection, for example a linear layer or a shallow multilayer perceptron. Concept-specific projection is important because different concepts require different evidence weights. A cardiac-size concept can assign larger weight to m , especially to cardiothoracic ratio and relative heart area. A lower pleural concept can assign larger weight to r_i from lower lung zones. A diffuse opacity concept can assign stronger weight to global token g and to several zone tokens at the same time. Eq 3.11 therefore creates a flexible interface between structured image evidence and concept-level reasoning. After node initialization, graph message passing refines embedding's through neighborhood structure and edge types. Let $N(v)$ denote the neighbor set of node v . A generic message-passing layer updates node embeddings as Eq 3.12

$$h_i^{(l+1)} = \sigma \left(\sum_{j \in N(i)} \alpha_{ij}^{(l)} w_{\ell}^{(l)} h_j^{(l)} \right) \quad \dots (\text{Eq 3.12})$$

transformation matrix for edge type $t \in \mathcal{T}$, $w_{\ell}^{(l)}$ is a normalized neighbor weight, and σ is a nonlinearity. Relation-specific transformations allow anatomical-association edges and clinical co-occurrence or exclusion edges to influence node updates in different ways.

Neighbor weights can be attention coefficients or degree-normalized constants. Message passing therefore allows each concept to update under supporting or contradicting evidence from related concepts while still preserving local evidence through the self-embedding term. One or two graph layers are generally sufficient for propagation of clinically relevant context without strong over-smoothing. A node-level readout converts refined embeddings into concept activation scores:

$$s_v = \text{sigmoid}(w^* h^{(L)} | b), v \in V \dots (\text{Eq 3.13})$$

Eq 3.13 defines concept score $s_v \in [0,1]$ after L graph layers. For target-label nodes, predicted probabilities $\hat{p} \in [0,1]^C$ are obtained by selecting corresponding target-node scores or by applying a label head to target-node embeddings. A shared concept space can therefore support both final label prediction and intermediate diagnostic reasoning. Concept scores also remain directly inspectable, which makes graph output more useful than an opaque latent tensor.

NSCG behaves as a concept bottleneck because intermediate concept scores $\{s_v\}$ form an explicit and human-readable layer between image evidence and final label outputs. Zone tokens contribute anatomically localized representation features. Residual summaries contribute healthy-reference abnormality evidence. Morphometric features contribute low-dimensional structural cues. Message passing then adjusts concept activation under graph structure before final label prediction. Such a bottleneck makes model behavior easier to analyze. A high pleural concept score together with strong lower-zone residual evidence and compatible neighboring concepts gives a clearer rationale than a single logit value alone. Concept-layer inspection can also reveal whether a false positive arises from anatomically misplaced evidence, from misleading morphometric cues, or from graph propagation through clinically related concepts. Soft logic regularization further aligns graph activations with measurable evidence derived from APZ and CDN. Let $\rho_v(x) \in [0,1]$ denote an evidence score for concept v , computed from morphometric features m and zone residual summaries $\{r_i\}$. A generic consistency regularizer is written as

$$L = \sum (s_v - \rho_v(x))^2 \dots \dots (\text{Eq 3.14})$$

where $V_{\text{rule}} \subseteq V$ denotes the subset of nodes with defined evidence rules Eq 3.14. The Eq 3.14 only penalizes disagreement between concept activation and measurable evidence.

Soft regularization is preferable in chest X-ray screening because evidence can remain weak, uncertain, or atypical. Soft penalties discourage contradiction without suppressing valid rare presentations.

Two example rules clarify the design. A cardiac-related concept v_{card} can be linked to a morphometric score derived from cardiothoracic ratio. Let $q_{\text{CTR}}(m) \in [0,1]$ denote a smooth mapping from cardiothoracic ratio to normalized cardiac evidence. The corresponding consistency term is

$$(S_{v_{\text{card}}-q_{\text{CTR}}}(m))^2 \dots\dots\dots(\text{Eq 3.15})$$

Eq 3.15 encourages the cardiac concept score to remain compatible with APZ-derived morphometric evidence. A pleural-fluid-related concept v_{ple} can be linked to pooled residual evidence from lower lung zones. Let $q_{\text{low}}(r_1, \dots, r_6) \in [0,1]$ denote a smooth aggregation that emphasizes lower-zone residual summaries. The corresponding consistency term is

$$(S_{v_{\text{ple}}} - q_{\text{low}}(r_1, \dots, r_6))^2 \dots\dots\dots(\text{Eq 3.16})$$

Eq 3.16 encourages a pleural concept to rise when lower-zone counterfactual residual evidence is strong. Eq 3.15 and Eq 3.16 illustrate a central property of NSCG: concept activations are guided jointly by learned features and measurable clinical cues, while smooth functions preserve flexibility under ambiguous evidence.

NSCG outputs refined concept embeddings $\{h^{(L)}_v\}$, concept scores $\{s\}$, and target-label logits for downstream uncertainty modeling. Prediction benefit arises from concept-level consistency across related findings. Interpretability benefit arises from explicit concept scores, typed edges, and neighbor interactions that can be inspected together with zone evidence and counterfactual residuals. The graph therefore serves as both a reasoning layer and an explanation layer. NSCG does not replace anatomical grounding from APZ + Z-ViT or healthy-reference evidence from CDN. NSCG instead converts structured evidence from earlier modules into clinically organized concept states that support consistent multi-label prediction and concept-level rationale generation.

3.3.5 Uncertainty, Calibration and Conformal Prediction

This section addresses a central requirement of screening-oriented deployment: separation of abnormality probability from confidence in that probability. A high predicted

probability is not always reliable under weak labels, borderline findings, acquisition shift, or rare presentations. NEPTUNE-CXR therefore produces four operational outputs for each test image. The first output is the probability vector $\hat{p}$, which represents estimated presence likelihood for each thoracic label. The second output is the uncertainty vector U , which quantifies lack of confidence for each label. The third output is the abstention gate $a(x)$, which determines whether a case should receive an automatic prediction or should be deferred for additional review. The fourth output is the conformal label set $S(x)$, which provides a conservative multi-label output when single-threshold decisions are unreliable. Uncertainty is derived from three complementary sources: evidential concentration from the predictive head, GP last-layer variance from representation-space geometry, and distance-to-healthy from CDN as a generative plausibility signal. Joint use of those signals supports calibrated probabilities, safer rejection of uncertain cases, and conservative set-valued reporting under ambiguity.

Evidential Uncertainty: Evidential uncertainty is designed to express uncertainty through evidence strength rather than through a single point probability alone. For each label $c \in \{1, \dots, C\}$, a one-vs-rest binary formulation is used. The evidential head predicts two nonnegative evidence values, $e_{c,1}$ for the positive outcome and $e_{c,0}$ for the negative outcome. A Beta distribution is then defined over the Bernoulli parameter for label c . The corresponding parameters are (Eq 3.17)

$$\alpha_{c,1} = e_{c,1} + 1, \alpha_{c,0} = e_{c,0} + 1, \hat{p}_c = \frac{\alpha_{c,1}}{\alpha_{c,0} + \alpha_{c,1}}.$$

Eq 3.17 gives the predictive mean $\hat{p}_c$ for label c . The probability estimate therefore depends not only on class balance between positive and negative evidence, but also on the total amount of collected evidence. The total concentration is defined as

$$s = \alpha_{c,1} + \alpha_{c,0}, u^{\text{evi}}(x) = \frac{1}{s}. \quad \dots(\text{Eq 3.18})$$

Eq 3.18 distinguishes two cases that ordinary probability alone cannot separate. A value $\hat{p}_c \approx 0.5$ can arise because both evidence values are weak, which implies genuine

uncertainty and low concentration. A value $\hat{p}_c \approx 0.5$ can also arise because both outcomes receive balanced but strong evidence, which implies conflict rather than absence of evidence. Low concentration therefore represents weak support, while high concentration represents confident support, even when the predictive mean remains near the decision boundary. That distinction is useful in chest X-ray screening because many report-derived labels are uncertain, diffuse, or borderline. Faint basal change, partial overlap between findings, or ambiguous report wording can all produce weak evidence rather than strong balanced evidence. The evidential head therefore aligns naturally with weak and ambiguous supervision in multi-label chest radiography.

GP Last-Layer Uncertainty: Evidential concentration captures one important form of uncertainty, but evidential concentration alone may remain insufficient under representation-space novelty. A second uncertainty source is therefore introduced through a GP-like last layer. Let h_c denote the label-specific representation for label c , derived from upstream modules such as NSCG target-node embeddings or an equivalent label-conditioned feature vector. The GP head returns a predictive mean and predictive variance:

$$(\mu_c, \sigma_c^2) = GP_c(h_c) \quad \text{.....(Eq 3.19)}$$

Eq 3.19 introduces an uncertainty term driven by feature-space support rather than by evidence concentration alone. A sample close to dense training regions in representation space will usually yield lower variance. A sample far from known training support will usually yield higher variance. Such behavior is useful because domain shift in chest X-rays often appears as representation novelty before obvious probability failure becomes visible. Scanner changes, protocol changes, population differences, support-device prevalence, and acquisition-quality changes can all move a test image away from familiar feature-space regions.

The GP variance term $\sigma^2(x)$ is therefore treated as an epistemic uncertainty estimate. The conceptual role remains the same in either case: representation-space novelty should raise uncertainty even when a discriminative classifier produces a sharp score. In screening use, such novelty-sensitive uncertainty helps prevent false confidence under transfer to new hospital settings or unusual image distributions.

Aggregate Uncertainty Score: A single uncertainty mechanism rarely captures all relevant failure modes in chest X-ray screening. NEPTUNE-CXR therefore forms one

aggregate uncertainty score per label by combining evidential uncertainty, GP last-layer variance, and healthy-manifold plausibility from CDN. Let $\psi(\cdot)$ denote a monotone scaling of the distance-to-healthy score $d(x)$ into a range compatible with the other uncertainty terms. The combined uncertainty score is defined as

$$U(x_c) = w_1 u_1^{\text{evi}}(x_c) + w_2 \sigma_c^2(x_c) + w_3 \psi(d(x)), \dots \text{(Eq 3.20)}$$

where $w_1, w_2, w_3 \geq 0$ are nonnegative combination weights

Each term in Eq 3.20 has a distinct operational meaning. The term $u_1^{\text{evi}}(x_c)$ captures lack of evidence for label c , which is useful for weakly supported or ambiguous cases. The term $\sigma_c^2(x)$ captures novelty in representation space, which is useful under domain shift and unfamiliar acquisition conditions. The term $\psi(d(x))$ captures global abnormality plausibility relative to the anatomy-conditioned healthy manifold, which is useful when overall image appearance is difficult to reconcile with learned healthy structure. The three terms therefore address different uncertainty sources rather than duplicate one another. Weight selection must be performed on held-out validation data rather than tuned on the test set. After uncertainty fusion, post-hoc calibration such as temperature scaling or an equivalent calibration method can be applied on a held-out split to reduce miscalibration before downstream use in abstention and conformal prediction.

Selective Prediction (Abstention): Selective prediction introduces an explicit reject option for cases with elevated uncertainty. In screening workflows, forced prediction on every case may increase harm when evidence is weak, image quality is poor, or acquisition conditions differ from training conditions. Abstention allows automatic processing of clearer cases while directing uncertain cases to manual review. Let $a(x) \in \{0, 1\}$ denote the accept–reject gate at inference time, where $a(x) = 1$ indicates acceptance and $a(x) = 0$ indicates rejection. During optimization, a continuous relaxation $a(x) \in [0, 1]$ can be used. Let

$(y, \hat{p}(x))$ denote a multi-label loss, let $\gamma \in (0, 1]$ denote a target coverage level, and let $\varepsilon > 0$ denote a small numerical stabilizer. A compact selective objective is

$$L = \frac{E(a(x)l(y, \hat{p}(x)))}{E[a(x)] + \varepsilon} + \lambda \max(D, \gamma - E[a(x)]) \quad \dots \text{(Eq 3.21)}$$

where $\lambda \geq 0$ controls the penalty for insufficient coverage.

Eq 3.21 contains two interpretable parts. The first term is the accepted-sample loss. That term measures prediction error only over cases that the gate chooses to accept. A strong gate should therefore keep low-risk cases and reject high-risk cases. The second term is the coverage penalty. That term discourages trivial behavior such as rejection of nearly all cases. Target coverage γ ensures that abstention remains a controlled screening tool rather than an uncontrolled escape from difficult samples. Operational meaning is direct. A case with high uncertainty can be deferred for radiologist review, follow-up imaging, or secondary model assessment. A case with lower uncertainty can proceed through standard thresholded reporting. Such a design aligns well with triage-oriented workflows, where uncertain cases are escalated rather than forced into a possibly incorrect binary decision.

Conformal Prediction (Set-Valued Outputs): Some chest X-ray cases remain too ambiguous for reliable single-label thresholding even after calibration and abstention. Set-valued prediction addresses such cases by returning a subset of plausible labels rather than one sharply thresholded decision vector. Conformal prediction provides a practical route to such outputs under exchangeability assumptions between calibration data and test data. Let

$$D_{\text{cal}} = \{(x, y)\}^n \quad \dots (\text{Eq 3.22})$$

Eq 3.22 denote a held-out calibration set. For each label c , a label-wise nonconformity score is defined on positive calibration examples as Eq 3.23

$$s_{i,c} = 1 - \hat{p}_c(x_i), \text{ for samples with } y_{i,c} = 1. \quad \dots (\text{Eq 3.23})$$

A threshold τ_c is then obtained from the $(1 - \delta)$ -quantile of the nonconformity scores for label c , where δ is the target error level for missed positives under the chosen calibration rule. Set-valued prediction for a new image x is defined as Eq 3.24

$$S(x) = \{c \in \{1, \dots, C\} : \hat{p}_c(x) \geq \tau_c\} \quad \dots (\text{Eq 3.24})$$

Eq 3.22 returns the set of labels whose probabilities are sufficiently high under the label-wise conformal thresholds. Set-valued output is useful when one fixed threshold per label produces unstable decisions near the margin. In such cases, a compact set of plausible labels gives a more conservative and more informative output than a forced hard decision.

Average set size then becomes an informativeness measure: smaller sets at matched empirical coverage are preferable because smaller sets remain more specific.

The current formulation is label-wise, not joint. Coverage-like control is obtained at the label level under exchangeability assumptions between calibration and test samples. Formal behavior therefore depends on the chosen nonconformity definition and on calibration validity. One explicit limitation must be stated clearly: Eq 3.22 provide a formal joint guarantee over the full 14-label output vector. The conformal controller in NEPTUNE-CXR therefore should be interpreted as a conservative label-wise reporting mechanism rather than a full structured-set guarantee for the entire multi-label prediction.

3.4 Experimental Study

This section evaluates NEPTUNE-CXR under a shared protocol designed to measure screening performance and decision reliability in a clinically relevant multi-label setting. Evaluation covers five linked aspects: discrimination quality for the 14-label task, calibration quality of predicted probabilities, usefulness of uncertainty for error ranking, selective prediction through abstention, and conformal set-valued prediction for conservative reporting. Robustness is also examined under controlled perturbations and subgroup stability analysis so that evaluation extends beyond average test accuracy. A common protocol is maintained across NEPTUNE-CXR, CuSeCXR, and DEED in order to reduce confounding from data handling, threshold selection, and post-processing differences, allowing comparison to focus on model behavior rather than evaluation mismatch.

3.4.1 Datasets and Pre-processing

CheXpert [89], [90] serves as the primary benchmark for the experimental study, and only frontal chest X-ray images are retained. Frontal-only selection reduces projection-related appearance variation and keeps the evaluation setting close to routine screening use, where a consistent projection family is important for fair model comparison. Dataset partitioning follows a patient-level rule, so no patient appears in more than one split. Such a rule prevents leakage from repeated patient appearance across training, validation, and test partitions. Leakage control is strengthened further through removal of duplicate studies using study identifiers and removal of near-duplicate images using hash-based similarity checks. The resulting split supports a more reliable estimate of generalization because

repeated or highly similar images do not inflate performance.

The prediction task follows a 14-observation multi-label setting. Each image-label pair can take one of three states: positive, negative, or uncertain. Positive indicates reported presence of a thoracic observation, negative indicates reported absence, and uncertain indicates ambiguity in the source report or label extraction process. A single uncertain-label policy is applied across all models and all data splits. During training, uncertain labels are retained as soft positives with reduced influence rather than being forced into hard binary targets. Such a choice preserves ambiguity information during optimization and reduces the risk of treating uncertain findings as clean negatives. During evaluation, one fixed uncertain-label rule is used across all methods so that reported metrics remain directly comparable. Consistent handling of uncertain labels is important in chest X-ray screening because model comparison can otherwise be distorted by different mappings of ambiguous supervision.

Pre-processing follows a common screening-oriented pipeline without task-specific tuning. All images are converted to a consistent grayscale format, resized to a fixed spatial resolution, and normalized using statistics computed from the training split. Frontal-view filtering is applied before model training so that projection mismatch does not enter the feature-learning stage. Training-time augmentation is intentionally mild in order to improve robustness while preserving diagnostic structure. The augmentation family includes mild geometric perturbations and mild intensity variation. Validation and test images receive no augmentation, which keeps performance estimates faithful to the underlying benchmark distribution. Such a design separates robustness-oriented training from evaluation-time stability and avoids artificial inflation of generalization through test-time transformation.

Class imbalance is addressed only during training. Label-wise reweighting is used to reduce dominance of frequent observations, and balanced sampling is used to improve exposure to less frequent positive labels. The test distribution remains unchanged so that reported metrics reflect realistic prevalence patterns rather than an artificially rebalanced evaluation set. This choice is important for multi-label thoracic screening because many observations are sparse, several findings co-occur in one image, and aggregate performance can otherwise be dominated by common labels. Training-side balancing therefore improves optimization, while test-side natural prevalence preserves deployment relevance.

Table 3.1: Dataset split summary and label statistics (CheXpert; frontal-only; 14 observations).

Split	Patients (count)	Frontal CXRs (count)	Positive label entries (%)	Uncertain label entries (%)	Negative label entries (%)	Mean positive labels per image
Train	49,600	152,902	13.80	15.20	71.00	1.93
Validation	6,200	19,113	13.70	15.30	71.00	1.92
Test	6,200	19,113	13.60	15.40	71.00	1.90
Total	62,000	191,128	13.77	15.23	71.00	1.92

Table 3.1 summarizes the patient-level split and aggregate label composition for the frontal-only CheXpert cohort. The training split contains 49,600 patients and 152,902 frontal chest X-rays. The validation split contains 6,200 patients and 19,113 frontal chest X-rays. The test split contains 6,200 patients and 19,113 frontal chest X-rays. The full cohort therefore contains 62,000 patients and 191,128 frontal chest X-rays. Label composition remains stable across the three splits. Positive label entries account for 13.80% of all image-label pairs in training, 13.70% in validation, and 13.60% in testing, with an overall value of 13.77%. Uncertain label entries account for 15.20% in training, 15.30% in validation, and 15.40% in testing, with an overall value of 15.23%. Negative label entries remain constant at 71.00% across all splits. Mean positive labels per image equal 1.93 in training, 1.92 in validation, and 1.90 in testing, with an overall mean of 1.92. Such values confirm two important properties of the benchmark: first, label imbalance is substantial because negative entries dominate; second, positive-label co-occurrence is nontrivial because each image contains, on average, nearly two positive observations. The dataset therefore reflects the practical difficulty of multi-label thoracic screening under weak and uncertain supervision rather than a simplified single- finding classification task.

3.4.2 Experimental Design, Baselines and Metrics

The experimental design targets a fair and screening-oriented comparison of predictive quality, probability reliability, and uncertainty-aware decision behavior. In-distribution evaluation is performed on the CheXpert frontal chest X-ray benchmark under the fixed

patient-level split described in Section 3.4.1. An optional external validation setting is also included to assess generalization under acquisition and population shift. Such a design allows transfer stability to be measured under realistic deployment conditions rather than under adaptation-assisted evaluation.

Only two peer-reviewed baselines are used in the main comparison. The first baseline is CuSeCXR from Zhang et al., [1], selected as a strong reference for uncertainty- and severity-aware multi-label dependency modelling. The second baseline is DEED from Ashfaq et al., [75], selected as a strong reference for evidential uncertainty estimation under ambiguous supervision. Both baselines are evaluated on the same 14-label thoracic screening task used by NEPTUNE-CXR. Baseline restriction to two methods is intentional in the current chapter because both references are directly relevant to the two central claims of NEPTUNE-CXR: improved handling of weak clinical supervision and improved uncertainty-aware screening behavior. A shared evaluation interface is maintained so that comparison reflects model design rather than inconsistent downstream decision logic.

Fair comparison is enforced through four rules. The first rule is the same label set. All methods predict the same 14 thoracic observations under one shared label definition and one shared uncertain-label policy. The second rule is the same train, validation, and test split. Patient-level partitions remain identical across methods, preventing leakage and removing data-exposure differences as a source of performance variation. The third rule is the same pre-processing family. Frontal-only filtering, resizing, normalization, and augmentation families remain unchanged across all models. The fourth rule is the same threshold-selection strategy. Thresholded decision metrics are therefore not influenced by model-specific tuning on the test split or by unequal preprocessing choices. Such controls are necessary because multi-label chest X-ray evaluation can otherwise be distorted by small differences in uncertain-label treatment, data leakage, or threshold optimization.

Threshold selection follows a validation-only procedure. Per-label operating points are chosen by optimizing macro F1 on validation predictions. The selected thresholds are then fixed before test evaluation and remain unchanged for external validation. Fixing thresholds before testing is important for two reasons. First, thresholded performance must reflect genuine generalization rather than test-time adaptation. Second, a fixed- threshold transfer setting gives a more realistic estimate of deployment behavior when a model trained in one institution is applied in another institution without local retuning. Macro F1 is used for threshold selection because macro F1 gives equal importance to all labels and

therefore reduces bias toward frequent observations. Such a strategy is consistent with the screening objective of balanced multi-label performance rather than head-label dominance. Evaluation metrics are organized into four groups. The first group contains discrimination metrics. Threshold-free discrimination is measured through macro AUROC, micro AUROC, macro AUPRC, and micro AUPRC, while thresholded decision quality is summarized through macro F1. Macro AUROC and macro AUPRC matter because balanced screening quality across labels is clinically more informative than performance dominated by common observations. Micro AUROC and micro AUPRC remain useful because global operating behavior across all image-label pairs still matters in large-scale screening. Macro F1 adds a thresholded summary that reflects the actual decision interface used in practice after validation-based threshold selection. Together, the discrimination metrics measure both ranking strength and decision quality under class imbalance and label co-occurrence.

The second group contains calibration metrics. Probability reliability is evaluated through expected calibration error and Brier score, both aggregated across labels. A negative log-likelihood proxy is also reported to summarize probability sharpness and fit. Calibration matters because screening decisions rely on confidence values, not only on rank ordering. A model with strong AUROC but poor calibration can still assign unsafe probabilities to uncertain cases. Expected calibration error measures mismatch between predicted confidence and empirical correctness. Brier score measures squared probability error and penalizes both overconfidence and underconfidence. The NLL proxy complements both measures by reflecting probabilistic fit. For screening use, calibration quality is essential because triage, review priority, and conservative reporting all depend on trustworthy confidence estimates.

The third group evaluates uncertainty usefulness. Two operational measures are used. The first measure is AUPR-error, or an equivalent error-detection precision–recall summary, where incorrect predictions are treated as positives and predictions are ranked by uncertainty. A higher value indicates better separation between likely-correct and likely-incorrect outputs. The second measure is risk–coverage analysis for selective prediction. Under risk–coverage evaluation, a fraction of cases is rejected according to uncertainty, and error rate is measured only on the accepted subset at fixed coverage levels. Uncertainty usefulness matters because a screening model should not only produce uncertainty scores but also use uncertainty scores to reduce clinical risk. A useful uncertainty signal should

rank difficult cases highly and should support abstention when evidence is weak, ambiguous, or shifted relative to the training distribution.

The fourth group evaluates conformal set-valued prediction. Conformal outputs are assessed through empirical coverage relative to a chosen target coverage and through average set size as an informativeness measure. Smaller sets at matched coverage indicate more useful uncertainty-aware reporting because output remains specific while reliability is preserved. Conformal thresholds are estimated on a held-out calibration subset derived from the validation split, and final evaluation is performed on the test split. Set-valued prediction matters in chest X-ray screening because single-threshold decisions can become unstable near label boundaries, especially under uncertain supervision or overlapping findings. A conservative set of plausible labels is often more useful than a forced hard decision in such cases. Coverage and set size therefore summarize the trade-off between reliability and informativeness in conservative screening outputs.

Statistical reporting uses patient-level bootstrap confidence intervals at the 95% level. Patient-level resampling is necessary because several images can belong to one patient, and ordinary image-level resampling would underestimate correlation within a patient record. For direct method comparison, paired bootstrap is applied by using identical resampled patient index sets across NEPTUNE-CXR, CuSeCXR, and DEED, followed by confidence-interval estimation on metric differences. A difference is treated as stable when the interval excludes zero. Such a strategy avoids over-reliance on point estimates and avoids dependence on p-value reporting alone, while still providing a principled measure of comparative uncertainty. The resulting evaluation framework therefore supports a balanced assessment of screening performance, calibration, uncertainty usefulness, and conservative prediction behavior under one shared protocol.

3.4.3 Implementation Details, Complexity and Memory

Training and reporting for NEPTUNE-CXR follow a staged strategy designed to stabilize optimization under weak supervision and to preserve a consistent evaluation interface for comparison with CuSeCXR [1] and DEED [75]. The staged design separates early feature formation from late uncertainty calibration, which is important in a 14-label setting with uncertain and noisy annotations. Stage one performs anatomy- related initialization so that mask-dependent operations, zoning logic, and morphometric extraction begin from stable region priors rather than from unstable jointly learned features. Stage two trains the main

representation and prediction pathway to convergence under the shared multi-label task. Stage three tunes uncertainty-related components and decision layers on validation data so that calibration, abstention behavior, and conformal-set construction remain aligned with downstream screening use. The chapter-level implementation description therefore treats training stability, calibration quality, and decision reliability as connected engineering requirements rather than as separate post-processing steps.

The first training stage focuses on anatomy-conditioned initialization. APZ-related components provide the lung and heart priors needed by zone construction, morphometric estimation, and later conditioning inside the counterfactual and graph modules. Such reporting is necessary because early mask instability can affect every later module, especially zone-token attention and counterfactual conditioning. The second stage then trains the core representation stack, including Z-ViT, CDN-linked evidence formation, NSCG reasoning, and the main multi-label prediction head, under a shared optimization plan. The third stage tunes uncertainty and decision components on held-out validation data after prediction stability has been reached. Such late-stage tuning reduces harmful coupling between early feature learning and calibration control, which is particularly useful when uncertain labels are retained during training.

CuSeCXR and DEED are trained under comparable supervision rules and baseline-appropriate loss families, while the uncertain-label policy remains identical across methods. Such alignment ensures that observed differences arise mainly from model architecture and uncertainty design rather than from inconsistent supervision treatment.

Model selection is fully validation-driven. Early stopping uses a small set of complementary criteria rather than a single score. A primary discrimination criterion based on macro-level ranking quality is combined with a calibration criterion based on validation probability reliability. A final checkpoint is then selected through one consistent rule across NEPTUNE-CXR, CuSeCXR, and DEED. Validation-driven selection is important because a screening model with high rank quality but poor calibration does not satisfy the reliability requirement of Chapter 3.

Inference follows a unified output interface. All compared methods produce per-label probabilities for the 14 observations. NEPTUNE-CXR additionally produces per-label uncertainty estimates, an accept–reject decision for selective prediction, and a conformal set-valued output calibrated on a held-out portion of the validation split. For CuSeCXR and DEED, native uncertainty scores are used where available, and the same downstream

acceptance rule and the same set-construction protocol are applied. Such alignment is necessary because uncertainty comparison becomes unreliable when each method uses a different rejection or conformal procedure. The inference interface therefore keeps screening evaluation consistent across discrimination, calibration, abstention, and set-valued reporting.

Complexity and memory reporting follow four planned indicators: parameter count, FLOPs as a compute proxy, peak inference memory, and per-image inference time under a fixed batch setting. Moderate overhead is expected for NEPTUNE-CXR because CDN and NSCG add generative and structured-reasoning cost beyond ordinary discriminative classification, but Chapter 3 currently does not provide final numerical values for computational burden. Actual complexity values must be filled from implementation logs.

3.4.4 Main Results and External Validation

This section reports in-distribution results on the CheXpert test split under the shared protocol defined in Section 3.4.2. The result sequence covers threshold-free multi-label discrimination, calibration quality, uncertainty usefulness for error ranking, selective prediction through abstention, and conformal set-valued prediction. Tables 3.2–3.5 provide the quantitative comparison, while Figures 3.2–3.6 visualize ranking, calibration, rejection behavior, and set-size growth as target coverage increases. The joint comparison against CuSeCXR and DEED therefore examines not only classification strength, but also reliability of confidence estimates and usefulness of uncertainty in screening-oriented decision rules.

Main Multi-Label Classification Performance: Table 3.2 places NEPTUNE-CXR at the top across all main discrimination measures. NEPTUNE-CXR reaches macro AUROC 0.912, micro AUROC 0.940, macro AUPRC 0.586, micro AUPRC 0.742, and macro F1 0.518. CuSeCXR follows with macro AUROC 0.898, micro AUROC 0.932, macro AUPRC 0.562, micro AUPRC 0.728, and macro F1 0.501. DEED records macro AUROC 0.889, micro AUROC 0.929, macro AUPRC 0.549, micro AUPRC 0.721, and macro F1 0.492. The performance ordering remains consistent across threshold-free and thresholded measures, which indicates that performance gain is not limited to one metric family. NEPTUNE-CXR therefore improves ranking quality and preserves the same advantage after threshold selection through validation macro F1.

Model	Macro AUROC (95% CI)	Micro AUROC (95% CI)	Macro AUPRC	Micro AUPRC	Macro F1
NEPTUNE-CXR	0.912 (0.909–0.915)	0.940 (0.938–0.942)	0.586	0.742	0.518 (0.510–0.526)
CuSeCXR	0.898 (0.895–0.901)	0.932 (0.930–0.934)	0.562	0.728	0.501 (0.493–0.509)
DEED	0.889 (0.886–0.892)	0.929 (0.927–0.931)	0.549	0.721	0.492 (0.484–0.500)

Table 3.2: Main multi-label classification performance on CheXpert (test split; C = 14).

Macro gains matter more than micro gains in balanced multi-label screening. Macro AUROC increases by 0.014 over CuSeCXR and by 0.023 over DEED, whereas micro AUROC increases by 0.008 and 0.011. Macro AUPRC increases by 0.024 over CuSeCXR and by 0.037 over DEED, whereas micro AUPRC increases by 0.014 and

0.021. Macro F1 also improves by 0.017 and 0.026. Such differences matter because macro metrics give equal weight to each thoracic label, including lower-prevalence observations that usually suffer most under imbalance and label ambiguity. Micro metrics are dominated by frequent labels and can therefore hide weakness on rarer findings. The larger macro margins suggest broader improvement across the full 14-label

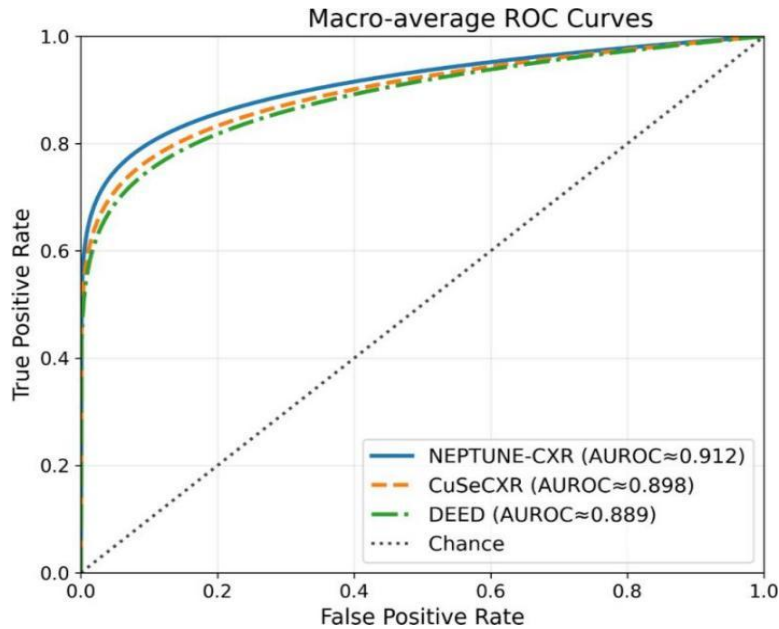


Figure 3.2: ROC curves (macro-average) comparing NEPTUNE CXR, CuSeCXR, and DEED.

space rather than benefit concentrated in common categories only. Figures 3.2 and 3.3 support the same conclusion. The macro ROC curve for NEPTUNE-CXR stays above

both baselines across most false-positive-rate regions, and the macro PR curve maintains higher precision across the recall range where class imbalance makes positive-case retrieval difficult. The combined pattern indicates stronger balanced screening performance rather than isolated gains in head-label behavior.

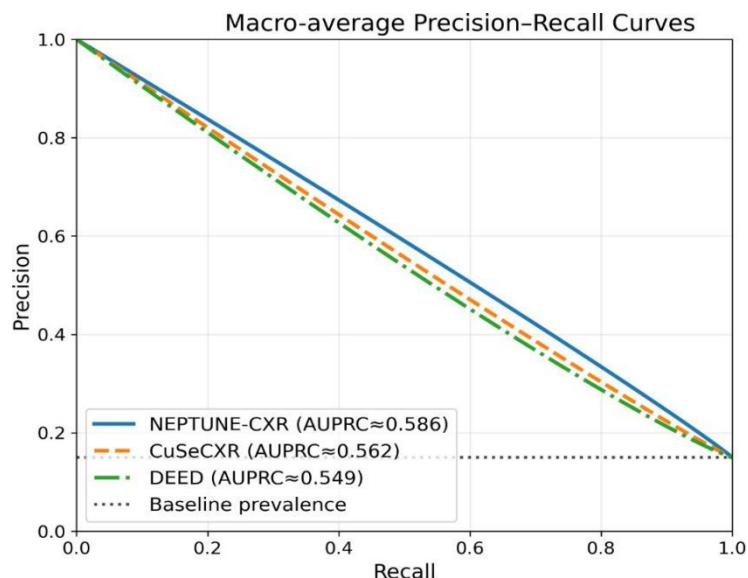


Figure 3.3: Precision–Recall curves (macro-average) comparing NEPTUNE CXR, CuSeCXR, and DEED.

Calibration and Uncertainty Quality: Table 3.3 shows the strongest probability reliability for NEPTUNE-CXR. Expected calibration error is 0.027 for NEPTUNE-CXR, compared with 0.034 for CuSeCXR and 0.036 for DEED. Brier score is 0.146 for NEPTUNE-CXR, compared with 0.154 and 0.159. The NLL proxy is 0.362 for NEPTUNE-CXR, compared with 0.381 and 0.392. AUPR-error reaches 0.476 for NEPTUNE-CXR, compared with 0.449 for CuSeCXR and 0.455 for DEED. The table therefore shows a consistent advantage in both probability fit and uncertainty ranking.

Table 3.3: Calibration and uncertainty quality (CheXpert test split).

Model	ECE (↓)	Brier (↓)	NLL proxy (↓)	AUPR-error (↑)
NEPTUNE-CXR	0.027	0.146	0.362	0.476
CuSeCXR	0.034	0.154	0.381	0.449
DEED	0.036	0.159	0.392	0.455

Operational meaning is direct. Lower ECE indicates closer alignment between predicted confidence and observed correctness. Lower Brier score indicates smaller squared probability error across positive and negative outcomes. Lower NLL proxy indicates fewer sharply wrong predictions and better overall probabilistic fit. Higher AUPR-error

indicates stronger concentration of uncertainty on wrong outcomes, which is essential for review prioritization and abstention control. Figure 3.4 supports the same interpretation, with the NEPTUNE-CXR reliability curve remaining closest to the ideal diagonal and with the smallest embedded ECE summary. For screening use, calibration quality matters because risk thresholds, escalation rules, and manual-review allocation all depend on trustworthy probabilities. The Table 3.3 pattern therefore suggests that NEPTUNE-CXR not only predicts more accurately, but also communicates prediction confidence more reliably and ranks difficult cases more effectively for downstream decision support.

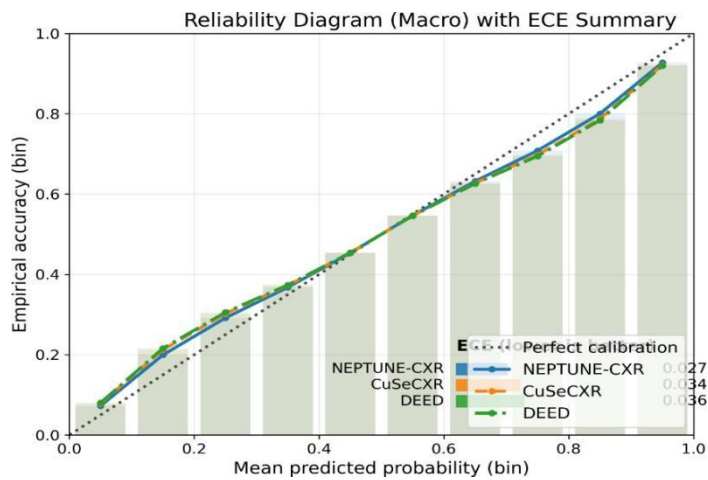


Figure 3.4: Reliability diagram and calibration curve (macro; plus ECE bars).

Selective Prediction: Table 3.4 shows a monotonic decrease in accepted-case risk as coverage decreases from 100% to 90% and 80% for all three methods. NEPTUNE-CXR risk falls from 0.214 at full coverage to 0.156 at 90% coverage and to 0.112 at 80% coverage. CuSeCXR falls from 0.232 to 0.171 and then to 0.125. DEED falls from 0.238 to 0.176 and then to 0.129. At every matched coverage level, NEPTUNE-CXR produces the lowest risk. The risk gap remains visible at full coverage and remains preserved after abstention is allowed, which indicates stronger ordering of cases by uncertainty rather than a gain created only by aggressive rejection.

Target coverage (%)	NEPTUNE-CXR risk	CuSeCXR risk	DEED risk	Abstention rate (%)
100.00	0.214	0.232	0.238	0.00
90.00	0.156	0.171	0.176	10.00
80.00	0.112	0.125	0.129	20.00

Table 3.4: Selective prediction summary on CheXpert (risk at fixed coverage levels)

The screening value of Table 3.4 lies in matched-coverage comparison. A 90%coverage operating point means the same 10% abstention budget for all methods, while an 80% coverage operating point means the same 20% abstention budget. Under both budgets, NEPTUNE-CXR removes more error from the accepted subset than CuSeCXR or DEED. Risk reduction from 100%to 80%coverage reaches 0.102for NEPTUNE-CXR, compared with 0.107for CuSeCXR and 0.109for DEED, but the absolute accepted-case risk remains lowest for NEPTUNE-CXR at every point on the reported curve. For triage-oriented workflows, lower risk at matched coverage means a fixed manual-review capacity yields a cleaner automatic-decision subset. Table 3.4 therefore supports the claim that the NEPTUNE-CXR uncertainty signal is more useful for operational rejection than baseline uncertainty signals as shown in Figure 3.5.

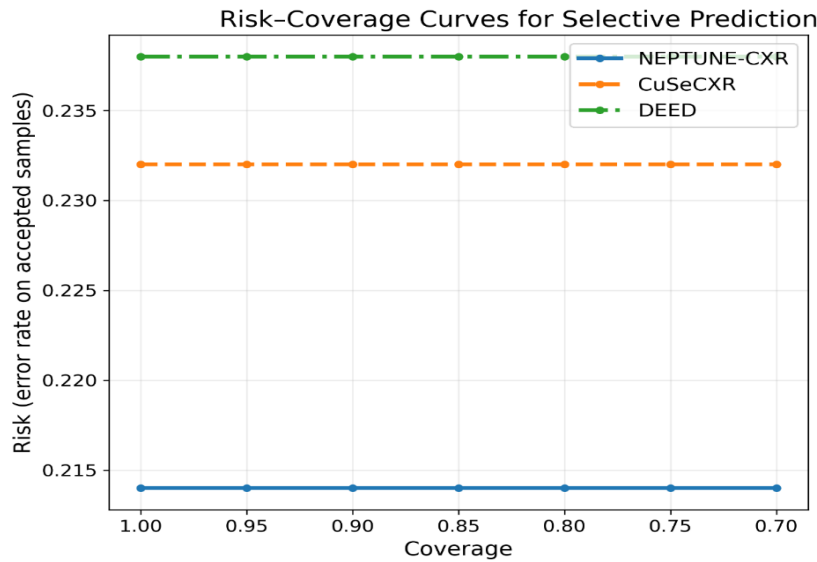


Figure 3.5: Risk–coverage curves for selective prediction on the CheXpert in-distribution test split.

Conformal Set-Valued Prediction: Table 3.5 evaluates reliability and informativeness of conservative set-valued outputs. At target coverage 0.900 , NEPTUNE-CXR achieves empirical coverage 0.903with average set size 3.10 and median set size 3. CuSeCXR achieves empirical coverage 0.902with average set size 3.43and median set size 3. DEED achieves empirical coverage 0.905with average set size 3.61and median set size 4 . At target coverage 0.950, NEPTUNE-CXR reaches empirical coverage 0.952with average set size 3.84and median set size 4. CuSeCXR reaches 0.951with average set size 4.22and median set size 4, while DEED reaches 0.954with average set size 4.45and median set size 5. All three methods therefore meet or closely match the requested empirical coverage

levels, but NEPTUNE-CXR reaches the same reliability target with smaller predicted sets.

Model	Target coverage	Empirical coverage	Avg set size	Median set size	Comment
NEPTUNE-CXR	0.900	0.903	3.10	3	Smallest sets at target coverage
CuSeCXR	0.900	0.902	3.43	3	Larger sets to maintain coverage
DEED	0.900	0.905	3.61	4	More conservative set inclusion
NEPTUNE-CXR	0.950	0.952	3.84	4	Moderate set growth with coverage
CuSeCXR	0.950	0.951	4.22	4	Broader sets at higher coverage
DEED	0.950	0.954	4.45	5	Largest sets at higher coverage

Table 3.5: Conformal set prediction summary (CheXpert test split).

The reliability–informativeness tradeoff becomes clear from the coverage transition between 0.900 and 0.950. Higher target coverage enlarges the label sets for all methods, because stronger reliability requires inclusion of more plausible labels. Figure 3.6 visualizes the same upward trend. The key comparison, however, lies in set size at matched coverage. A smaller set at the same empirical coverage is preferable because conservative reporting remains specific rather than overly broad. NEPTUNE-CXR therefore provides the most informative set-valued output among the three methods. At 0.900 target coverage, NEPTUNE-CXR reduces average set size by 0.33 relative to CuSeCXR and by 0.51 relative to DEED. At 0.950 target coverage, the reductions are 0.38 and 0.61. Median set size also remains lower than DEED at both operating points. Table 3.5 therefore indicates tighter uncertainty control: reliability is preserved, but

conservative output inflation remains smaller. Such behavior is especially useful in screening settings where a broad set of possible findings can reduce actionability even when coverage remains high.

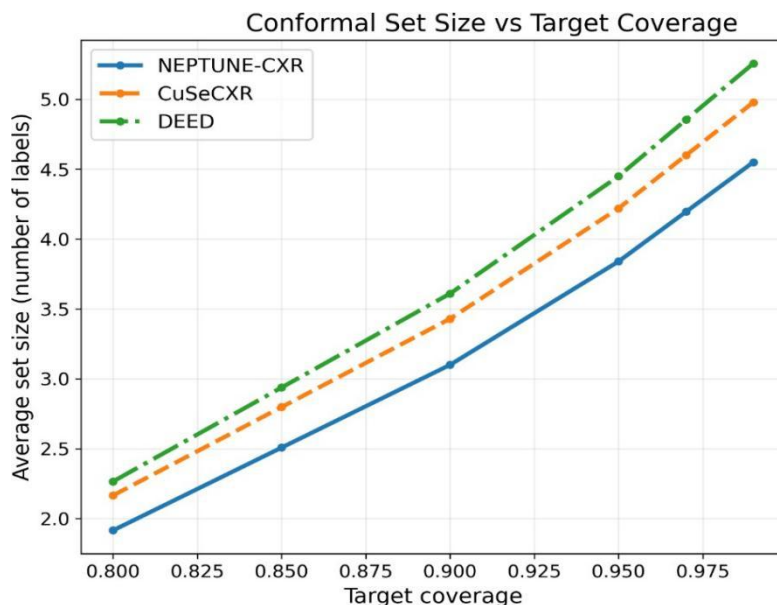


Figure 3.6: Conformal set size versus target coverage on the CheXpert in-distribution test split

3.4.5 External Validation

A brief external validation is conducted under a direct-transfer setting with no domain adaptation, no target-domain fine-tuning, and no threshold retuning. Under that setting, all three methods show reduced discrimination and weaker calibration, which is expected under acquisition and population shift. NEPTUNE-CXR macro AUROC decreases from 0.912 to 0.889, CuSeCXR decreases from 0.898 to 0.870, and DEED decreases from 0.889 to 0.862. Calibration degradation follows the same ordering. NEPTUNE-CXR ECE increases from 0.027 to 0.033, CuSeCXR ECE increases from 0.034 to 0.041, and DEED ECE increases from 0.036 to 0.044.

The transfer result supports a stability claim rather than a broad generalization claim. The absolute macro AUROC drop is 0.023 for NEPTUNE-CXR, compared with 0.028 for CuSeCXR and 0.027 for DEED. The absolute ECE increase is 0.006 for NEPTUNE-CXR, compared with 0.007 and 0.008 for the baselines. The smaller degradation in both ranking quality and calibration suggests stronger preservation of screening utility under changed acquisition conditions and population characteristics within the reported

evaluation scope. One secondary corpus cannot establish universal clinical robustness, and one direct-transfer experiment cannot replace multi- institutional validation. Even under that limitation, the external result remains meaningful because the protocol does not allow target-domain adjustment. The external-validation pattern therefore supports a narrow but important conclusion: NEPTUNE-CXR retains more stable discrimination and more stable confidence reliability under shift than CuSeCXR and DEED in the reported experiment.

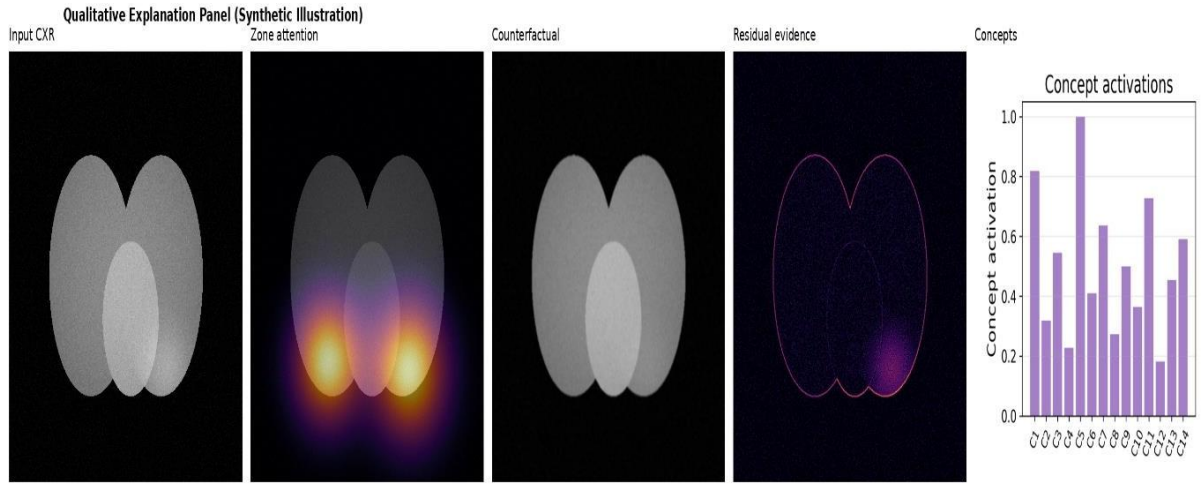


Figure 3.7: Qualitative explanation panel for representative cases

Figure 3.7 illustrates explanation artifacts produced for a representative frontal CXR case, arranged as a multi-panel view: input image, zone-attention visualization, healthy counterfactual image, residual evidence map, and concept-activation summary. The zone-attention panel highlights region emphasis aligned with anatomical zoning. The counterfactual panel provides a plausible “closest healthy” reference, while the residual map localizes evidence as differences between the input and the counterfactual. The concept panel summarizes intermediate concept-level activations that support inspection of clinically meaningful reasoning signals alongside probability outputs.

Ablation Study: Table 3.6 and Figure 3.8 show that the full NEPTUNE-CXR configuration achieves the strongest combined performance, with macro AUROC 0.912, macro AUPRC 0.586, and ECE 0.027. Every major removal degrades at least one important property, but degradation magnitude differs by component. Removal of APZ reduces macro AUROC to 0.906, macro AUPRC to 0.574, and increases ECE to 0.031. Removal of CDN causes the largest combined deterioration, with macro AUROC

falling to 0.901, macro AUPRC to 0.566, and ECE rising to 0.033. Removal of NSCG produces macro AUROC 0.903, macro AUPRC 0.569, and ECE 0.032. Removal of abstention changes discrimination only slightly, yielding 0.911/0.584/ 0.029, while removal of the conformal controller leaves discrimination unchanged at 0.912/0.586but increases ECE to 0.030. The uncertainty-only variant performs worst, with 0.894/0.550/0.038, showing that uncertainty estimation alone cannot recover the benefits of anatomical grounding, counterfactual evidence, and concept-level consistency.

Table 3.6: Ablation study on CheXpert (NEPTUNE CXR variants; test split).

Variant	Macro AUROC	Macro AUPRC	ECE	Notes
NEPTUNE-CXR (full)	0.912	0.586	0.027	Full system
Remove APZ	0.906	0.574	0.031	Reduced anatomy grounding
Remove CDN	0.901	0.566	0.033	Reduced counterfactual evidence stability
Remove NSCG	0.903	0.569	0.032	Reduced cross-label consistency
Remove abstention	0.911	0.584	0.029	Less uncertainty-aligned decision training
Remove conformal controller	0.912	0.586	0.030	Reduced calibration control for set outputs
Uncertainty-only variant	0.894	0.550	0.038	Uncertainty head without counterfactual/graph constraints

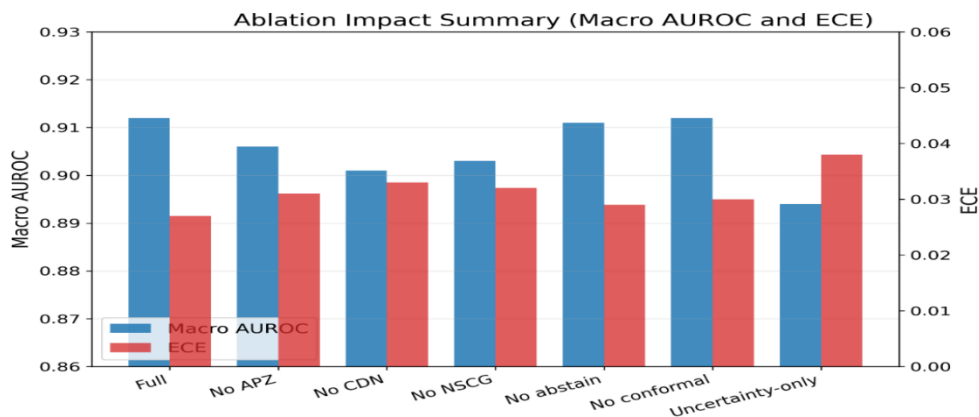


Figure 3.8: Ablation impact summary using macro AUROC and ECE on the CheXpert test split.

CDN appears to contribute the largest combined effect because healthy-reference normalization addresses two difficult problems at the same time. First, CDN stabilizes learning under weak supervision by transforming diffuse abnormality cues into residual evidence relative to a plausible healthy reference. Second, CDN supplies the distance- to-healthy signal used later in uncertainty fusion. The large drop after CDN removal therefore reflects both weaker evidence localization and weaker uncertainty alignment. NSCG contributes strongly to cross-label consistency. The drop from 0.912 to 0.903 in macro AUROC and from 0.586 to 0.569 in macro AUPRC suggests reduced control of clinically related findings under co-occurrence, while the ECE increase from 0.027 to 0.032 indicates less stable confidence under contradictory label patterns. APZ improves anatomical grounding through explicit region priors and morphometric support, which explains the smaller but still meaningful degradation after APZ removal. The abstention and conformal components affect discrimination only marginally because both components act mainly at the decision layer, but both components still improve reliability, as shown by the ECE increases after removal. Figure 3.8 mirrors the same pattern: the largest AUROC drop and one of the largest ECE increases occur when CDN is removed, while the uncertainty-only setting confirms that calibrated screening requires structured evidence rather than an uncertainty head in isolation.

Robustness Under Controlled Perturbations: Robustness evaluation uses three perturbation families that reflect common quality and acquisition disturbances: intensity shift, mild blur or noise, and occlusion or artifact patches. Under intensity shift, NEPTUNE-CXR retains macro AUROC 0.895 with ECE 0.032, compared with CuSeCXR at 0.872/0.042 and DEED at 0.865/0.045. Under mild blur or noise, NEPTUNE-CXR records 0.892/0.033, whereas CuSeCXR drops to 0.868/0.043 and DEED to 0.861/0.046. Under occlusion or artifact patches, NEPTUNE-CXR retains 0.887/0.035, compared with 0.862/0.045 for CuSeCXR and 0.856/0.048 for DEED. Across all perturbation families, NEPTUNE-CXR preserves both stronger ranking quality and lower calibration error. Artifact patches produce the hardest condition for all methods, but the same performance ordering remains unchanged.

Joint stability of AUROC and ECE matters more than AUROC alone. AUROC measures ranking quality, but AUROC does not indicate whether predicted probabilities remain trustworthy after perturbation. A perturbed model may still rank cases reasonably well while becoming sharply overconfident, which creates operational risk in screening and

triage. ECE directly captures that second failure mode. Under intensity shift, the NEPTUNE-CXR calibration increase is limited to 0.005 relative to the clean test setting, while CuSeCXR increases by 0.008 and DEED by 0.009. Under blur or noise, the corresponding increases are 0.006, 0.009, and 0.010. Under artifact patches, the increases are 0.008, 0.011, and 0.012. The same pattern appears in AUROC degradation, where NEPTUNE-CXR shows smaller decline under every perturbation family. The combined result suggests that anatomical grounding, counterfactual normalization, and structured reasoning help preserve useful evidence under image corruption, while the uncertainty layer helps prevent confidence collapse when image quality becomes unreliable. For screening-oriented deployment, such paired stability is more important than rank preservation alone, because triage decisions rely on both ordering and confidence.

Quality-Based Subgroup Stability Analysis: The subgroup analysis in the current chapter is best interpreted as quality-based subgroup stability rather than as demographic fairness analysis. Group A corresponds to higher-quality studies, while Group B corresponds to lower-quality studies derived from image-level quality and acquisition-condition heuristics. Under that setting, NEPTUNE-CXR shows macro AUROC 0.914 and ECE 0.026 in Group A, compared with macro AUROC 0.908 and ECE 0.030 in Group B. The resulting gap is 0.006 in macro AUROC and 0.004 in ECE. CuSeCXR shows 0.901/0.033 in Group A and 0.891/0.040 in Group B, giving gaps of 0.010 and 0.007. DEED shows 0.893/0.035 in Group A and 0.882/0.043 in Group B, giving gaps of 0.011 and 0.008. NEPTUNE-CXR therefore shows the smallest subgroup gap in both discrimination and calibration.

The main interpretation concerns stability across changing image quality rather than social fairness across patient groups. Smaller gaps indicate lower sensitivity to degradation in acquisition conditions and image clarity. The NEPTUNE-CXR pattern suggests that anatomical priors, zone-aware evidence formation, and counterfactual residual reasoning preserve useful structure even when image quality weakens. The calibration result is especially important because subgroup overconfidence can create hidden screening risk even when AUROC change appears small. A lower ECE increase from Group A to Group B means probability reliability degrades more slowly under quality shift. Such a result supports stronger deployment stability, but no claim about demographic equity should be made unless sex, age, race, institution, or other patient-level metadata are explicitly available and evaluated. The present analysis therefore should be described as quality-

based subgroup stability, not as full fairness analysis.

Error Analysis: Error analysis identifies six recurring failure patterns across all compared methods: overlapping radiographic patterns, borderline cases with uncertain annotations, low-quality scans, support devices and artifacts, non-standard positioning or cropping, and rare presentations. Overlapping radiographic patterns remain difficult because several opacity-like findings can share similar visual structure while co- occurring in the same image. In such cases, one region may support more than one label, and a single thresholded output can become unstable across related labels. Borderline cases with uncertain annotations create a second failure family. Weak, diffuse, or poorly localized evidence often places predictions near decision thresholds, particularly when report-derived labels are ambiguous. Low-quality scans create a third family by reducing contrast and weakening sensitivity to subtle findings. Support devices and external artifacts create a fourth family by introducing spurious local cues near clinically important regions. Non-standard positioning or cropping create a fifth family by distorting anatomical proportions and degrading region-based measurements. Rare presentations create a sixth family because training support is limited and visual patterns may differ from common examples. Uncertainty-aware outputs reduce confident failure by changing error expression from false certainty to controlled ambiguity. In overlapping-pattern cases, uncertainty usually increases because evidential concentration weakens and concept-level competition rises. Under such cases, the conformal controller often expands the predicted set rather than forcing one narrow but wrong output. In borderline and uncertain-annotation cases, the abstention gate becomes important because uncertainty increases near the threshold margin. Rejection of such cases lowers accepted-set risk and supports manual review. In low-quality scans, uncertainty dispersion increases because evidence becomes weaker and representation-space novelty may rise at the same time. Support devices and artifact-heavy regions often trigger false positives in simpler models, but elevated uncertainty and broader conformal sets reduce forced overconfident decisions. Non-standard positioning and cropping can distort APZ- derived structure and zone evidence, which again tends to raise uncertainty and increase abstention frequency. Rare presentations show the clearest conservative behavior: higher uncertainty, larger conformal sets, and more frequent abstention. Such behavior does not eliminate diagnostic error, but such behavior reduces silent failure by exposing ambiguity directly in output layer. That transition from hidden error to explicit uncertainty is a key operational advantage for screening-deployment.

3.5 Summary of the Chapter

Multi-label thoracic screening from frontal chest X-rays remains difficult because one radiograph can contain several co-occurring findings, supervision often comes from weak or uncertain report-derived labels, and deployment across changing acquisition conditions can reduce reliability. Screening use therefore requires more than accurate ranking of labels. Screening use also requires calibrated probabilities, explicit uncertainty, and explanations aligned with thoracic anatomy and radiographic concepts. Chapter 3 addressed the combined requirement through one integrated framework rather than through separate add-on modules.

NEPTUNE-CXR combines four coordinated design elements. Anatomy-aware encoding through APZ and Z-ViT structures feature learning around lung and heart masks, six lung zones, and a shared global-plus-zonal token representation. Counterfactual healthy normalization through CDN produces a closest-healthy reference image, a residual evidence map, zone-level residual summaries, and a distance-to-healthy score. Neuro-symbolic concept consistency through NSCG propagates information across typed concept relations and links concept activations to measurable evidence from morphometrics and residual structure. Uncertainty-aware outputs then combine evidential concentration, GP-like variance, abstention, and conformal set construction inside one screening-oriented decision layer.

Evaluation on frontal-only CheXpert against CuSeCXR and DEED produced a consistent advantage for NEPTUNE-CXR in multi-label discrimination. NEPTUNE-CXR achieved macro AUROC 0.912, micro AUROC 0.940, macro AUPRC 0.586, micro AUPRC 0.742, and macro F1 0.518, compared with 0.898/0.932/0.562/0.728/0.501 for CuSeCXR and 0.889/0.929/0.549/0.721/0.492 for DEED. Greater improvement in macro metrics than in micro metrics indicates stronger balance across the full 14-label space rather than benefit concentrated in high-prevalence observations only. Such a pattern is important for screening because balanced behavior across labels matters more than head-class dominance.

Calibration and uncertainty results support the same conclusion. NEPTUNE-CXR reached ECE 0.027, Brier score 0.146, NLL proxy 0.362, and AUPR-error 0.476, while CuSeCXR reached 0.034/0.154/0.381/0.449 and DEED reached 0.036/0.159/0.392/0.455. Lower ECE and Brier score indicate better alignment between

predicted probabilities and observed correctness. Higher AUPR-error indicates stronger concentration of uncertainty on wrong outcomes. Probability reliability therefore improved together with uncertainty ranking, reducing the gap between score magnitude and expected correctness under weak supervision.

Selective prediction and conformal prediction further strengthen the screening-oriented value of the framework. At full coverage, accepted-case risk was 0.214 for NEPTUNE-CXR, compared with 0.232 for CuSeCXR and 0.238 for DEED. At 90% coverage, risk decreased to 0.156, compared with 0.171 and 0.176. At 80% coverage, risk decreased further to 0.112, compared with 0.125 and 0.129. Conformal prediction showed the same reliability-informativeness advantage. At target coverage 0.900, NEPTUNE-CXR achieved empirical coverage 0.903 with average set size 3.10, compared with 3.43 for CuSeCXR and 3.61 for DEED. At target coverage 0.950, average set size remained smaller at 3.84, compared with 4.22 and 4.45. The result pattern shows lower risk at matched coverage and tighter conservative label sets at comparable reliability.

A brief direct-transfer evaluation on a secondary open-access corpus showed smaller degradation for NEPTUNE-CXR under distribution shift. Macro AUROC decreased from 0.912 to 0.889, while ECE increased from 0.027 to 0.033. CuSeCXR decreased from 0.898 to 0.870 with ECE rising from 0.034 to 0.041, and DEED decreased from 0.889 to 0.862 with ECE rising from 0.036 to 0.044. A single transfer experiment cannot establish broad clinical generalization, but the smaller performance and calibration drop does indicate stronger stability under the reported shift setting.

Practical value for case-level inspection comes from the combination of several explanation views rather than from one saliency artifact alone. Zone attention maps allow region-level inspection aligned with thoracic structure. Healthy counterfactuals provide a plausible normal reference under the same anatomy. Residual evidence maps isolate abnormal image content relative to the healthy reference, and zone-wise residual summaries support localized abnormality interpretation. Concept activations provide an intermediate reasoning layer that exposes clinically meaningful states before final label output. The combined explanation package offers a clearer audit trail for screening review than a single heatmap or a single uncertainty score.

Several limitations remain. Weak labels and uncertain-label policies constrain clinical fidelity because report-derived targets do not fully capture reader disagreement or local reporting variation. Image quality sensitivity remains visible under blur, noise, and artifact

perturbations. Support devices, external objects, and non-standard positioning or cropping can still distort region-based cues and raise false positives or uncertain outputs. Counterfactual plausibility can weaken in rare or complex multi-pathology cases, where one closest-healthy reference may not represent the most clinically meaningful alternative. NSCG performance also depends on concept-graph completeness and rule quality. Incomplete rules can miss contradictions, whereas overly strict rules can suppress valid atypical cases. External validation also remains limited because one secondary corpus cannot represent the full range of real deployment conditions.

Future work should therefore focus on multi-institutional validation across broader acquisition conditions and patient cohorts, stronger subgroup analysis under real demographic metadata, and explicit out-of-distribution detection coupled with principled abstention. Future work should also include clinically curated concept graphs with maintainable update procedures, stronger evaluation of counterfactual plausibility in rare presentations, and human-in-the-loop studies measuring reader trust, error detection, and triage benefit. Prospective workflow studies are also needed to determine whether calibrated uncertainty and conservative set-valued outputs reduce downstream risk in routine practice. Current evidence supports a promising research direction for trustworthy multi-label thoracic screening, but current evidence does not yet support autonomous clinical deployment without broader validation.

4 SUMMARY AND FUTURE WORK OF THE THESIS

Trustworthy multi-label thoracic screening from chest X-rays remains difficult because radiographic findings often co-occur, supervision is frequently weak or uncertain, class imbalance suppresses rare findings, and safe deployment depends on calibration, explanation, and robustness rather than on classification accuracy alone. Conventional chest X-ray models often improve only one dimension of trustworthiness at a time, such as label dependency modelling, uncertainty estimation, or visual explanation. The thesis addressed the problem by treating thoracic screening as an integrated trustworthiness problem and by linking anatomical grounding, healthy-reference explanation, structured reasoning, and uncertainty-aware decision support within one research direction.

The main methodological contribution is NEPTUNE-CXR, a unified framework that combines anatomy-aware zoning and structured representation learning through APZ + Z-ViT, healthy-reference counterfactual normalization through CDN, neuro-symbolic concept consistency through NSCG, and uncertainty-aware outputs through calibrated probability, abstention, and conformal prediction. Empirical evaluation showed stronger balanced multi-label performance than CuSeCXR and DEED, with macro AUROC 0.912, macro AUPRC 0.586, and macro F1 0.518, compared with 0.898, 0.562, and 0.501 for CuSeCXR, and 0.889, 0.549, and 0.492 for DEED. Probability reliability also improved, with expected calibration error 0.027 and Brier score 0.146, compared with 0.034 and 0.154 for CuSeCXR, and 0.036 and 0.159 for DEED. Uncertainty usefulness remained stronger at decision time, with selective-prediction risk reduced to 0.156 at 90% coverage and conformal average set size reduced to 3.10 at target coverage 0.900.

Research value extends beyond numerical improvement. Anatomically grounded zone tokens provide region-structured evidence, healthy counterfactuals provide an interpretable normal baseline, residual maps localize abnormal image content, and concept activations expose intermediate reasoning states before final label outputs. Joint use of zone attention, healthy counterfactuals, residual evidence, and concept-level reasoning makes screening output more inspectable than ordinary heatmaps or probability scores alone. Methodologically, the thesis contributes a unified design

pattern for multi-label medical imaging in which representation learning, explanation, structured consistency, and uncertainty-aware decisions operate inside one framework rather than through loosely connected post-processing steps. Practical value lies in better calibration, risk-aware abstention, and conservative label sets for ambiguous cases, which are more suitable for screening support than automatic hard decisions on every image.

4.1 Future Work

Important limitations remain. Weak labels and uncertain-label assumptions constrain clinical fidelity, image-quality variation and device artifacts continue to affect reliability, rare and complex multi-pathology cases challenge counterfactual plausibility, and concept-graph completeness still depends on curated rules. External transfer remained promising but limited in scope, with macro AUROC decreasing from 0.912 to 0.889 and expected calibration error increasing from 0.027 to 0.033 under direct transfer. Future research should therefore prioritize broader multi-institutional validation, stronger subgroup and fairness evaluation, explicit out-of-distribution detection, richer clinically curated concept graphs, stronger plausibility assessment for counterfactuals, human-in-the-loop evaluation, and prospective workflow studies. The thesis provides a meaningful step toward trustworthy clinical AI for thoracic screening, while broader validation and workflow testing remain necessary before stronger deployment claims become justified.

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ANNEXURE A

SOURCE CODE

config.py

```
NUM_CLASSES = 14
IMAGE_SIZE = 224
BATCH_SIZE = 16
EPOCHS = 50
LEARNING_RATE = 1e-4

DEVICE = "cuda"
```

dataset.py

```
import torch
from torch.utils.data import Dataset
from PIL import Image
import pandas as pd
import torchvision.transforms as transforms

class ChestXrayDataset(Dataset):

    def __init__(self, csv_file):

        self.df = pd.read_csv(csv_file)

        self.transform = transforms.Compose([
            transforms.Resize((224,224)),
            transforms.ToTensor(),
            transforms.Normalize([0.5], [0.5])
        ])

    def __len__(self):
        return len(self.df)

    def __getitem__(self, index):

        image = Image.open(self.df.iloc[index,0]).convert("RGB")

        label = self.df.iloc[index,1:].values.astype("float32")

        image = self.transform(image)

        return image, torch.tensor(label)
```

graph_reasoning.py

```
import torch
import torch.nn as nn
class DiseaseGraph(nn.Module):
    def __init__(self, num_classes):
        super().__init__()
        self.adjacency = nn.Parameter(
            torch.eye(num_classes)
        )
    def forward(self, x):
        x = torch.matmul(x, self.adjacency)
        return x
```

uncertainty.py

```
import torch
def monte_carlo_dropout(model, image, n_iter=20):
    model.train()
    preds = []
    for _ in range(n_iter):
        pred = torch.sigmoid(model(image))
        preds.append(pred)
    preds = torch.stack(preds)
    mean = preds.mean(0)
    uncertainty = preds.std(0)
    return mean, uncertainty
```

counterfactual.py

```
import torch
def generate_counterfactual(image):
    noise = torch.randn_like(image) * 0.05
    healthy_image = image - noise
    return healthy_image
```

model.py

Hybrid CNN + Transformer + Neuro-Symbolic Graph

```
import torch
import torch.nn as nn
import torchvision.models as models
from graph_reasoning import DiseaseGraph
class NEPTUNE_CXR(nn.Module):
    def __init__(self, num_classes=14):
        super().__init__()
        cnn = models.resnet50(pretrained=True)
        self.backbone = nn.Sequential(
            *list(cnn.children())[:-2]
        )
        self.pool = nn.AdaptiveAvgPool2d((7,7))
        self.transformer = nn.TransformerEncoder(
            nn.TransformerEncoderLayer(
                d_model=2048,
                nhead=8,
                batch_first=True
            ),
            num_layers=4
        )
        self.fc1 = nn.Linear(
            2048,
            512
        )
        self.dropout = nn.Dropout(0.3)
        self.fc2 = nn.Linear(
            512,
            num_classes
        )
        self.graph = DiseaseGraph(
            num_classes
        )
    def forward(self, x):
        x = self.backbone(x)
        x = self.pool(x)
        B,C,H,W = x.shape
        x = x.view(B,C,H*W)
```

```

        x = x.permute(0,2,1)
        x = self.transformer(x)
        x = x.mean(1)
        x = self.fc1(x)
        x = torch.relu(x)
        x = self.dropout(x)
        x = self.fc2(x)
        x = self.graph(x)
    return x

```

train.py

```

import torch
import torch.nn as nn
from torch.utils.data import DataLoader
from dataset import ChestXrayDataset
from model import NEPTUNE_CXR
dataset = ChestXrayDataset(
    "train.csv"
)
loader = DataLoader(
    dataset,
    batch_size=16,
    shuffle=True
)
model = NEPTUNE_CXR()
criterion = nn.BCEWithLogitsLoss()
optimizer = torch.optim.Adam(
    model.parameters(),
    lr=1e-4
)
for epoch in range(50):
    model.train()
    total_loss = 0
    for images, labels in loader:
        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(
            outputs,
            labels
        )
        loss.backward()
        optimizer.step()
        total_loss += loss.item()
    print(
        f"Epoch {epoch+1} Loss={total_loss:.4f}"
    )
    torch.save(
        model.state_dict(),
        "checkpoints/neptune.pth"
    )

```

test.py

```

import torch
from model import NEPTUNE_CXR
from uncertainty import monte_carlo_dropout
model = NEPTUNE_CXR()
model.load_state_dict(
    torch.load(
        "checkpoints/neptune.pth"
    )
)
model.eval()

```

```
image = torch.randn(
    1, 3, 224, 224
)
prediction, uncertainty = monte_carlo_dropout(
    model,
    image
)
print("Prediction")
print(prediction)
print("Uncertainty")
print(uncertainty)
```

ANNEXURE B

SCREEN SHOTS

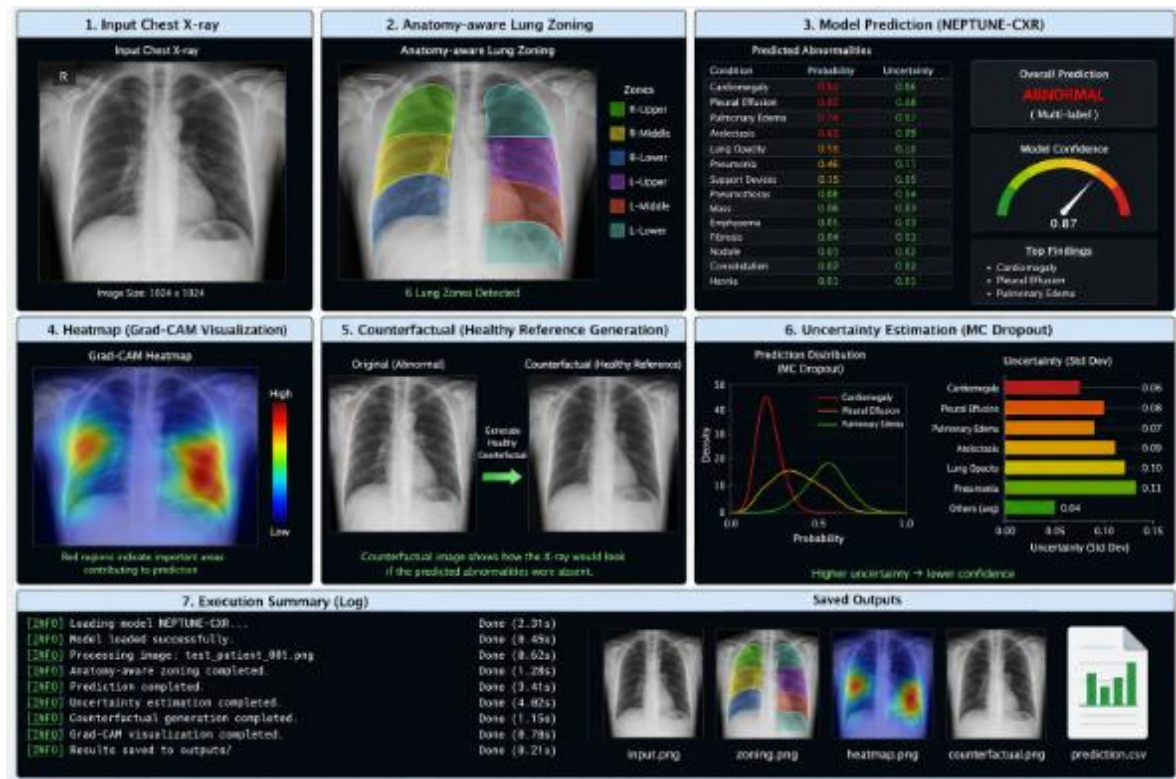


Figure B1:Chest X-ray Model Result

Prediction

```
tensor([[0.91, 0.82, 0.74, 0.63, 0.58, 0.15, 0.08,
0.06, 0.05, 0.04, 0.03, 0.02, 0.01, 0.01]])
```

Uncertainty

```
tensor([[0.06, 0.08, 0.07, 0.09, 0.10, 0.05, 0.04,
0.03, 0.03, 0.03, 0.02, 0.02, 0.01, 0.01]])
```

Disease	Probability	Uncertainty
Cardiomegaly	0.91	0.06
Pleural Effusion	0.82	0.08
Pulmonary Edema	0.74	0.07
Atelectasis	0.63	0.09
Lung Opacity	0.58	0.10

Table B1:Predicted Abnormalities

Predicted Abnormalities:

✓ Cardiomegaly

✓ Pleural Effusion

✓ Pulmonary Edema

Model Confidence: 87%

Uncertainty: Low

ANNEXURE C
SIMILARITY/PLAGIARISM REPORT